1 Fig. 10 shows a sketch of the curve $y = x^2 - 4x + 3$. The point A on the curve has x-coordinate 4. At point B the curve crosses the x-axis.

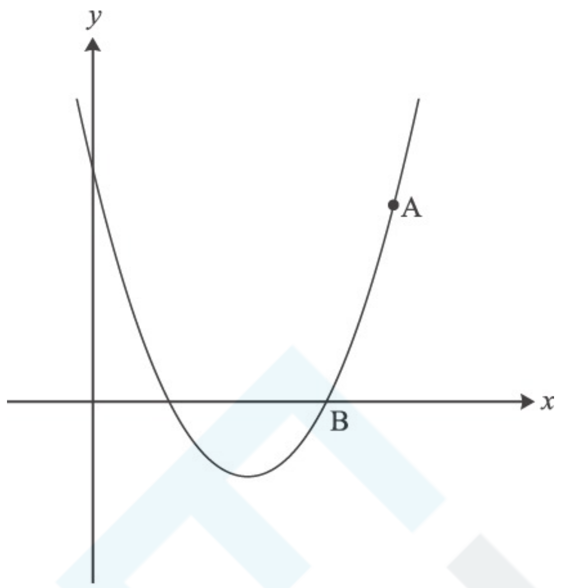


Fig. 10

(i) Use calculus to find the equation of the normal to the curve at A and show that this normal intersects the x-axis at C (16, 0).

[6]

(ii) Find the area of the region ABC bounded by the curve, the normal at A and the x-axis.

[5]

2 Differentiate $2x^3 + 9x^2 - 24x$. Hence find the set of values of x for which the function $f(x) = 2x^3 + 9x^2 - 24x$ is increasing.

[4]

3 A and B are points on the curve $y = 4\sqrt{x}$. Point A has coordinates (9, 12) and point B has x-coordinate 9.5. Find the gradient of the chord AB.

The gradient of AB is an approximation to the gradient of the curve at A. State the x-coordinate of a point C on the curve such that the gradient of AC is a closer approximation.

[3]

4 Find $\int 30x^{\frac{3}{2}}\,dx$.

[3]

5 Fig. 9 shows a sketch of the curve $y = x^3 - 3x^2 - 22x + 24$ and the line $y = 6x + 24$.

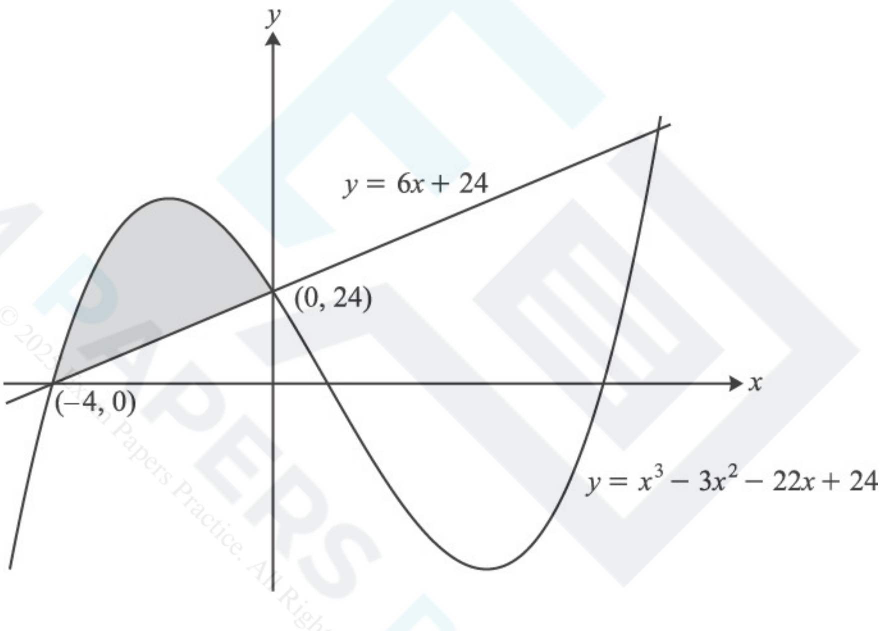


Fig. 9

(i) Differentiate $y = x^3 - 3x^2 - 22x + 24$ and hence find the x-coordinates of the turning points of the curve. Give your answers to 2 decimal places.

[4]

(ii) You are given that the line and the curve intersect when $x = 0$ and when $x = -4$. Find algebraically the x -coordinate of the other point of intersection.

[3]

(iii) Use calculus to find the area of the region bounded by the curve and the line $y = 6x + 24$ for $-4 \leqslant x \leqslant 0$, shown shaded on Fig. 9.

[4]

6

The gradient of a curve is given by $\frac{dy}{dx} = \frac{18}{x^3} + 2$. The curve passes through the point (3, 6). Find the equation of the curve.

[5]

7

Find $\frac{dy}{dx}$ when

(i) $y = 2x^{-5}$,

[2]

(ii) $y = \sqrt[3]{x}$.

[3]

8 Oskar is designing a building. Fig. 12 shows his design for the end wall and the curve of the roof. The units for x and y are metres.

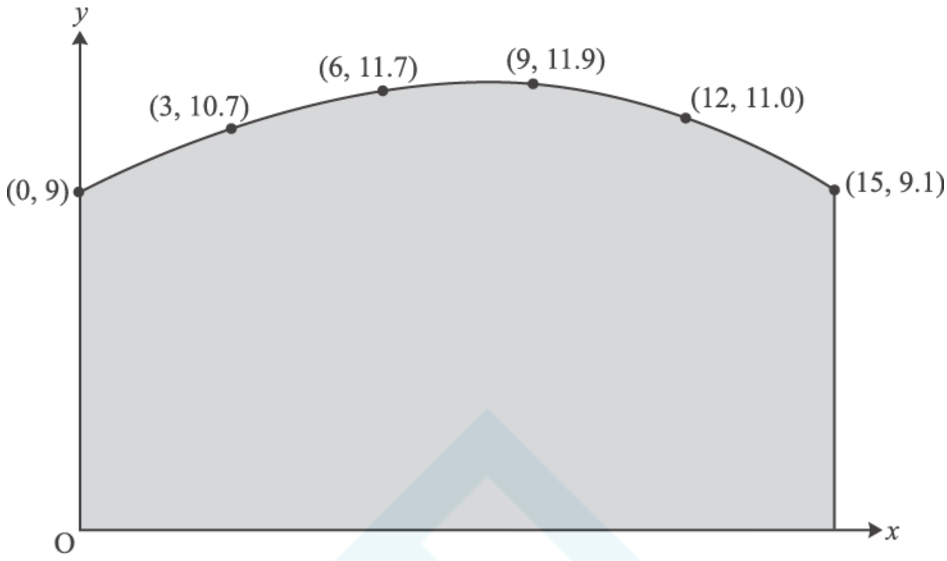


Fig. 12

(i) Use the trapezium rule with 5 strips to estimate the area of the end wall of the building.

[4]

(ii) Oskar now uses the equation $y = -0.001x^3 - 0.025x^2 + 0.6x + 9$, for $0 \leq x \leq 15$, to model the curve of the roof.

(*A*) Calculate the difference between the height of the roof when $x = 12$ given by this model and the data shown in Fig. 12.

[2]

(*B*) Use integration to find the area of the end wall given by this model.

[4]

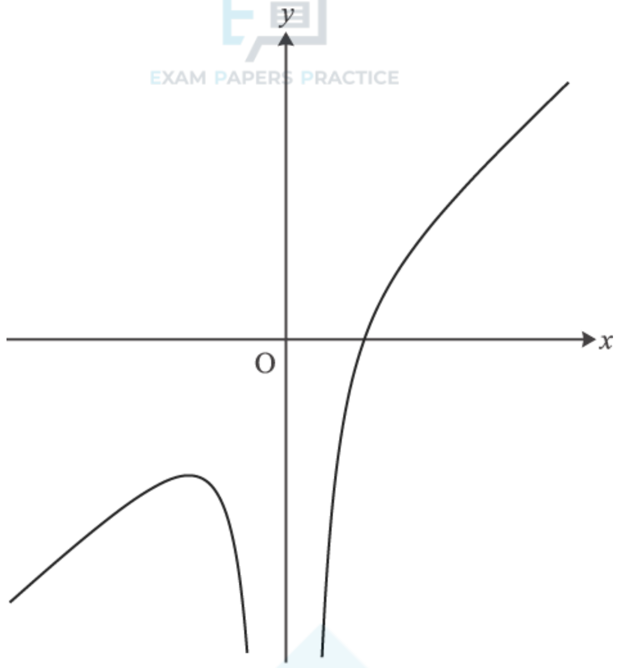


Fig. 11

Fig. 11 shows a sketch of the curve with equation $y = x - \frac{4}{x^2}$.

(i) Find $\frac{dy}{dx}$ and show that $\frac{d^2y}{dx^2} = -\frac{24}{x^4}$.

[3]

(ii) Hence find the coordinates of the stationary point on the curve. Verify that the stationary point is a maximum.

[5]

(iii) Find the equation of the normal to the curve when $x = -1$. Give your answer in the form $ax + by + c = 0$.

[5]

10 The points P (2, 3.6) and Q (2.2, 2.4) lie on the curve $y = f(x)$. Use P and Q to estimate the gradient of the curve at the point where $x = 2$.

[2]

11

Find $\int 7x^{\frac{5}{2}}\mathrm{d}x$.

[3]

12

The gradient of a curve is given by $\frac{\mathrm{d}y}{\mathrm{d}x} = 4x + 3$. The curve passes through the point (2, 9).

(i) Find the equation of the tangent to the curve at the point (2, 9).

[3]

(ii) Find the equation of the curve and the coordinates of its points of intersection with the x-axis. Find also the coordinates of the minimum point of this curve.

[7]

(iii) Find the equation of the curve after it has been stretched parallel to the x-axis with scale factor $\frac{1}{2}$. Write down the coordinates of the minimum point of the transformed curve.

[3]

13 Use calculus to find the set of values of x for which $x^3 - 6x$ is an increasing function.

[5]

14

(i) Calculate the gradient of the chord of the curve $y = x^2 - 2x$ joining the points at which the values of x are 5 and 5.1.

[2]

(ii) Given that $f(x) = x^2 - 2x$, find and simplify $\frac{f(5+h) - f(5)}{h}$.

[4]

(iii) Use your result in part (ii) to find the gradient of the curve $y = x^2 - 2x$ at the point where $x = 5$, showing your reasoning.

[2]

(iv) Find the equation of the tangent to the curve $y = x^2 - 2x$ at the point where $x = 5$.
Find the area of the triangle formed by this tangent and the coordinate axes.

[5]

15 Fig. 9 shows the cross-section of a straight, horizontal tunnel. The x-axis from 0 to 6 represents the floor of the tunnel.

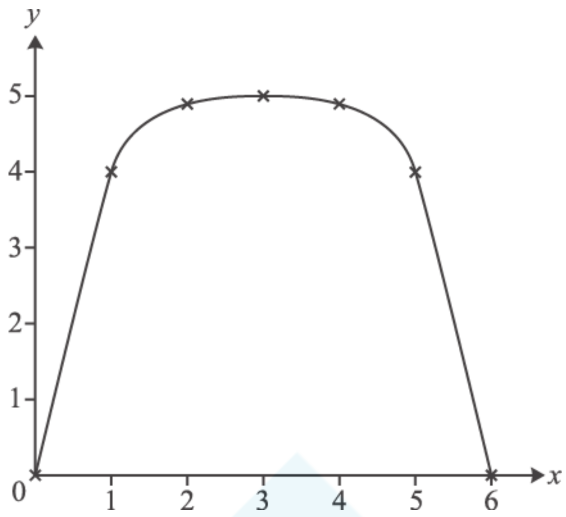


Fig. 9

With axes as shown, and units in metres, the roof of the tunnel passes through the points shown in the table.

x	0	1	2	3	4	5	6
y	0	4.0	4.9	5.0	4.9	4.0	0

The length of the tunnel is 50 m.

(i) Use the trapezium rule with 6 strips to estimate the area of cross-section of the tunnel. Hence estimate the volume of earth removed in digging the tunnel.

[4]

(ii) An engineer models the height of the roof of the tunnel using the curve . This curve is symmetrical about x = 3.

(A) Show that, according to this model, a vehicle of rectangular cross-section which is 3.6 m wide and 4.4 m high would not be able to pass through the tunnel.

[2]

(B) Use integration to calculate the area of the cross-section given by this model. Hence obtain another estimate of the volume of earth removed in digging the tunnel.

[5]

(i) Find $\frac{dy}{dx}$ when $y = 6\sqrt{x}$.

[2]

(ii) Find $\int \frac{12}{x^2}\,dx$.

[3]

17 Fig. 9 shows the line $y = x$ and the curve $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = \frac{1}{2}(\mathrm{e}^x - 1)$. The line and the curve intersect at the origin and at the point P(a, a).

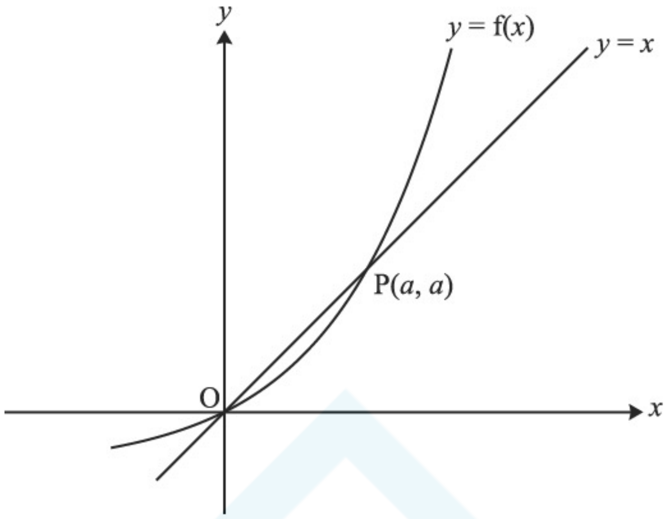


Fig. 9

(i) Show that $\mathrm{e}^a = 1 + 2a$.

[1]

(ii) Show that the area of the region enclosed by the curve, the x-axis and the line $x = a$ is $\frac{1}{2}a$. Hence find, in terms of a, the area enclosed by the curve and the line $y = x$.

[6]

(iii) Show that the inverse function of $\mathrm{f}(x)$ is $\mathrm{g}(x)$, where $\mathrm{g}(x) = \ln(1 + 2x)$. Add a sketch of $y = \mathrm{g}(x)$ to the copy of Fig. 9.

[5]

(iv) Find the derivatives of $\mathrm{f}(x)$ and $\mathrm{g}(x)$. Hence verify that $\mathrm{g}'(a) = \dfrac{1}{\mathrm{f}'(a)}$.

Give a geometrical interpretation of this result.

[7]

18 Fig. 8 shows parts of the curves $y = \mathrm{f}(x)$ and $y = \mathrm{g}(x)$, where $\mathrm{f}(x) = \tan x$ and $\mathrm{g}(x) = 1 + \mathrm{f}(x - \frac{1}{4}\pi)$.

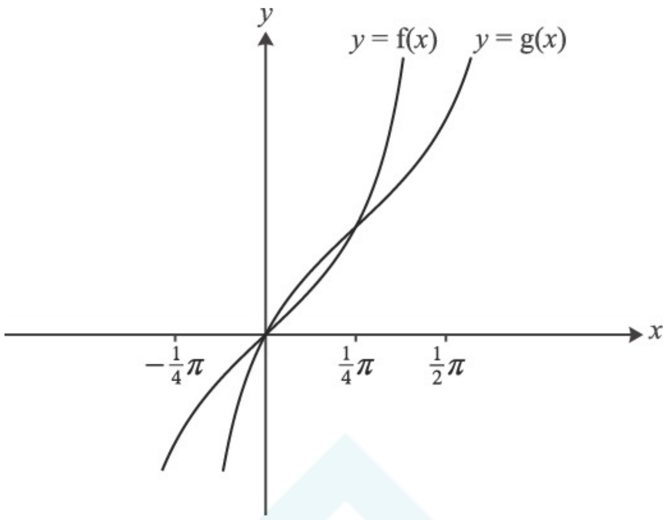


Fig. 8

(i) Describe a sequence of two transformations which maps the curve $y = \mathrm{f}(x)$ to the curve $y = \mathrm{g}(x)$.

[4]

It can be shown that $\mathrm{g}(x) = \dfrac{2\sin x}{\sin x + \cos x}$.

(ii) Show that $\mathrm{g}'(x) = \dfrac{2}{(\sin x + \cos x)^2}$. Hence verify that the gradient of $y = \mathrm{g}(x)$ at the point $(\frac{1}{4}\pi, 1)$ is the same as that of $y = \mathrm{f}(x)$ at the origin.

[7]

(iii) By writing $\tan x = \dfrac{\sin x}{\cos x}$ and using the substitution $u = \cos x$, show that $\displaystyle\int_0^{\frac{1}{4}\pi} \mathrm{f}(x)\,\mathrm{d}x = \int_{\frac{1}{\sqrt{2}}}^{1} \frac{1}{u}\,\mathrm{d}u$.

Evaluate this integral exactly.

[4]

(iv) Hence find the exact area of the region enclosed by the curve $y = \mathrm{g}(x)$, the x-axis and the lines

$x = \frac{1}{4}\pi$ and $x = \frac{1}{2}\pi$.

[2]

19

Evaluate $\int_0^3 x(x+1)^{-\frac{1}{2}}\,dx$, giving your answer as an exact fraction.

[5]

20

The driving force F newtons and velocity v km s^{-1} of a car at time t seconds are related by the equation $F = \frac{25}{v}$.

(i) Find $\frac{dF}{dv}$.

[2]

(ii) Find $\frac{dF}{dt}$ when $v = 50$ and $\frac{dv}{dt} = 1.5$.

[3]

21 A curve has equation $x^2 + 2y^2 = 4x$.

(i) By differentiating implicitly, find $\frac{dy}{dx}$ in terms of x and y.

[3]

(ii) Hence find the exact coordinates of the stationary points of the curve. [You need not determine their nature.]

[3]

22

(i) Given that $y = e^{-x} \sin 2x$, find $\frac{dy}{dx}$.

[3]

(ii) Hence show that the curve $y = e^{-x} \sin 2x$ has a stationary point when $x = \frac{1}{2}$ arctan 2.

[3]

23 Fig. 9 shows the curve with equation $y^3 = \dfrac{x^3}{2x-1}$. It has an asymptote $x = a$ and turning point P.

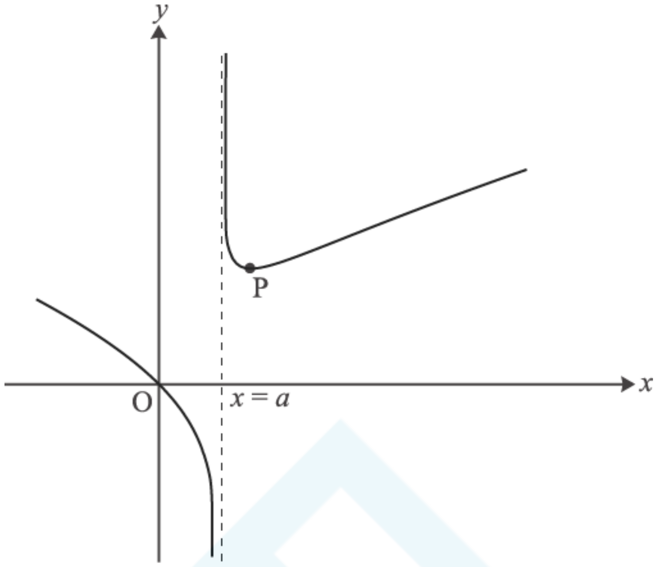


Fig. 9

(i) Write down the value of *a*.

[1]

(ii) Show that $\dfrac{dy}{dx} = \dfrac{4x^3 - 3x^2}{3y^2(2x-1)^2}$.

Hence find the coordinates of the turning point P, giving the *y*-coordinate to 3 significant figures.

[9]

(iii) Show that the substitution $u = 2x - 1$ transforms $\displaystyle\int \frac{x}{\sqrt[3]{2x-1}}\,dx$ to $\displaystyle\frac{1}{4}\int\left(u^{\frac{2}{3}} + u^{-\frac{1}{3}}\right)du$.

Hence find the exact area of the region enclosed by the curve $y^3 = \dfrac{x^3}{2x-1}$, the *x*-axis and the lines $x = 1$ and $x = 4.5$.

[8]

24 Fig. 8 shows the curve $y = f(x)$, where $f(x) = (1 - x)e^{2x}$, with its turning point P.

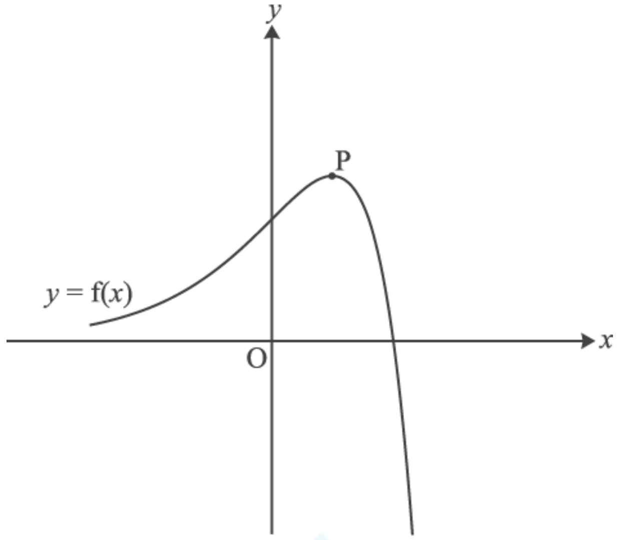


Fig. 8

(i) Write down the coordinates of the intercepts of $y = f(x)$ with the x- and y-axes.

[2]

(ii) Find the exact coordinates of the turning point P.

[6]

(iii) Show that the exact area of the region enclosed by the curve and the x- and y-axes is $\frac{1}{4}\left(e^2 - 3\right)$.

[5]

The function g(x) is defined by $g(x) = 3f\left(\frac{1}{2}x\right)$.

(iv) Express g(x) in terms of x.
Sketch the curve $y = g(x)$ on the copy of Fig. 8, indicating the coordinates of its intercepts with the x- and y-axes and of its turning point.

[4]

(v) Write down the exact area of the region enclosed by the curve $y = g(x)$ and the x- and y-axes.

[1]

25

Using a suitable substitution or otherwise, show that $\int_0^{\frac{1}{2}\pi} \frac{\sin 2x}{3 + \cos 2x} dx = \frac{1}{2} \ln 2$.

[5]

26

Given that $y = \ln\left(\sqrt{\frac{2x-1}{2x+1}}\right)$, show that $\frac{dy}{dx} = \frac{1}{2x-1} - \frac{1}{2x+1}$

[4]

27 Water flows into a bowl at a constant rate of 10 $cm^3 s^{-1}$ (see Fig. 4).

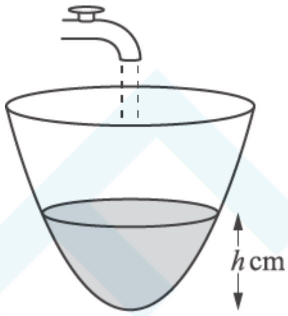


Fig. 4

When the depth of water in the bowl is h cm, the volume of water is V cm^3, where $V = \pi h^2$. Find the rate at which the depth is increasing at the instant in time when the depth is 5 cm.

[5]

28

The function f(x) is defined by f(x) = 1 – 2 sin$x$ for $-\frac{1}{2}\pi \leqslant x \leqslant \frac{1}{2}\pi$. Fig. 3 shows the curve y = f(x).

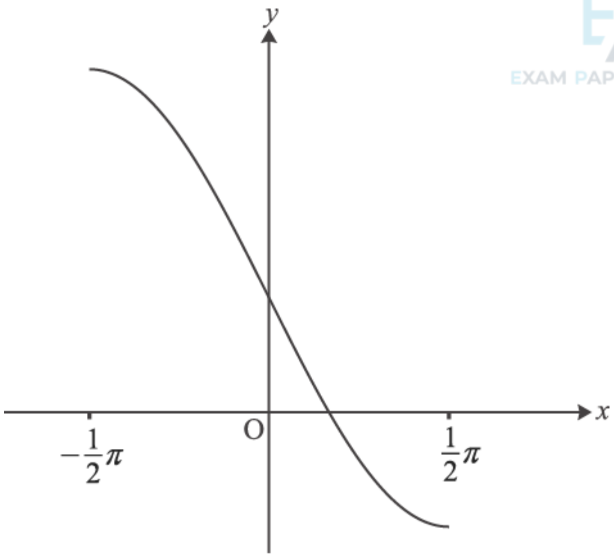


Fig. 3

(i) Write down the range of the function f(x).

[2]

(ii) Find the inverse function $f^{-1}(x)$.

[3]

(iii) Find $f'(0)$. Hence write down the gradient of $y = f^{-1}(x)$ at the point (1, 0).

[3]

29 Fig. 9 shows the curve $y = xe^{-2x}$ together with the straight line $y = mx$, where m is a constant, with $0 < m < 1$. The curve and the line meet at O and P. The dashed line is the tangent at P.

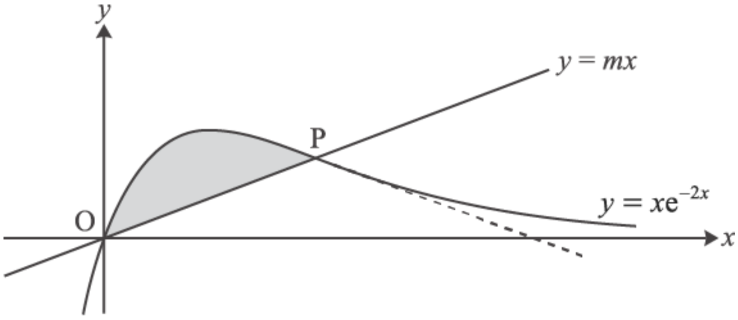


Fig. 9

(i) Show that the x–coordinate of P is $-\frac{1}{2}\ln m$.

[3]

(ii) Find, in terms of m, the gradient of the tangent to the curve at P.

[4]

You are given that OP and this tangent are equally inclined to the x-axis.

(iii) Show that $m = e^{-2}$, and find the exact coordinates of P.

[4]

(iv) Find the exact area of the shaded region between the line OP and the curve.

[7]

Fig. 8 shows the curve $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = \dfrac{x}{\sqrt{2+x^2}}$.

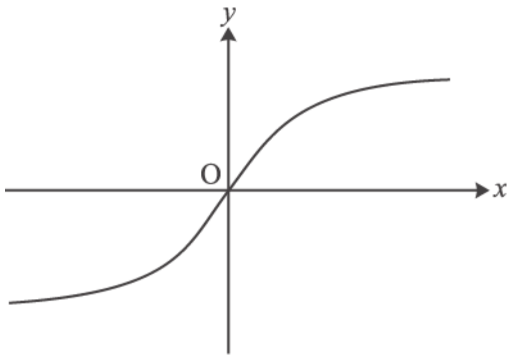


Fig. 8

(i) Show algebraically that f (x) is an odd function. Interpret this result geometrically.

[3]

(ii) Show that $\mathrm{f}'(x) = \dfrac{2}{(2+x^2)^{\frac{3}{2}}}$. Hence find the exact gradient of the curve at the origin.

[5]

(iii) Find the exact area of the region bounded by the curve, the x-axis and the line $x = 1$.

[4]

(iv)

A Show that if $y = \dfrac{x}{\sqrt{2+x^2}}$, then $\dfrac{1}{y^2} = \dfrac{2}{x^2} + 1$.

[2]

B Differentiate $\dfrac{1}{y^2} = \dfrac{2}{x^2} + 1$ implicitly to show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2y^3}{x^3}$. Explain why this expression cannot be used to find the gradient of the curve at the origin.

[4]

31

A spherical balloon of radius r cm has volume V cm^3, where $V = \frac{4}{3}\pi r^3$. The balloon is inflated at a constant rate of 10 cm^3 s^{-1}. Find the rate of increase of r when $r = 8$.

[5]

32

Find the exact gradient of the curve $y = \ln(1 - \cos 2x)$ at the point with x-coordinat $\frac{1}{6}\pi$.

[5]

33

Evaluate $\int_0^{\frac{1}{6}\pi} (1 - \sin 3x)\mathrm{d}x$ giving your answer in exact form.

[3]

34 Fig. 9 shows the curve $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (\mathrm{e}^x - 2)^2 - 1, \ x \in \mathbb{R}.$$

The curve crosses the x-axis at O and P, and has a turning point at Q.

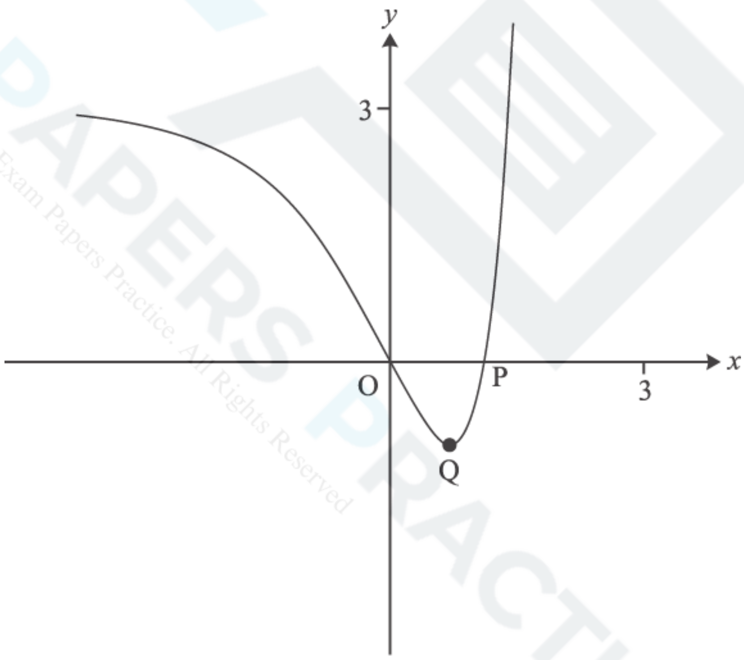


Fig. 9

(i) Find the exact x-coordinate of P.

[2]

(ii) Show that the x-coordinate of Q is ln 2 and find its y-coordinate.

[4]

(iii) Find the exact area of the region enclosed by the curve and the x-axis.

[5]

The domain of f(x) is now restricted to $x \geq \ln 2$.

(iv) Find the inverse function $f^{-1}(x)$. Write down its domain and range, and sketch its graph on the copy of Fig. 9.

[7]

35

Fig. 8 shows the line $y = 1$ and the curve $y = f(x)$, where $f(x) = \frac{(x-2)^2}{x}$. The curve touches the x-axis at P(2, 0) and has another turning point at the point Q.

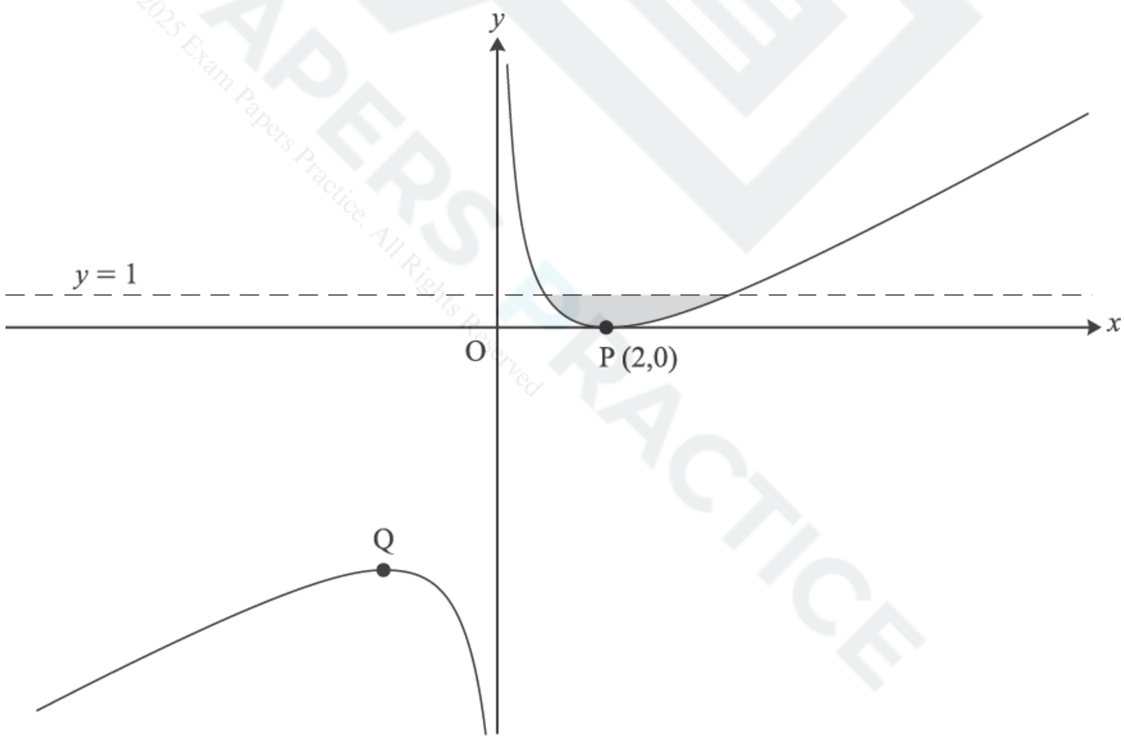


Fig. 8

(i) Show that $f'(x) = 1 - \frac{4}{x^2}$, and find $f''(x)$.

Hence find the coordinates of Q and, using $f''(x)$, verify that it is a maximum point.

[7]

(ii) Verify that the line $y = 1$ meets the curve $y = f(x)$ at the points with x-coordinates 1 and 4. Hence find the exact area of the shaded region enclosed by the line and the curve.

[6]

The curve y = f(x) is now transformed by a translation with vector $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$. The resulting curve has equation $y = g(x)$

(iii) Show that $g(x) = \frac{x^2 - 3x}{x + 1}$.

[3]

(iv) Without further calculation, write down the value of $\int_0^3 g(x)\,dx$, justifying your answer.

[2]

36 A curve has implicit equation $y^2 + 2x \ln y = x^2$.

Verify that the point (1, 1) lies on the curve, and find the gradient of the curve at this point.

[6]

37 Fig. 4 shows a cone with its axis vertical. The angle between the axis and the slant edge is 45°. Water is poured into the cone at a constant rate of 5cm^3 per second. At time t seconds, the height of the water surface above the vertex O of the cone is h cm, and the volume of water in the cone is V cm^3.

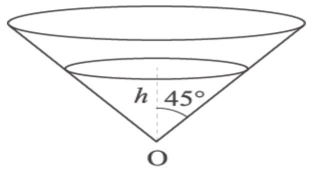


Fig. 4

Find V in terms of h.

Hence find the rate at which the height of water is increasing when the height is 10 cm.

[You are given that the volume V of a cone of height h and radius r is $V = \frac{1}{3}\pi r^2 h$].

[5]

38 Find the exact value of $\int_1^2 x^3 \ln x \, dx$ ln x dx.

[5]

39 Find $\int \sqrt[3]{2x-1}\, dx$.

[4]

40 Fig. 1 shows part of the curve $y = e^{2x} \cos x$.

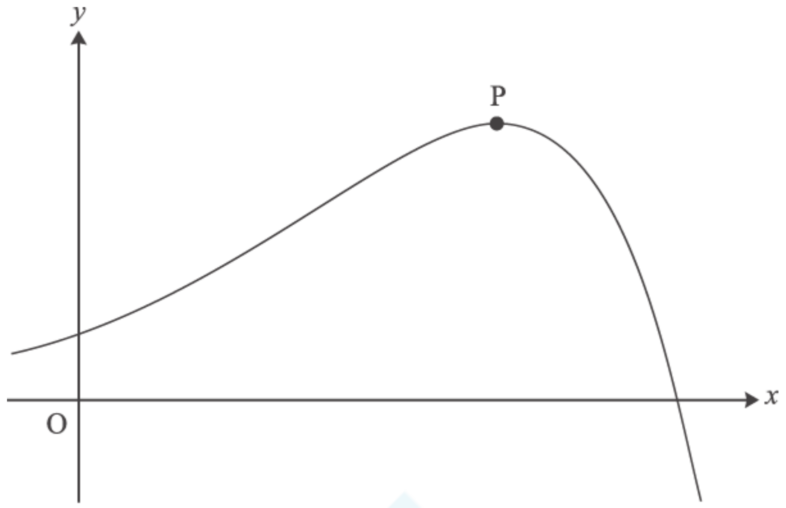


Fig. 1

Find the coordinates of the turning point P.

[6]

41 Fig. 9 shows the curve $y = f(x)$, where $f(x) = e^{2x} + k\,e^{-2x}$ and k is a constant greater than 1.

The curve crosses the y-axis at P and has a turning point Q.

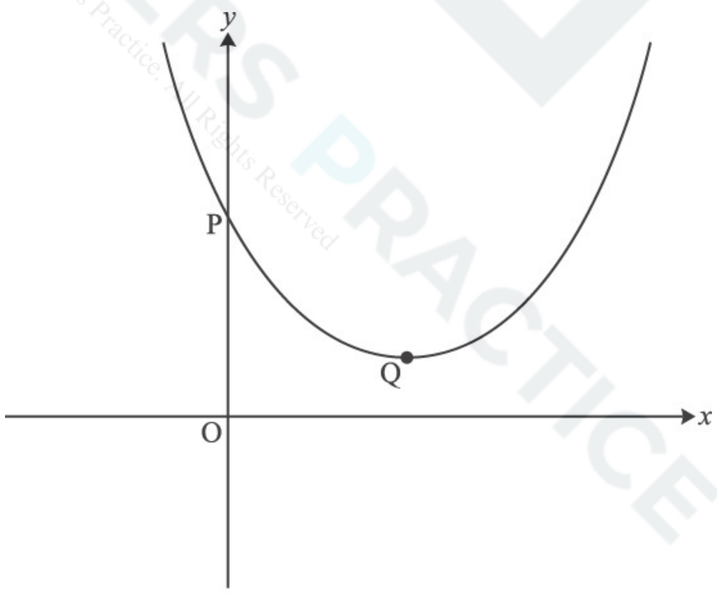


Fig. 9

(i) Find the y-coordinate of P in terms of k.

[1]

(ii) Show that the x-coordinate of Q is $\frac{1}{4}\ln k$, and find the y-coordinate in its simplest form.

[5]

(iii) Find, in terms of k, the area of the region enclosed by the curve, the x-axis, the y-axis and the line $x = \frac{1}{2}\ln k$. Give your answer in the form $ak + b$.

[4]

The function g(x) is defined by $\mathrm{g}(x) = \mathrm{f}(x + \frac{1}{4}\ln k)$.

(iv)

A Show that $\mathrm{g}(x) = \sqrt{k}\,(\mathrm{e}^{2x} + \mathrm{e}^{-2x})$.

[3]

B Hence show that g(x) is an even function.

[2]

C Deduce, with reasons, a geometrical property of the curve y = f(x).

[3]

Fig. 8 shows the curve $y = \frac{x}{\sqrt{x+4}}$ and the line $x = 5$. The curve has an asymptote l.

The tangent to the curve at the origin O crosses the line l at P and the line $x = 5$ at Q.

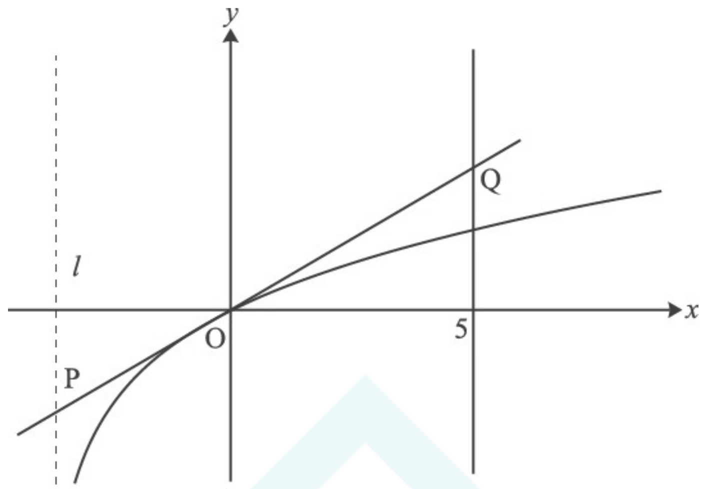


Fig. 8

(i) Show that for this curve $\frac{dy}{dx} = \frac{x+8}{2(x+4)^{\frac{3}{2}}}$.

[5]

(ii) Find the coordinates of the point P.

[4]

(iii) Using integration by substitution, find the exact area of the region enclosed by the curve, the tangent OQ and the line $x = 5$.

[9]

43

Fig. 6 shows part of the curve $\sin 2y = x - 1$. P is the point with coordinates $(1.5, \frac{1}{12}\pi)$ on the curve.

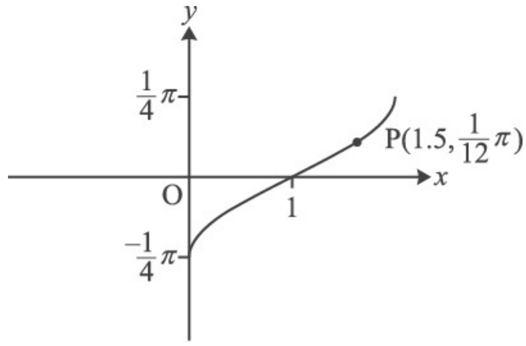


Fig. 6

(i) Find $\frac{dy}{dx}$ in terms of y.

Hence find the exact gradient of the curve $\sin 2y = x - 1$ at the point P.

[4]

The part of the curve shown is the image of the curve $y = \arcsin x$ under a sequence of two geometrical transformations.

(ii) Find y in terms of x for the curve $\sin 2y = x - 1$.

Hence describe fully the sequence of transformations.

[4]

44 The volume V m^3 of a pile of grain of height h metres is modelled by the equation

$$V = 4\sqrt{h^3+1} - 4.$$

(i) Find $\frac{dV}{dh}$ when $h = 2$.

[4]

At a certain time, the height of the pile is 2 metres, and grain is being added so that the volume is increasing at a rate of 0.4 m^3 per minute.

(ii) Find the rate at which the height is increasing at this time.

[3]

45 Find $\int_1^4 x^{-\frac{1}{2}} \ln x \, dx$, giving your answer in an exact form.

[5]

46 Find the exact value of $\int_0^{\frac{1}{2}\pi} (1 + \cos \frac{1}{2}x) \, dx$.

[3]

47 The growth of a tree is modelled by the differential equation

$$10\frac{dh}{dt} = 20 - h$$

where h is its height in metres and the time t is in years. It is assumed that the tree is grown from seed, so that h = 0 when t = 0.

(i) Write down the value of h for which $\frac{dh}{dt} = 0$, and interpret this in terms of the growth of the tree.

[1]

(ii) Verify that $h = 20\,(1 - e^{-0.1t})$ satisfies this differential equation and its initial condition.

[5]

The alternative differential equation

$$200\frac{dh}{dt} = 400 - h^2$$

is proposed to model the growth of the tree. As before, $h = 0$ when $t = 0$.

(i) Using partial fractions, show by integration that the solution to the alternative differential equation is

$$h = \frac{20\left(1 - e^{-0.2t}\right)}{1 + e^{-0.2t}}$$

[9]

(ii) What does this solution indicate about the long-term height of the tree?

[1]

(iii) After a year, the tree has grown to a height of 2 m. Which model fits this information better?

[3]

48 The parametric equations of a curve are

$$x = \sin\theta, \quad y = \sin 2\theta, \quad \text{for } 0 \leqslant \theta \leqslant 2\pi$$

(i) Find the exact value of the gradient of the curve at the point where $\theta = \frac{1}{6}\pi$.

[4]

(ii) Show that the cartesian equation of the curve is $y^2 = 4x^2 - 4x^4$.

[3]

49 The motion of a particle is modelled by the differential equation

$$v\frac{\mathrm{d}v}{\mathrm{d}x} + 4x = 0,$$

where x is its displacement from a fixed point, and v is its velocity.

Initially $x = 1$ and $v = 4$.

(i) Solve the differential equation to show that $v^2 = 20 - 4x^2$.

[4]

Now consider motion for which $x = \cos 2t + 2\sin 2t$, where x is the displacement from a fixed point at time t.

(ii) Verify that, when $t = 0$, $x = 1$. Use the fact that $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ to verify that when $t = 0$, $v = 4$.

[4]

(iii) Express x in the form $R\cos(2t - \alpha)$, where R and α are constants to be determined, and obtain the corresponding expression for v. Hence or otherwise verify that, for this motion too, $v^2 = 20 - 4x^2$.

[7]

(iv) Use your answers to part **(iii)** to find the maximum value of x, and the earliest time at which x reaches this maximum value.

[3]

(i) Express $\frac{x}{(1+x)(1-2x)}$ in partial fractions.

[3]

(ii) Hence use binomial expansions to show that $\frac{x}{(1+x)(1-2x)} = ax + bx^2 + \ldots$, where a and b are constants to be determined.

State the set of values of x for which the expansion is valid.

[5]

51 Fig. 8.1 shows an upright cylindrical barrel containing water. The water is leaking out of a hole in the side of the barrel.

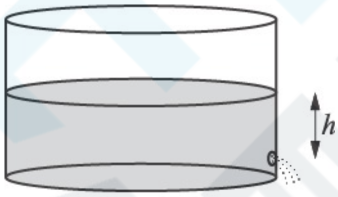


Fig. 8.1

The height of the water surface above the hole t seconds after opening the hole is h metres, where

$$\frac{dh}{dt} = -A\sqrt{h}$$

and where A is a positive constant. Initially the water surface is 1 metre above the hole.

(i) Verify that the solution to this differential equation is

$$h = \left(1 - \tfrac{1}{2}At\right)^2$$

[3]

The water stops leaking when $h = 0$. This occurs after 20 seconds.

(ii) Find the value of A, and the time when the height of the water surface above the hole is 0.5 m.

[4]

Fig. 8.2 shows a similar situation with a different barrel; h is in metres.

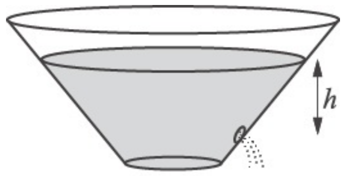


Fig. 8.2

For this barrel,

$$\frac{dh}{dt} = -B\frac{\sqrt{h}}{(1+h)^2}$$

where B is a positive constant. When $t = 0$, $h = 1$.

(iii) Solve this differential equation, and hence show that

$$h^{\frac{1}{2}}(30 + 20h + 6h^2) = 56 - 15Bt$$

[7]

(iv) Given that $h = 0$ when $t = 20$, find B.

Find also the time when the height of the water surface above the hole is 0.5 m.

[4]

52 A curve has parametric equations $x = e^{3t}$, $y = te^{2t}$.

(i) Find $\frac{dy}{dx}$ in terms of t. Hence find the exact gradient of the curve at the point with parameter $t = 1$.

[4]

(ii) Find the cartesian equation of the curve in the form $y = ax^b \ln x$, where a and b are constants to be determined.

[3]

53

Express $\frac{3x}{(2-x)(4+x^2)}$ in partial fractions.

[5]

54 A drug is administered by an intravenous drip. The concentration, x, ofthe drug in the blood is measured as a fraction of its maximum level. The drug concentration after t hours is modelled by the differential equation

$$\frac{dx}{dt} = k(1 + x - 2x^2),$$

where $0 \leq x < 1$, and k is a positive constant. Initially, $x = 0$.

(i) Express $\frac{1}{(1+2x)(1-x)}$ in partial fractions.

[3]

(ii) Hence solve the differential equation to show that $\frac{1+2x}{1-x} = e^{3kt}$.

[7]

(iii) After 1 hour the drug concentration reaches 75% of its maximum value and so $x = 0.75$.

Find the value of k, and the time taken for the drug concentration to reach 90% of its maximum value.

[3]

(iv) Rearrange the equation in part (ii) to show that $x = \frac{1 - e^{-3kt}}{1 + 2e^{-3kt}}$.

Verify that in the long term the drug concentration approaches its maximum value.

[5]

55 Solve the equation $\frac{5x}{2x+1} - \frac{3}{x+1} = 1$.

[5]

(i) Show that $\frac{1}{2+x}+\frac{1}{2-x}=\frac{4}{(2+x)(2-x)}$.

[1]

In a chemical reaction, the time t minutes taken for a mass x mg of a substance to be produced is modelled by the equation

$$t=\ln\left(\frac{2+x}{2-x}\right).$$

[2]

(ii) Show that when $t = 0$, $x = 0$.

[2]

(iii) Show that the rate of change of x is proportional to the product of $(2 + x)$ and $(2 – x)$, and find the constant of proportionality.

[4]

(iv) Show that $x=\frac{2(1-e^{-t})}{1+e^{-t}}$.

Hence determine the long-term mass of the substance predicted by this model.

[4]

In another chemical reaction, the mass x mg at time t minutes is modelled by the differential equation

$$\frac{dx}{dt}=k(2+x)(2-x)e^{-t},$$

where k is a positive constant, and $x = 0$ when $t = 0$.

(v) Show by integration that, for this reaction, $\ln\left(\frac{2+x}{2-x}\right)=4k(1-e^{-t})$.

[5]

(vi) Given that the long-term mass of this substance is 1.85 mg, find the value of k.

[2]

57

P is a general point on the curve with parametric equations $x = 2t,\ y = \frac{2}{t}$. This is shown in Fig. 6. The tangent at P intersects the x - and y-axes at the points Q and R respectively.

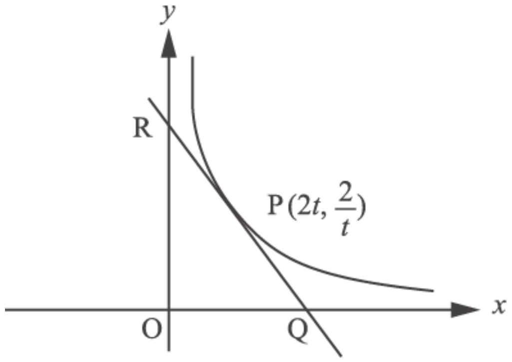


Fig. 6

Show that the area of the triangle OQR, where O is the origin, is independent of t.

[7]

58

(i) Differentiate $12\sqrt[3]{x}$.

[2]

(ii) Integrate $\frac{6}{x^3}$.

[3]

59

(a) Sketch the graph of $y = \frac{1}{x} + a$, where a is a positive constant.

- State the equations of the horizontal and vertical asymptotes.
- Give the coordinates of any points where the graph crosses the axes.

[4]

(b) Find the equation of the normal to the curve $y = \frac{1}{x} + 2$ at the point where $x = 2$. **[5]**

(c) Find the coordinates of the point where this normal meets the curve again. **[3]**

60 Evaluate $\int_0^{\frac{\pi}{12}} \cos 3x \, dx$, giving your answer in exact form. **[3]**

61 Fig. 12 shows the curve $2x^3 + y^3 = 5y$.

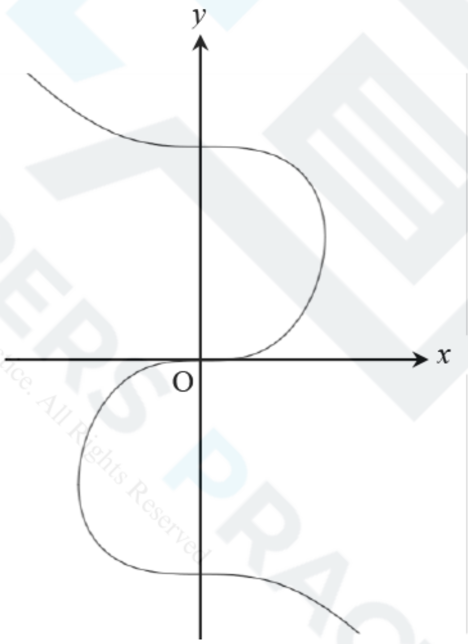


Fig. 12

(a) Find the gradient of the curve $2x^3 + y^3 = 5y$ at the point (1, 2), giving your answer in exact form. **[4]**

(b) Show that all the stationary points of the curve lie on the *y*-axis. **[2]**

62

Evaluate $\int_0^1 \frac{1}{1+\sqrt{x}}\,dx$, giving your answer in the form $a + b \ln c$, where a, b and c are integers. **[6]**

63 In a chemical reaction, the mass m grams of a chemical at time t minutes is modelled by the differential equation

$$\frac{dm}{dt} = \frac{m}{t(1+2t)}.$$

At time 1 minute, the mass of the chemical is 1 gram.

(a) Solve the differential equation to show that $m = \frac{3t}{(1+2t)}$. **[8]**

(b) Hence

(i) find the time when the mass is 1.25 grams, **[2]**

(ii) show what happens to the mass of the chemical as t becomes large. **[2]**

64 Find $\int \left(x^2 + \frac{1}{x^2} \right) dx$. **[3]**

65 Show that the area of the region bounded by the curve $y = 3x^{-\frac{3}{2}}$, the lines $x = 1$, $x = 3$ and the x-axis is $6 - 2\sqrt{3}$. **[5]**

66 In an experiment, the temperature of a hot liquid is measured every minute. The difference between the temperature of the hot liquid and room temperature is D °C at time t minutes.

Fig. 8 shows the experimental data.

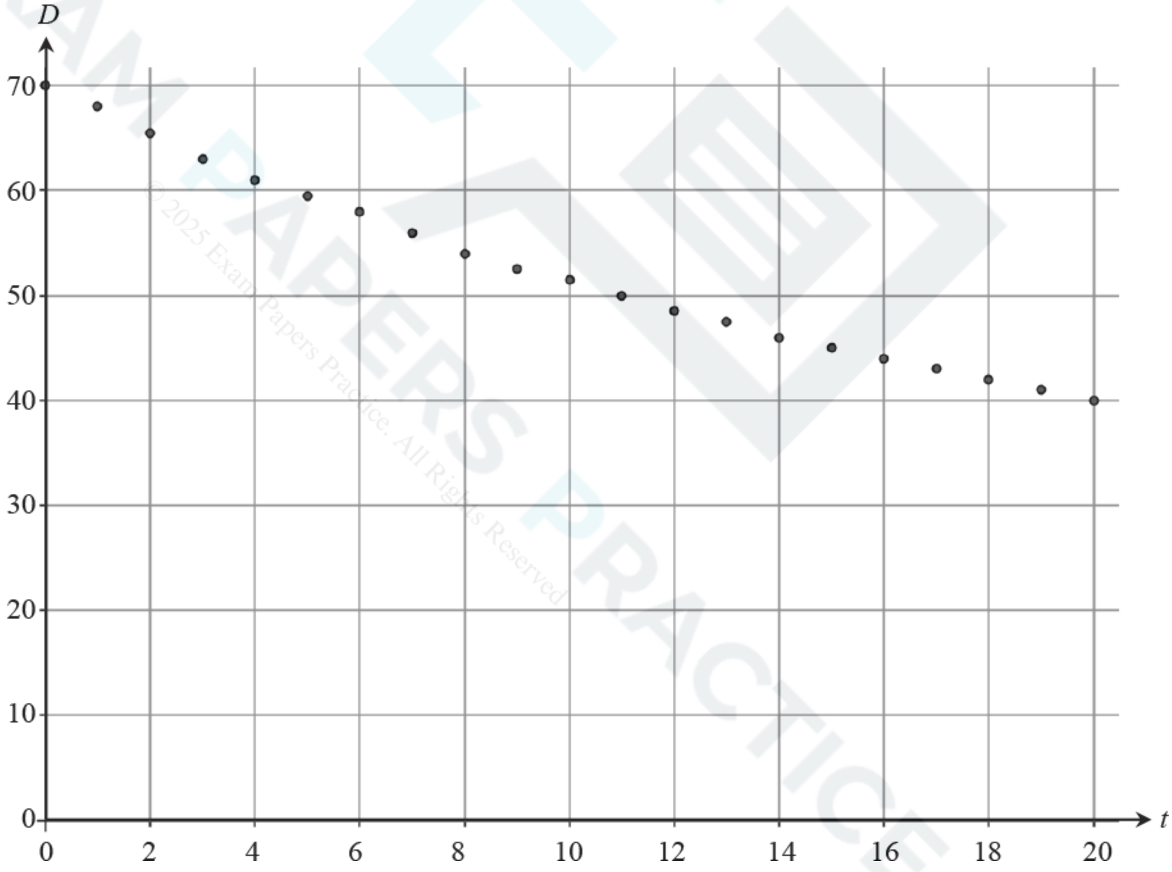


Fig. 8

It is thought that the model $D = 70e^{-0.03t}$ might fit the data.

(a) Write down the derivative of $e^{-0.03t}$. **[1]**

(b) Explain how you know that $70e^{-0.03t}$ is a decreasing function of *t*. **[1]**

(c) Calculate the value of $70e^{-0.03t}$ when

(i) $t = 0$, **[1]**

(ii) $t = 20$. **[1]**

(d) Using your answers to parts **(b)** and **(c)**, discuss how well the model $D = 70e^{-0.03t}$ fits the data. **[3]**

67 Differentiate the following.

(a) $\sqrt{1-3x^2}$ [3]

(b) $\dfrac{x^2}{3x+2}$ [3]

68 Find $\int x^2 e^{2x}\, dx$. **[7]**

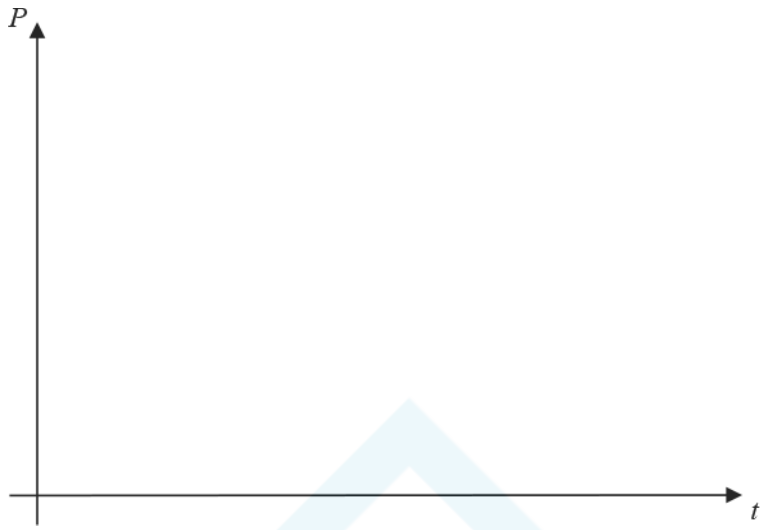
69 In a certain region, the populations of grey squirrels, P_G and red squirrels P_R, at time t years are modelled by the equations:

$$P_G = 10000(1 - e^{-kt})$$

$$P_R = 20000e^{-kt}$$

where $t \geq 0$ and k is a positive constant.

(a) (i) On the axes below, sketch the graphs of P_G and P_R on the same axes.



(ii) Give the equations of any asymptotes. **[4]**

(b) What does the model predict about the long term population of

- grey squirrels
- red squirrels? **[2]**

Grey squirrels and red squirrels compete for food and space. Grey squirrels are larger and more successful than red squirrels.

(c) Comment on the validity of the model given by the equations, giving a reason for your answer. **[1]**

(d) Show that, according to the model, the rate of decrease of the population of red squirrels is always double the rate of increase of the population of grey squirrels. **[4]**

(e) When $t = 3$, the numbers of grey and red squirrels are equal. Find the value of k. **[4]**

70 Fig. 11 shows the curve with parametric equations

$$x = 2\cos\theta,\ y = \sin\theta,\ 0 \le \theta \le 2\pi.$$

The point P has parameter $\frac{1}{4}\pi$. The tangent at P to the curve meets the axes at A and B.

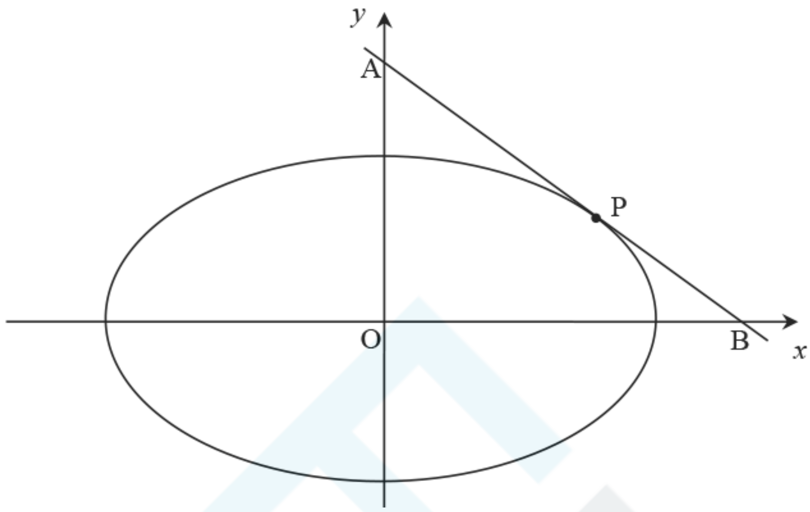


Fig.11

(a) Show that the equation of the line AB is $x + 2y = 2\sqrt{2}$. [6]

(b) Determine the area of the triangle AOB. **[3]**

71 Fig. 6 shows the curve with equation $y = x^4 - 6x^2 + 4x + 5$.

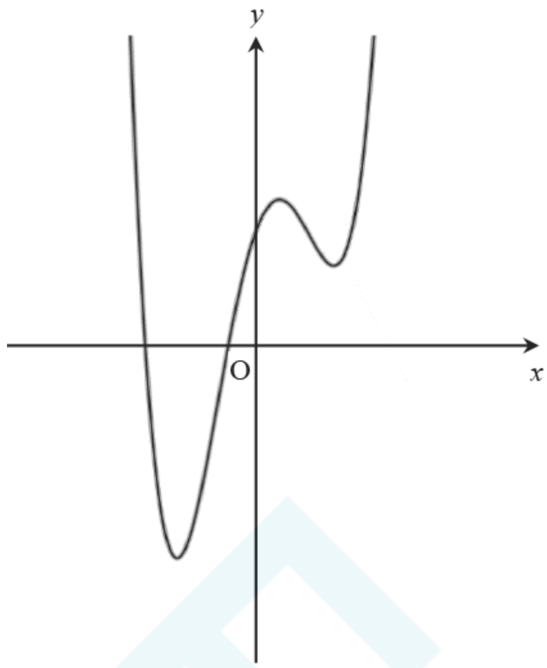


Fig. 6

Find the coordinates of the points of inflection. **[5]**

72 The function f(x) is defined by $f(x) = x^4 + x^3 - 2x^2 - 4x - 2$.

(a) Show that $x = -1$ is a root of $f(x) = 0$. **[1]**

(b) Show that another root of $f(x) = 0$ lies between $x = 1$ and $x = 2$. **[2]**

(c) Show that $f(x) = (x + 1)g(x)$, where $g(x) = x^3 + ax + b$ and a and b are integers to be determined. **[3]**

(d) Without further calculation, explain why $g(x) = 0$ has a root between $x = 1$ and $x = 2$. **[1]**

(e) Use the Newton-Raphson formula to show that an iteration formula for finding roots of g(x) = 0 may be written

$$x_{n+1} = \frac{2x_n^{\ 3} + 2}{3x_n^{\ 2} - 2}$$

Determine the root of g(x) = 0 which lies between x = 1 and x = 2 correct to 4 significant figures. **[3]**

73 The curve $y = f(x)$ is defined by the function $f(x) = e^{-x} \sin x$ with domain $0 \le x \le 4\pi$.

(a) (i) Show that the x-coordinates of the stationary points of the curve $y = f(x)$, when arranged in increasing order, form an arithmetic sequence.

(ii) Show that the corresponding y-coordinates form a geometric sequence. **[9]**

(b) Would the result still hold with a larger domain? Give reasons for your answer **[1]**

74 **[3]**

Given that $y = 6x + 3 - \frac{5}{x^2}$ find $\frac{dy}{dx}$.

75 Prove that $f(x) = x^3 - 3x^2 + 6x + 5$ is an increasing function for all real values of x. **[5]**

76 A curve passes through the point (4, 122) and its gradient is given by

$$\frac{dy}{dx} = 1 - \frac{4}{\sqrt{x}} + 6x^2.$$

Find the equation of the curve. **[5]**

77 **In this question you must show detailed reasoning.**

Find the total area of the shaded regions shown in Fig. 8, bounded by the line $x = -1$, the x-axis and the curve $y = x^3(x-3)$. **[6]**

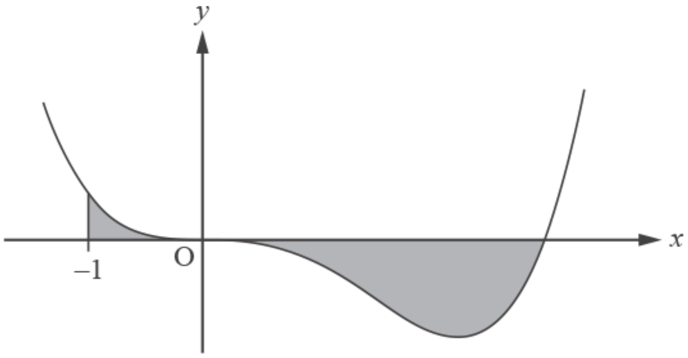


Fig. 8

78 **(a)** Sketch the curve $y = e^{2x}$. **[2]**

(b) Describe fully the transformation that maps the curve $y = e^x$ onto the curve $y = e^{2x}$. **[2]**

(c) Find the equation of the tangent to $y = e^{2x}$ at the point where $x = 3$, giving your answer in the form $y = e^a (bx + c)$ where a, b and c are integers. **[6]**

79 Differentiate $(3x^2 + 5)^4$. **[3]**

80 **In this question you must show detailed reasoning.**

The shaded region in Fig. 2 is bounded by the curves $y = \sin 3x$ and $y = 3 \sin 3x$.

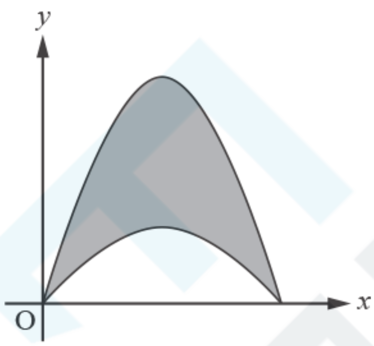


Fig. 2

Show that the area of this region is $\frac{4}{3}$. **[4]**

81 (a) Solve the differential equation

$$\frac{dy}{dx} = y(1+y)(1-x),$$

given that $y = 1$ when $x = 1$. Give your answer in the form $y = f(x)$, where f is a function to be determined. **[9]**

(b) By considering the sign of $\frac{dy}{dx}$ near (1, 1), or otherwise, show that this point is a maximum point on the curve $y = f(x)$. **[3]**

82 Prove from first principles that the derivative of x^3 is $3x^2$. **[5]**

83 **In this question you must show detailed reasoning.**

(a) For the curve $y = x^4 - 3x^2 + 2$, find the equation of the normal at the point (1, 0). **[4]**

(b) For the curve $y = x^4 - 3x^2 + c$, L is the normal at the point where $x = 1$. Determine the value of c for which L passes through the origin. **[3]**

84

Find $\int 12e^{3x}\,dx$. [2]

85

Find $\int 18x(3x+1)^7\,dx$.

You may wish to use the substitution $u = (3x + 1)$. [6]

86 **In this question you must show detailed reasoning.**

A curve has parametric equations

$$x = \cos t - 3t \text{ and } y = 3t - 4\cos t - \sin 2t, \text{ for } 0 \leq t \leq \pi.$$

Show that the gradient of the curve is always negative. **[7]**

87 When a particular type of glass is filled to a depth of x cm with champagne, the volume of champagne in the glass is modelled by the formula

$$V = kx^2 - \frac{x^3}{3},$$

where $V = 36$ when $x = 3$.

(a) Find the value of k. **[1]**

When full, a glass contains 140 cm^3 of champagne and the depth is 7.5 cm.

(b) Determine whether the model works well for this value. **[2]**

When Leonie pours champagne into a glass of this type, the volume in the glass after t seconds is modelled by the formula

$$V = 140(1 - e^{-0.2t}).$$

(c) (i) Verify that after 5 seconds the depth of champagne is between 5.20 cm and 5.21 cm. **[2]**

(ii) Find the rate at which the depth of champagne is increasing 5 seconds after she starts pouring. **[4]**

(iii) According to the model, would it be faster for Leonie to pour two half glasses of 70 cm^3 or one full glass of 140 cm^3 of champagne? **[2]**

88 Fig. 15 shows the graph of $f(x) = 2x + \frac{1}{x} + \ln x - 4$.

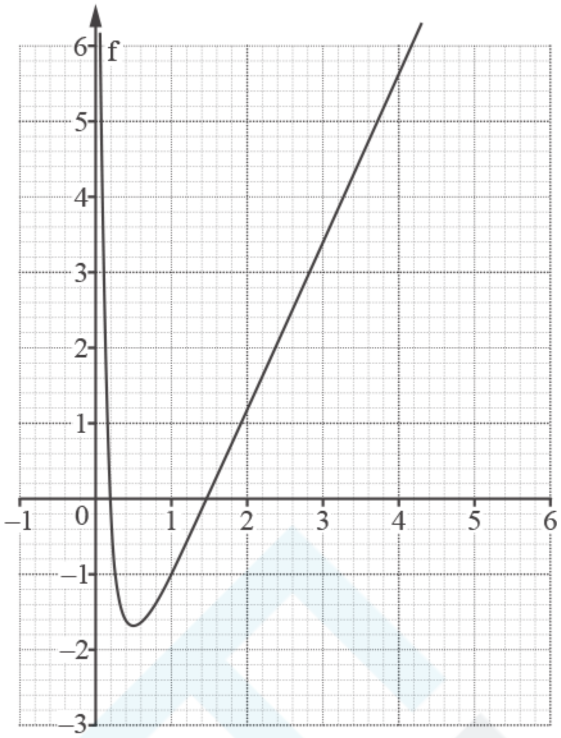


Fig. 15

(a) Show that the equation

$$2x + \frac{1}{x} + \ln x - 4 = 0$$

has a root, α, such that $0.1 < \alpha < 0.9$. **[2]**

(b) Obtain the following Newton-Raphson iteration for the equation in part **(a)**.

$$x_{r+1} = x_r - \frac{2x_r^3 + x_r + x_r^2(\ln x_r - 4)}{2x_r^2 - 1 + x_r}$$

[3]

(c) Explain why this iteration fails to find α using each of the following starting values.

(i) $x_0 = 0.4$ **[2]**

(ii) $x_0 = 0.5$ **[2]**

(iii) $x_0 = 0.6$ **[2]**

89 (i) Find $\int_1^5 4x\,dx$ [3]

(ii) Find $\int 6x^{\frac{1}{2}}\,dx$ [2]

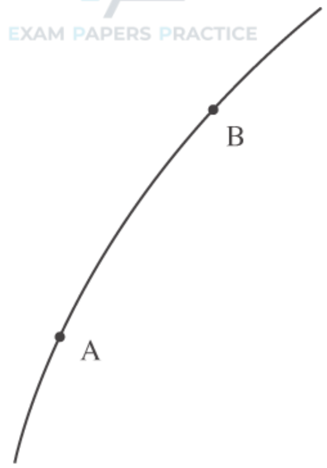


Fig. 3

Fig. 3 shows two points A and B on the curve $y = \log_{10}x$. At A, $x = 0.1$ and at B, $x = 0.2$.

(i) Calculate the gradient of the chord AB. **[2]**

(ii) The gradient of the chord AB gives an estimate for the gradient of the curve at A. On the copy of Fig. 3 below, mark a point C on the curve such that the gradient of the chord AC would give a better estimate. **[1]**

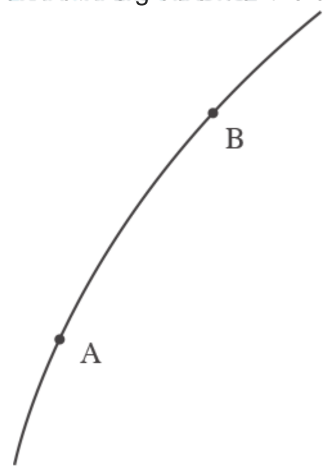


Fig. 3

91 Find the equation of the normal to the curve $y = 2x^3$ at the point on the curve where $x = 2$. Give your answer in the form $ax + by = c$. **[5]**

92 A curve passes through the point (2, 10) and has gradient $\frac{dy}{dx} = 12x^3 - 7$. Find the equation of the curve. **[5]**

93 The standard formulae for the volume V and total surface area A of a solid cylinder of radius r and height h are

$$V = \pi r^2 h \text{ and } A = 2\pi r^2 + 2\pi rh.$$

You are given that $V = 400$.

(i) Show that $A = 2\pi r^2 + \dfrac{800}{r}$. **[2]**

(ii) Find $\dfrac{dA}{dr}$ and $\dfrac{d^2A}{dr^2}$. **[4]**

(iii) Hence find the value of r which gives the minimum surface area. Find also the value of the surface area in this case. **[4]**

94

Differentiate $\dfrac{1}{(5-2x^3)^2}$. [3]

95 A curve has equation $3x^{\frac{2}{3}} + 2y^{\frac{1}{3}} = 7$.

(i) By differentiating implicitly, find $\frac{dy}{dx}$ in terms of x and y. [3]

(ii) Hence find the gradient of the curve at the point with coordinates (1, 8). [2]

96 The equation of a curve is $y = \arcsin \frac{1}{2}x$.

(i) Express each of x and $\frac{dx}{dy}$ in terms of y. [2]

(ii) A point is moving on the curve, and has coordinates (x, y) at time t. When $x = 1$, the value of $\frac{dx}{dt}$ is 2.

Find the exact value of $\frac{dy}{dt}$ at this instant. [5]

97 Fig. 8 shows part of the curve $y = \frac{\cos x}{2 - \sin x}$. The curve intersects the x- and y-axes at A and C respectively, and has a turning point at B.

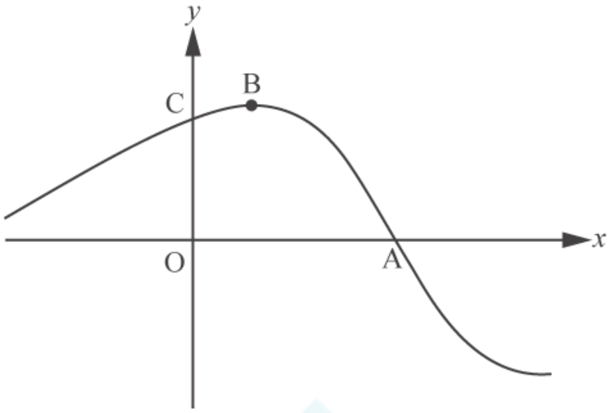


Fig. 8

(i) Write down the coordinates of A and C. **[2]**

.

(ii) Find $\frac{dy}{dx}$ and the exact coordinates of B. **[7]**

(iii) (*A*) Using integration by substitution, or otherwise, find the exact area of the region enclosed by the curve, the y-axis and the positive x-axis. **[4]**

(*B*) The line $x = k$ divides this region into two parts of equal area. Show that $k = \arcsin(2 - \sqrt{2})$. **[5]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	at A $y = 3$	B1		
		i	$\frac{dy}{dx} = 2x - 4$	B1		
		i	their $\frac{dy}{dx} = 2 \times 4 - 4$	M1*	must follow from attempt at differentiation	
		i	grad of normal = $^{-1}/_{\text{their4}}$	M1dep*		
		i	$y - 3 = (^{-}\frac{1}{4}) \times (x - 4)$ oe isw	A1		
		i	substitution of $y = 0$ and completion to given result with at least 1 correct intierim step www	A1	or substitution of $x = 16$ to obtain $y = 0$ **Examiner's Comments** This was done extremely well, with the majority of even the weakest candidates scoring full marks. A few wrote $2x - 4 = 0$ to incorrectly obtain $m = 2$ and made no further progress, and a very small minority tried to answer the question without using calculus and working backwards.	correct interim step may occur before substitution
		ii	at B, $x = 3$	B1	may be embedded	
		ii	$F[x] = \frac{x^3}{3} - \frac{4x^2}{2} + 3x$	M1*	condone one error, must be three terms, ignore + c	
		ii	F[4] – F[their 3]	M1*dep	dependent on integration attempted	
		ii	area of triangle = 18 soi	B1		may be embedded in final answer

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Area of region = $19\frac{1}{3}$ oe isw	A1	19.3 or better **Examiner's Comments** Nearly all candidates identified the coordinates of B correctly. However, most — as if by rote — subtracted the equation of the line from the equation of the curve and then integrated. Some candidates integrated the equation of the curve correctly, but used the wrong limits (usually 3 to 16) and made no further progress, and of those that did adopt the correct approach, a large number were unable to find the area of the triangle correctly (½ × 12 × 4 was common).	
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
2			$6x^2 + 18x - 24$	B1		
			their $6x^2 + 18x - 24 = 0$ or >0 or $\leqslant 0$	M1		or sketch of $y = 6x^2 + 18x - 24$ with attempt to find x -intercepts
			-4 and $+1$ identified oe	A1		
			$x < -4$ and $x > 1$ cao	A1	or $x \leqslant -4$ and $x \leq 1$ **Examiner's Comments** Most candidates differentiated correctly and identified the correct values of x. The final mark was often lost, either due to a misunderstanding of what had been found — answer given as –4 < x < 1 or poor notation — answer given as –4 > x > 1. Those who used a graphical approach with the derivative generally scored full marks. A few candidates missed the last term out, converted the first plus sign to a minus sign or failed to multiply 2 by 3 correctly, and lost the first mark.	if **B0M0** then **SC2** for fully correct answer
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3			$\text{gradient} = \dfrac{4\sqrt{9.5}-12}{9.5-9}$	M1		$4\sqrt{38}-24\,4\sqrt{38}-24$
			0.6577 to 0.66	A1	or 0.657656…isw	
			$9 < x_C < 9.5$	B1	**Examiner's Comments** This was done very well. Some candidates lost the second mark through premature rounding or simply giving the answer as 0.6. Only a few calculated the reciprocal of the gradient (which didn't score) and nearly all gave an appropriate value for x_c. A few candidates differentiated and substituted values in the derivative.	allow $8.53 \leq x_c < 9$
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4			$kx^{\frac{5}{2}}$	M1		
			k = 12	A1		
			+ c	A1	**Examiner's Comments** The majority of candidates scored full marks on this question. However, a significant minority omitted "+ c" and lost an easy mark. Similarly, some candidates failed to simplify $\frac{30}{5/2}$ correctly or didn't try to, and thus lost a mark. Occasionally $\frac{30^{5/2}}{5/2}$ was seen, which of course scored 0.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5		i	$3x^2 - 6x - 22$	M1	condone one incorrect term, but must be three terms	condone "$y =$"
		i	their $y' = 0$ soi	M1	at least one term correct in their y'	may be implied by use of e.g. quadratic formula, completing square, attempt to factorise
		i	3.89	A1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	–1.89	A1	if **A0A0, SC1** for $\frac{3 \pm 5\sqrt{3}}{3}$ or $1 \pm {}^{5}/_{\sqrt{3}}$ or better, or both decimal answers given to a different accuracy or from truncation __Examiner's Comments__ Nearly all candidates differentiated successfully and set their derivative to zero. Over 60% of candidates went on to score full marks, although a few candidates made an error (usually $2x^2$ but occasionally + 24 was retained). However, a significant minority attempted unsuccessfully to factorise the quadratic and then gave up and a surprising number were unable to use the quadratic formula correctly. Very few candidates appeared to check their answers. Some candidates lost an easy mark by leaving their answers in an exact form or by quoting a different precision. Occasionally, candidates found the second derivative and set this equal to zero. A significant minority wasted time either by finding the associated *y*-values or by determining the nature of the turning points, neither of which were required.	3.886751346 and –1.886751346
		ii	$x^3 - 3x^2 - 22x + 24 = 6x + 24$	M1	may be implied by $x^3 - 3x^2 - 28x$ [= 0]	
		ii	$x^3 - 3x^2 - 28x$ [= 0]	M1	may be implied by $x^2 - 3x - 28$ [= 0]	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	other point when $x = 7$ isw	A1	dependent on award of both **M** marks <u>Examiner's Comments</u> This was very well answered by most candidates. Well over 80% earned the first mark and most went on to score full marks. Occasionally, candidates slipped up when collecting like terms and a few made a sign error when factorising. The minority who failed to score either omitted the question altogether, or set $6x + 24$ equal to the derivative.	ignore other values of x
		iii	$\mathrm{F}[x] = \frac{x^4}{4} - \frac{3x^3}{3} - \frac{22x^2}{2} + 24x$	M1*	allow for three terms correct; condone + c	**alternative method** **M1** for $\int ((x^3 - 3x^2 - 22x + 24) - (6x + 24))dx$ may be implied by 2nd **M1**
		iii	F[0] – F[–4]	M1dep	allow 0 – F[–4], condone – F[–4], but do not allow F[–4] only	**M1*** for $\mathrm{F}[x] = \frac{x^4}{4} - \frac{3x^3}{3} - \frac{28x^2}{2}$ condone one error in integration
		iii	area of triangle = 48	B1		**M1dep** for F[0] – F[–4]

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	area required = 96 from fully correct working	A1	**A0** for – 96, ignore units, <u>Examiner's Comments</u> This question was accessible to most candidates, although a significant minority scored zero. Many candidates found the area of the triangle using ½×base×height. Most of those who used a base of –4 realised that a negative area was impossible and so removed the minus sign. Some used integration and more often than not were successful – sometimes after 'losing' a minus sign. Most candidates also integrated successfully, but some made no further progress, as they ignored the upper limit and then 'airbrushed' the minus sign. A good proportion of those who did integrate successfully then made errors with the arithmetic. Some candidates earned two marks by combining the equations and integrating correctly, but a similar proportion ignored the upper limit or made arithmetical slips.	no marks for 96 unsupported
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6			kx^{-2}	M1*		$k \neq 0$
			$-9x^{-2}$	A1	may be awarded later	no marks at all for responses based on "$mx + c$"
			$+ 2x + c$	M1*	c may appear at substitution stage	
			substitution of $x = 3$ and $y = 6$ in their expression following integration	M1dep	on award of *either* of previous **M1**s	e.g. $6 = k3^{-2} + 2 \times 3 + c$

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$c = 1$	A1	**A0** if spoiled by further working __Examiner's Comments__ Most candidates recognised this standard question and integrated successfully; the majority went on to score full marks. A few dropped the minus sign on the first term and ended up losing both A marks, a few made arithmetic or substitution errors: c = – 1 and c = 81 were the most common wrong answers. In a small number of cases the final mark was withheld because at no point did the candidate write “y =” in their solution. A small number of candidates spoiled fully correct answers by reverting to an answer based on $y = mx + c$ and an equally small number integrated successfully but used the original expression to evaluate c. Some candidates were unable to deal with the negative power successfully: variations of $\frac{18x}{x^3/3}$ and $\frac{18x^{-4}}{-4}$ were the most common errors. A significant minority of candidates (approximately 20%) failed to score because they multiplied by	for full marks, **must** see “y =” at some stage

Question			Answer/Indicative content	Marks	Part marks and guidance	
					x and added c.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7		i	$-10x^{-6}$ isw	B1	for – 10	if **B0B0** then **SC1** for $-5 \times 2x^{-5-1}$ or better soi
		i		B1	for x^{-6} ignore + c and y = <u>Examiner's Comments</u> The overwhelming majority of candidates scored full marks on this question. A few candidates omitted the minus sign, and others lost a mark because they calculated the power as – 5 – 1 = – 4. A small number of candidates integrated. Some of these did so incorrectly, obtaining the answer $\frac{-2x^{-6}}{-6}$.	
		ii	$y = x^{\frac{1}{3}}$ soi	B1	condone $y' = x^{\frac{1}{3}}$ if differentiation follows	
		ii	kx^{n-1}	M1	ft their fractional n	
		ii	$\frac{1}{3}x^{-\frac{2}{3}}$ isw	A1	ignore + c and y = <u>Examiner's Comments</u> Most candidates identified the correct power, and went on to differentiate correctly. However, a significant minority gave the new power as –⅓, and a few integrated instead of differentiating. In cases where candidates failed to identify ⅓ as the power, – 3 and $^{3}/_{2}$ were the most common errors.	allow 0.333 or better
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	$h = 3$ soi	B1		allow if used with 6 separate trapezia
		i	$\frac{3}{2}\{9+9.1+2(10.7+11.7+11.9+11.0)\}$	M1	basic shape of formula correct with their 3; omission of brackets may be recovered later; M0 if any *x*-values used (NB $y_0 = 9$ and $x_3 = 9$, so check position)	with 3, 4 or 5 *y*-values in middle bracket, eg $\frac{3}{2}\{9+2(10.7+11.7+11.9)+11.0\}$
		i	all *y*-values correctly placed in formula	B1	condone omission of outer brackets	
		i	163.05 or 163.1 or 163 isw	A1	answer only does not score **Examiner's Comments** This was very well done, with many candidates scoring full marks. The most common error was the omission of the outer brackets; occasionally $h = 5$ was seen, and occasionally *y* values were misplaced.	or **B1** + **B3** if 5 separate trapezia calculated to give correct answer NB 29.55 + 33.6 + 35.4 + 34.35 + 30.15
		ii	(*A*) $-0.001 = 12^3 - 0.025 \times 12^2 + 0.6 \times 12 + 9$ soi	M1	may be implied by 10.872, 10.87 or 10.9	NB allow misread if minus sign omitted in first term if consistent in (A) and (B). Lose A1 in this part only
		ii	± 0.128[m] or ± 12.8cm or ± 128mm isw	A1	B2 if unsupported **Examiner's Comments** Many correctly substituted $x = 12$, and showed their working so that even if arithmetic went astray, a method mark was still earned. A common error was to omit the minus sign from the first term. Strangely many candidates stopped there, or subtracted 10.872 from 12 instead of 11.	appropriate units must be stated if answer not given in metres
		ii	(*B*) $\mathrm{F}[x] = \frac{-0.001x^4}{4} - \frac{0.025x^3}{3} + \frac{0.6x^2}{2} + 9x$	M2	M1 if three terms correct; ignore + *c*	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	F(15) [– F(0)] soi	M1	dependent on at least two terms correct in F[*x*]	condone F (15) + 0
		ii	161.7 to 162	A1	A0 if a numerical value is assigned to *c* **Examiner's Comments** This was generally very well done. Occasionally candidates made a sign error or inserted an extra zero in one or both of the first two terms. Some left “9” untouched or used an upper limit of 12 instead of 15.	answer only does not score NB allow misread if minus sign omitted in first term if consistent in (A) and (B). 187.03…
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	$y' = 1 + 8x^{-3}$	M2	M1 for just $8x^{-3}$ or $1 - 8x^{-3}$	
		i	$y'' = -24x^{-4}$ oe	A1	**Examiner's Comments** Most knew what to do here, but $8x^{-3}$ and $1 - 8x^{-3}$ were often seen. Only a few candidates failed to show sufficient detail of their working to earn the third mark following a fully correct $\frac{dy}{dx}$	but not just $\frac{-24}{x^4}$ as AG
		ii	their $y' = 0$ soi	M1		
		ii	$x = -2$	A1	A0 if more than one x-value	$x = -2$ must have been correctly obtained for all marks after first M1
		ii	$y = -3$	A1	A0 if more than one y-value	
		ii	substitution of $x = -1$ in their y'	M1	or considering signs of gradient either side of -2 with negative x-values	condone any bracket error

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	< 0 or = – 1.5 oe correctly obtained isw	A1	signs for gradients identified to verify maximum **Examiner's Comments** There were many correct solutions, although even some of the better candidates neglected to find the corresponding value of y, or evaluated the second derivative as + 1.5, and concluded the stationary value must be a local maximum. A few obtained x = 2 following correct differentiation, but never bothered to look at the graph to realise that this must be wrong. It was surprising just how many candidates solved $8x^{-3} = 0$ to obtain $x = 2$, without realising that something must have gone wrong.	must follow from M1 A1 A0 M1 or better
		iii	$y = -5$ soi	B1		
		iii	substitution of $x = -1$ in their y^1	M1	may be implied by – 7	
		iii	grad normal = $^{-1}/_{\text{their} - 7}$	M1*	may be implied by e.g. $^{1}/_{7}$	
		iii	y – their (–5) = their $^{1}/_{7}(x - -1)$	M1dep*	or their (–5) = their $^{1}/_{7} \times (-1) + c$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$-x + 7y + 34 = 0$ oe	A1	allow e.g. $y-\frac{1}{7}x+\frac{34}{7}=0$ **Examiner's Comments** This was generally well done. Only a small minority of candidates did not understand how to obtain the gradient of the normal, and many obtained follow through marks, at least. Some candidates slipped up finding the value of *y*, and a few made sign errors when finishing off.	must see = 0 do not allow e.g. $y=\frac{x}{7}-\frac{34}{7}$
			Total	13		
10			$\frac{2.4-3.6}{2.2-2}$ oe – 6 cao	M1 A1	**Examiner's Comments** Many candidates scored full marks here. A few switched the values in the numerator round to obtain + 6 and lost both marks. A small minority found the reciprocal of the gradient, which didn't score, and a tiny minority wrote down the correct calculation, but obtained an incorrect answer. Some candidates needlessly went on to obtain the equation of the chord, and then differentiated it just to convince themselves that the gradient really was – 6, and a few went straight to finding the equation of the chord, and then left the answer embedded, which cost the accuracy mark.	M1 may be embedded eg in equation of straight line B2 if unsupported ignore subsequent work irrelevant to finding the gradient
			Total	2		

Question			Answer/Indicative content	Marks	Part marks and guidance	
11			$kx^{\frac{5}{2}+1}$	M1	*k* is any non-zero constant	
			$2x^{\frac{7}{2}}$ cao	A1		
			+ *c*	A1	**Examiner's Comments** This was done well. A surprisingly high number of candidates omitted " + c " and lost an easy mark, and a few candidates went astray when simplifying 7 ÷ , or didn't bother to simplify it at all. A very few differentiated instead of integrating.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12		i	$\left[\frac{dy}{dx}=\right] 4 \times 2 + 3$ or 11 isw	M1*		
		i	9 = their (4 × 2 + 3) × 2 + c	M1dep*	or y – 9 = their (4 × 2 + 3) × (x – 2)	
		i	y = 11x – 13 or y = 11x + c and c = – 13 stated isw	A1	or y – 9 = 11 (x – 2) isw **Examiner's Comments** The majority of candidates gained full marks on this question. A few found the gradient correctly and then went on to find the equation of the normal, and some candidates integrated and found the equation of the curve.	
		ii	$\frac{4x^2}{2}+3x$	M1*		
		ii	[y =] $2x^2 + 3x + c$	A1	must see “2” and “+ c”, may be earned later eg after attempt to find c	
		ii	9 = 2 × 22 + 3 × 2 + c	M1dep*	must include constant, which may be implied by answer	
		ii	$y = 2x^2 + 3x – 5$ cao	A1	allow first 4 marks for $y = 2x^2 + 3x + c$ and c = –5 stated	
		ii	(1, 0) *and* (–2.5, 0) oe cao	B1	or for x = 1, y = 0 and x = –2.5, y = 0	**B0** for just stating x = 1 and x = –2.5
		ii	$x=-\frac{3}{4}$	B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$y = -\dfrac{49}{8}$	B1	–6.125 or – 6⅛ **Examiner's Comments** Most candidates integrated successfully. A few omitted '+ *c*' and made little progress thereafter, but the majority successfully obtained the equation of the curve. Many candidates failed to give the coordinates in a correct form or transposed the signs, thus losing an accuracy mark. Most candidates used the given derivative to find the co-ordinates of the minimum point, but it was surprising how many made a sign error and then couldn't obtain the correct value for *y*. A significant number ofcandidates completed the square instead of using the derivative, and most lost accuracy.	
		iii	substitution to obtain [y =] f($2x$) in polynomial form	M1	f (x) must be the quadratic in x with linear and constant term obtained in part (ii), may be in factorised form	or their x = 1 → their 0.5 and their x = – 2.5 → their x = –1.25
		iii	$y = (2x – 1)(4x + 5)$ or y = or $y = 2\left(2x+\dfrac{3}{4}\right)^2 - \dfrac{49}{8}$	A1FT	must be simplified to one of these forms, **FT** their quadratic in$x$ with linear and constant term obtained in part (ii)	hence $y = (2x – 1)(4x + 5)$ FT their x-intercepts from their quadratic in x with linear and constant term obtained in part (ii)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\left(-\frac{3}{8},-\frac{49}{8}\right)$ oe	B1	or **FT** their (both non-zero) co-ordinates for minimum point or their quadratic in x with linear and constant term obtained in part (ii) **Examiner's Comments** Very few candidates realised that they needed to work with f(2x) to find the newequation. The majority of those who did adopt the correct approach often went wrong, usually with the first term. In order to earn the method mark by this approach examiners needed to see the substitution: many candidates just wrote down $y = 4x^2 + 6x - 5$ and failed to score. It may have been the case that the correct approach was being attempted, but this answer was also seen resulting from totally wrong working! A few candidates successfully worked with the images of the intercepts following the stretch, but often failed to simplify the answer correctly. A minority of candidates earned the third mark, either as a follow through mark or for a fully correct answer. However, many candidates multiplied the x value by 2, or halved the y value instead or as well.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13			$3x^2 - 6$ seen	B1		
			their $y' = 0$ or $y' > 0$ or $y' \geq 0$	M1	must be quadratic with at least one of only two terms correct	
			$\sqrt{2}$ and $-\sqrt{2}$ identified?	A1	may be implied by use with inequalities or by ± 1.41[4213562] to 3 sf or more	$\lvert x \rvert = \sqrt{2}$ implies **A1**
			$x < -\sqrt{2}$ or $x \leq -\sqrt{2}$ isw	A1	if **A1A0A0**, allow **SC1** for fully correct answer in decimal form to 3 sf or more	NB **just** $-\sqrt{2} > x > \sqrt{2}$ or $\sqrt{2} < x < -\sqrt{2}$ or $x > \pm\sqrt{2}$ implies the first **A1** then **A0A0**

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$x > \sqrt{2}$ or $x \geq \sqrt{2}$	A1	or **A2** for $\|x\| > \sqrt{2}$ or $\|x\| \geq \sqrt{2}$ Examiner's Comments The majority of candidates differentiated successfully and went on to identify $\pm\sqrt{2}$correctly. A few neglected the negative root, losing an easy mark. Thereafter candidates went astray in a variety of ways. Many candidates used incorrect forms when writing their inequalities. $x > \pm\sqrt{2}$ was seen frequently and many candidates combined their separate inequalities in illegal ways such as $\sqrt{2} < x < -\sqrt{2}$. These candidates were penalised if the correct inequalities were not seen first. Candidates should realise that it is good practise to write the two inequalities separately first, before any attempt is made to combine them. Some candidates decimalised $\pm\sqrt{2}$ were penalised for having a slight inaccuracy in their answer. A few candidates didn't differentiate at all, thereby ignoring the instruction to use calculus and so made no progress.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	$\dfrac{(5.1^2-10.2)-(5^2-10)}{5.1-5}$ oe	M1	condone omission of brackets	0 for 8.1 unsupported
		i	8.1	A1	**Examiner's Comments** The majority of candidates gained full marks on this question. A significant minority differentiated and substituted in the midpoint, or the endpoints of the chord and found the mean. Whilst these approaches do achieve the correct numerical answer, they nevertheless went unrewarded.	
		ii	$\dfrac{(5+h)^2-2(5+h)-\text{ their }15}{h}$ oe	M1	condone omission of brackets	
		ii	$25 + 10h + h^2 - 10 - 2h$ oe seen	M1	allow one sign error	
		ii	numerator is $8h + h^2$	A1		
		ii	$8 + h$ isw	A1	**Examiner's Comments** Many candidates clearly didn't understand the notation, and either produced expressions involving x and h, or "expanded brackets" and worked with $5f + fh$. A good number of candidates did understand what this question was about, and successfully substituted to obtain correct expressions. Some made sign errors or slips in arithmetic: $h + 12$ was a common wrong answer, and a few knew what the answer was supposed to be and "back-engineered" their incorrect work accordingly.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$h \to 0$	M1	may be embedded; allow eg "tends to 0"	**M0** for differentiation of $x^2 - 2x$ **M0** for following from part (i) **M0** for $h = 0$
		iii	their 8	A1	**FT** their $k + h$ from part (ii) **Examiner's Comments** Only a few candidates used the correct terminology or notation here. Some worked with $h = 0$ and a good number ignored part (ii) and differentiated. Neither approach scored.	
		iv	$y = 8x - 25$ isw	B1	or $y - 15 = 8(x - 5)$ isw or $y = 8x + c$ and $c = -25$ stated isw	
		iv	non-zero numerical value for x-intercept on their straight line found	M1		
		iv	[x =] 3.125 oe	A1	may be embedded in calculation for area	
		iv	$\frac{1}{2} \times$their non-zero y -intercept × their $\frac{25}{8}$	M1	condone arithmetic slips in finding values of intercepts	or integration and evaluation of their $\int_0^{25/8} (8x - 25)\mathrm{d}x$; lower limit must be 0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$\frac{625}{16}$ or $39\frac{1}{16}$ or 39.0625 isw	A1	accept rounded to 1 dp or better for **A1**; but **A0** if **final** answer negative **Examiner's Comments** Many candidates found the correct equation and went on to achieve full marks. Some didn't read the question carefully and used (5, 15) with (3.125, 0). A small number of candidates found the equation of the normal and were thus only able to access two method marks.	
			Total	13		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	$\frac{h}{2} \times (0 + 0 + 2[4 + 4.9 + 5 + 4.9 + 4])$ oe	M1	correct formula used with 4, 5 or 6 strips and numerical value for *h*; condone omission of zeros or omission of outer brackets for both **M** marks	allow eg ½ × 1 × (4 + 4 + 2[4.9 + 5 + 4.9]) ½ × 1 × (4 + 0 + 2[4 + 4.9 + 5 + 4.9]) (**NB** may be implied by 18.8 & 20.8 respectively)
		i	all non-zero *y*-values correctly placed	M1	**M0M0** if 1, 2, 3 or 6 used as *y*-values (these are *x* -values)	
		i	*h* = 1 used in formula or consistently with two triangles and four trapezia	B1	if **M0M0** allow **B1** for *h* = 1 and **B2** for 22.8 from area of 4 trapezia and 2 triangles and **B1** for 1140	
		i	area = 22.8 **and** volume = 1140 isw cao	A1	ignore units **Examiner's Comments** Most candidates used the Trapezium rule correctly and went on to score full marks. A few made bracket errors or misplaced the *y* -values. Even fewer successfully found the correct value for the area by splitting the area into separate triangles and rectangles. This approach is not recommended – most go wrong and fail to score.	if **M0M0B0** allow **SC4** for 22.8 and 1140 obtained correctly by other method
		ii	A substitution of $x = 1.2$ or 4.8 to find y	M1	allow substitution of $1.2 \le x \le 1.234$ or $4.766 \le x \le 4.8$	or **M1** for $y = 4.4$, $x = 1.234$ [or 4.766] and

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	y = 4.35 or 4.352 and correct comparison with 4.4 isw	A1	**Examiner's Comments** A minority worked out what to do here and used a correct value of x to find y, which was usually correctly compared with 4.4. However, many candidates misunderstood what was required, substitution of 3 or 3.6 were common errors. A few unsuccessfully tried to compare cross-sectional areas.	**A1** for comparison of 1.234 with 1.2 or 4.766 with 4.8 [so gap less than 3.6]
		ii	$\mathrm{F}[x]=\frac{5}{81}\left(\frac{108}{2}x^2-\frac{54}{3}x^3+\frac{12}{4}x^4-\frac{x^5}{5}\right)$ oe B	M2	**M1** for 3 correct terms; ignore $+c$	condone omission of $\frac{5}{81}$;
		ii	eg $\frac{10}{3}x^2-\frac{10}{9}x^3+\frac{5}{27}x^4-\frac{1}{81}x^5$		allow coefficients 3.333333…, 1.11111…, 0.185185…, 0.01234567…r.o.t to 2 sf or better or decimal equivalents in numerator: 6.6666…, 3.333333…, 0.74074…, 0.061728… r.o.t to 2 sf or better	**M0** if $\frac{5}{81}x$ seen outside braceket but next **M1** is still available; ignore subsequent attempt to evaluate c for first **M2**
		ii	F[6] – F[0] or 2 × (F[3] – F[0])	M1	dependent on at least two terms correctly integrated in bracket; condone omission of –F(0)	**M0** for non-zero lower limit
		ii	24	A1		24 unsupported does not score

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	1200	B1	**Examiner's Comments** Most candidates integrated successfully and substituted the correct limits to find the correct area. However, some made an error in one of the terms, and some omitted the factor of $^5/_{81}$, which cost the later accuracy marks A few candidates lost marks by substituting incorrect limits.	ignore units
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$kx^{\frac{1}{2}-1}$ or $kx^{-\frac{1}{2}}$ seen	M1	$k > 0$	**B2** for correct answer unsupported
		i	$3x^{-\frac{1}{2}}$ or $\frac{3}{\sqrt{x}}$ isw	A1	**A0** for eg $3x^{-\frac{1}{2}} + c$ **Examiner's Comments** This was done well. A small minority of candidates failed to score: most problems were caused by a failure to put the original function into index form correctly. Occasionally $3^{-½}$ was seen as a final answer.	
		ii	kx^{-2+1} or kx^{-1} oe seen	M1	for any non-zero k	**SC0** for $\frac{12}{2x}$ or $\frac{6}{x}$
		ii	$-12x^{-1}$ or $-\frac{12}{x}$ or $\frac{-12}{x}$ isw	A1		**A0** for $\frac{12}{-x}$ or $\frac{12x^{-1}}{-1}$
		ii	$+c$	A1	seen at least once following integration **Examiner's Comments** A few candidates differentiated or tried to integrate both the numerator and the denominator independently, but most knew what to do here and went on to score 2 or 3 marks. A significant minority of candidates neglected to add "+ c", thereby losing an easy mark.	do not allow MR for integration of $12x^2$
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	At P(a, a) g(a) = a so ½(e^a –1) = a			
		i	⇒ $e^a = 1 + 2a$*	B1	NB AG **Examiner's Comments** This mark was usually earned.	
		ii	$A = \int_0^a \frac{1}{2}(e^x - 1)\,dx$	M1	correct integral and limits	limits can be implied from subsequent work
		ii	$= \frac{1}{2}\left[e^x - x\right]_0^a$	B1	integral of $e^x - 1$ is $e^x - x$	
		ii	= ½ (e^a –a e^0)	A1		
		ii	= ½ (1 + 2a –a –1) = ½ a*	A1	NB AG	
		ii	area of triangle = ½ a^2	B1		
		ii	area between curve and line = ½ a^2–½ a	B1cao	mark final answer **Examiner's Comments** Virtually everyone scored M1 for writing down the correct integral and limits, but many candidates made a meal of trying to integrate ½ (e^x – 1) , with ¼ (e^x – 1)2 not an uncommon wrong answer. Having successfully negotiated this hurdle, using part (i) to derive ½ a was spotted by about 50% of the candidates. Quite a few candidates managed to recover to earn the final 2 marks for ½ (a^2 – a) (without incorrectly simplifying this to½ a!).	
		iii	y = ½(e^x – 1) swap x and y x = ½ (e^y – 1)			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Rightarrow 2x = e^y - 1$	M1	Attempt to invert — one valid step	merely swapping x and y is not 'one step'
		iii	$\Rightarrow 2x + 1 = e^y$	A1		
		iii	$\Rightarrow \ln(2x + 1) = y^*$ $\Rightarrow g(x) = \ln(2x + 1)$	A1	$y = \ln(2x + 1)$ or $g(x) = \ln(2x + 1)$ AG	apply a similar scheme if they start with $g(x)$ and invert to get $f(x)$. or $g\,f(x) = g((e^x - 1)/2)$ M1
		iii	Sketch: recognisable attempt to reflect in			
		iii	$y = x$	M1	through O and (a, a)	$= \ln(1 + e^x - 1) = \ln(e^x)$ A1 $= x$ A1
		iii	Good shape	A1	no obvious inflexion or TP, extends to third quadrant, without gradient becoming too negative **Examiner's Comments** Finding the inverse function proved to be an easy 3 marks for most candidates – candidates are clearly well practiced in this. The graphs were usually recognisable reflections in y = x, but only well drawn examples – without unnecessary maxima or inflections – were awarded the 'A' mark.	similar scheme for fg See appendix for examples
		iv	$f'(x) = \frac{1}{2} e^x$	B1		
		iv	$g'(x) = 2/(2x + 1)$	M1	$1/(2x + 1)$ (or $1/u$ with $u = 2x + 1$) …	
		iv		A1	… × 2 to get $2/(2x + 1)$	
		iv	$g'(a) = 2/(2a + 1)$, $f'(a) = \frac{1}{2} e^a$	B1	either $g'(a)$ or $f'(a)$ correct soi	
		iv	so $g'(a) = 2/e^a$ or $f'(a) = \frac{1}{2}(2a + 1)$	M1	substituting $e^a = 1 + 2a$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$= 1/(½e^a) = (2a + 1)/2$ [$= 1/f'(a)$] [$= 1/g'(a)$])	A1	establishing $f'(a) = 1/g'(a)$	either way round
		iv	tangents are reflections in $y = x$	B1	must mention tangents **Examiner's Comments** This proved to be more difficult, as intended for the final question in the paper. As with the integral, many candidates struggled to differentiate ½ $(e^x – 1)$ correctly, and equally many omitted the '2' in the numerator of the derivative of $\ln(1 + 2x)$. Once these were established correctly the substitution of $x = a$ and establishing of $f'(a) = 1/g'(a)$ was generally done well, though sometimes the arguments using the result in part (i) were either inconclusive or done 'backwards'. The final mark proved to be elusive for most, as we needed the word 'tangent' used here to provide a geometric interpretation of the reciprocal gradients.	
			Total	19		

Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	translation in the *x*-direction	M1	allow 'shift', 'move'	If just vectors given withhold **one** 'A' mark only
		i	of π/4 to the right	A1	oe (eg using vector)	'Translate $\begin{pmatrix}\pi/4\\1\end{pmatrix}$', is 4 marks; if this is followed by an additional incorrect transformation, SC M1M1A1A0
		i	translation in *y*–direction	M1	allow 'shift', 'move'	$\begin{pmatrix}\pi/4\\1\end{pmatrix}$ only is M2A1A0
		i	of 1 unit up.	A1	oe (eg using vector) **Examiner's Comments** We usually insist on the word 'translation' here, but in this case allowed 'move', 'shift', etc. A vector on its own does not in our view imply a translation. Occasionally, candidates clearly knew what the transformations were, but wrote the vectors incorrectly, for example the wrong way up. Nevertheless, this topic is usually well known and done well.	
		ii	$g(x)=\dfrac{2\sin x}{\sin x+\cos x}$			(Can deal with num and denom separately)
		ii	$g'(x)=\dfrac{(\sin x+\cos x)2\cos x-2\sin x(\cos x-\sin x)}{(\sin x+\cos x)^2}$	M1	Quotient (or product) rule consistent with their derivs	$\dfrac{vu'-uv'}{v^2}$; allow one slip, missing brackets
		ii	$=\dfrac{2\sin x\cos x+2\cos^2 x-2\sin x\cos x+2\sin^2 x}{(\sin x+\cos x)^2}$ $=\dfrac{2\cos^2 x+2\sin^2 x}{(\sin x+\cos x)^2}=\dfrac{2(\cos^2 x+\sin^2 x)}{(\sin x+\cos x)^2}$	A1	Correct expanded expression (could leave the '2' as a factor)	$\dfrac{uv'-vu'}{v^2}$
		ii	$=\dfrac{2}{(\sin x+\cos x)^2}$ *	A1	NB AG	must take out 2 as a factor or state $\sin^2 x+\cos^2 x=1$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	When $x = \pi/4$, $g'(\pi/4) = 2/(1/\sqrt{2} + 1/\sqrt{2})^2$	M1	substituting $\pi/4$ into correct deriv	
		ii	$= 1$	A1		
		ii	$f'(x) = \sec^2 x$	M1	o.e., e.g. $1/\cos^2 x$	
		ii	$f'(0) = \sec^2(0) = 1$, [so gradient the same here]	A1	**Examiner's Comments** The quotient rule is generally well known, and errors here usually stemmed from faulty derivatives or poor algebra. Brackets are not optional in an expression like this, and their removal was not always successfully achieved. We also needed evidence of the use of $\cos^2 x + \sin^2 x = 1$, either by its direct quotation or by factoring out the '2' in the numerator. The evaluation of $g'(x)$ was usually correct. With $f'(x)$, some used a quotient rule on $\sin x/\cos x$ rather than quoting the derivative of $\tan x = \sec^2 x$; we also got some occasional 'translation' arguments here which misunderstood the nature of the verification.	
		iii	$= \int_1^{1/\sqrt{2}} -\frac{1}{u}\,\mathrm{d}u$ let $u = \cos x$, $\mathrm{d}u = -\sin x\,\mathrm{d}x$ when $x = 0$, $u = 1$, when $x = \pi/4$, $u = 1/\sqrt{2}$			
		iii	$= \int_{1/\sqrt{2}}^{1} \frac{1}{u}\,\mathrm{d}u$ *	M1	substituting to get $\int -1/u$ ($\mathrm{d}u$)	ignore limits here, condone no du but not dx allow $\int 1/u.-\mathrm{d}u$
		iii	$= \int_{1/\sqrt{2}}^{1} \frac{1}{u}\,\mathrm{d}u$ *	A1	NB **AG**	but for A1 must deal correctly with the -ve sign by interchanging limits

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= [\ln u]_{1/\sqrt{2}}^{1}$	M1	[ln *u*]	
		iii	= ln 1 - ln (1/√2)			
		iii	= ln √2 = ln $2^{½}$ ln 2	A1	ln √2, ½ ln 2 or –ln(1/√2) **Examiner's Comments** This was a case where giving the transformed integral proved to be of doubtful value, as many candidates 'lost' the negative sign in their ∫-1/*u* d*u*, and placed the limits the wrong way round. It appears that the idea of swapping limits making the integral negative was not generally understood. The evaluation of the given integral with respect to *u* was more successfully done, though quite a few candidates approximated their final answer.	mark final answer
		iv	Area = area in part **(iii)** translated up 1 unit.	M1	soi from π/4 added	or
		iv	So = ½ ln 2 + 1 × π4 = ½ ln 2 + π/4.	A1cao	oe (as above) **Examiner's Comments** These marks were gained by candidates who managed to spot the rectangle of area added by the translation upwards of the graph of f(*x*).	
			Total	17		

Question			Answer/Indicative content	Marks	Part marks and guidance	
19			Let $u = 1 + x \Rightarrow$			
			$\int_0^3 x(1+x)^{-1/2}\,\mathrm{d}x = \int_1^4 (u-1)u^{-1/2}\,\mathrm{d}u$	M1	$\int (u-1)u^{-1/2}(\mathrm{d}u)$*	condone no du, missing bracket, ignore limits
			$= \int_1^4 (u^{1/2} - u^{-1/2})\,\mathrm{d}u$	A1	$\int (u^{1/2} - u^{-1/2})(\mathrm{d}u)$	
			$= \left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^4$	A1	$\left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]$ o.e.	e.g. $\left[\frac{u^{3/2}}{3/2} - \frac{u^{1/2}}{1/2}\right]$; ignore limits
			= (16/3 – 4) – (2/3 – 2)	M1dep	upper–lower dep 1st M1 and integration	with correct limits e.g. 1, 4 for u or 0, 3 for x
			$= 2\frac{2}{3}$	A1cao	or $2.\dot{6}$ but must be exact	or using $w = (1 + x)^{1/2} \Rightarrow$ $\int \frac{(w^2-1)2w}{w}(\mathrm{d}w)$ M1
			OR Let $u = x$, $v' = (1 + x)^{-1/2}$	M1		$= \int 2(w^2-1)(\mathrm{d}w)$ A1 $= \left[\frac{2}{3}w^3 - 2w\right]$ A1
			$\Rightarrow u' = 1$, $v = 2(1 + x)^{1/2}$	A1		upper–lower with correct limits (w = 1,2) M1
			$\Rightarrow \int_0^3 x(1+x)^{-1/2}\,\mathrm{d}x = \left[2x(1+x)^{1/2}\right]_0^3 - \int_0^3 2(1+x)^{1/2}\,\mathrm{d}x$	A1	ignore limits, condone no dx	8/3 A1 cao
			$= \left[2x(1+x)^{1/2} - \frac{4}{3}(1+x)^{3/2}\right]_0^3$	A1	ignore limits	*If $\int_1^4 (u-1)u^{-1/2}$ du done by parts:
			= (2 × 3 × 2 – 4 × 8/3) – (0 – 4/3)			$2u^{1/2}(u-1) - \int 2u^{1/2}$ du A1 $[2u^{1/2}(u-1) - 4u^{3/2}/3]$ A1

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$= 2\frac{2}{3}$	A1cao	or $2.\dot{6}$ but must be exact **Examiner's Comments** Most candidates used integration by substitution, though a significant minority used integration by parts. In general, the former were more successful, with the main difficulty being in expanding $(u - 1)u^{-1/2}$ as $u^{1/2} - u^{-1/2}$. Some proceeded from here using integration by parts, with mixed success. When parts were used, the most common error was in deriving $v = 2(1 + x)1/2$ from $v' = (1 + x)^{-1/2}$.	substituting correct limits M1 8/3 A1cao
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i	$\mathrm{d}F / \mathrm{d}v = -25\ v^{-2}$	M1	$\mathrm{d} / \mathrm{d}v(v^{-1}) = -v^{-2}$ soi	
		i		A1	$-25\ v^{-2}$ o.e mark final ans **Examiner's Comments** This was almost invariably correctly done. No candidates seemed to be put off by the rather excessive speed of the car. Occasionally, the quotient rule was seen, with errors in differentiating the '25'.	
		ii	When $v = 50$, $\mathrm{d}F/\mathrm{d}v = -25/50^2$ (= –0.01)	B1	$-25/50^2$	
		ii	$\frac{\mathrm{d}F}{\mathrm{d}t} = \frac{\mathrm{d}F}{\mathrm{d}v} . \frac{\mathrm{d}v}{\mathrm{d}t}$	M1	o.e.	e.g. $\frac{\mathrm{d}F}{\mathrm{d}v} = \frac{\mathrm{d}F}{\mathrm{d}t} / \frac{\mathrm{d}v}{\mathrm{d}t}$
		ii	= –0.01 × 1.5 = –0.015	A1cao	o.e. e.g. –3/200 isw **Examiner's Comments** Again, this was very well answered, provided part (i) was correct. Almost all candidates scored an M1 for the chain rule.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
21		i	$2x+4y\frac{dy}{dx}=4$	M1	$4y\frac{dy}{dx}$	Rearranging for *y* and differentiating explicitly is M0
		i		A1	correct equation	Ignore superfluous d*y*/d*x* = … unless used subsequently
		i	$\Rightarrow \quad \frac{dy}{dx}=\frac{4-2x}{4y}$	A1	o.e., but mark final answer **Examiner's Comments** This relatively simple implicit differentiation was very well done by almost all candidates.	
		ii	$\frac{dy}{dx}=0 \Rightarrow x=2$	B1dep	dep correct derivative	
		ii	$\Rightarrow 4 + 2y^2 = 8 \Rightarrow y^2 = 2$, $y = \sqrt{2}$ or $-\sqrt{2}$	B1B1	$\sqrt{2}, -\sqrt{2}$ **Examiner's Comments** Most candidates scored two out of three for the point (2, √2), but missed the y = .√2 solution. In a few cases, the denominator was set to zero, giving y = 0.	can isw, penalise inexact answers of ±1.41 or better once only –1 for extra solutions found from using $y = 0$
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	$y = e^{-x} \sin 2x$	M1	Product rule	u × their v' + v × their u'
		i	⇒ $dy/dx = e^{-x}.2\cos 2x + (e^{-x})\sin 2x$	B1	$d/dx(\sin 2x) = 2\cos 2x$	
		i		A1	Any correct expression **Examiner's Comments** This proved to be a straightforward start to the paper, with the large majority of candidates getting full marks. Of those who did not, the most common errors were in the derivative of sin 2x (getting cos 2x or ½ cos 2x) or e-x (omitting the negative sign).	but mark final answer
		ii	$dy/dx = 0$ when $2\cos 2x - \sin 2x = 0$	M1	ft their dy/dx but must eliminate e^{-x}	derivative must have 2 terms
		ii	′ $2 = \tan 2x$	M1	$\sin 2x / \cos 2x = \tan 2x$ used	substituting ½ arctan 2 into their deriv M0
		ii	′ $2x = \arctan 2$		[or $\tan^{-1}$]	(unless $\cos 2x = 1/\sqrt{5}$ and $\sin 2x = 2/\sqrt{5}$ found)
		ii	⇒ $x = ½ \arctan 2$*	A1	NB AG **Examiner's Comments** This part was somewhat less successful. Quite a few candidates just substituted the given answer into the derivative and claimed that this was zero.	must show previous step
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	$a = ½$	B1	allow $x = ½$ Examiner's Comments Nearly all candidates gained this mark for the asymptote.	
		ii	$y^3 = \frac{x^3}{2x-1}$ $\Rightarrow \quad 3y^2\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(2x-1)3x^2 - x^3.2}{(2x-1)^2}$	B1 M1 A1	$3y^2\mathrm{d}y/\mathrm{d}x$ Quotient (or product) rule consistent with their derivatives; (v du + udv)/v2 M0 correct RHS expression – condone missing bracket	
		ii	$= \frac{6x^3 - 3x^2 - 2x^3}{(2x-1)^2} = \frac{4x^3 - 3x^2}{(2x-1)^2}$	A1		
		ii	$\Rightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{4x^3 - 3x^2}{3y^2(2x-1)^2}$ *	A1	**NB AG** penalise omission of bracket in QR at this stage	
		ii	$\mathrm{d}y/\mathrm{d}x = 0$ when $4x^3 - 3x^2 = 0$	M1		
		ii	$\Rightarrow x^2(4x - 3) = 0,\ x = 0$ or $¾$	A1	if in addition $2x - 1 = 0$ giving $x = ½$, A0	
		ii	$y^3 = (¾)^3/½ = 27/32,$	M1	must use $x = ¾$; if (0, 0) given as an additional TP, then A0	
		ii	$y = 0.945$ (3sf)	A1	can infer M1 from answer in range 0.94 to 0.95 inclusive	
		ii	Additional suggestions $y = \frac{x}{(2x-1)^{1/3}}$ $\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(2x-1)^{1/3}.1 - x.(1/3)(2x-1)^{-2/3}.2}{(2x-1)^{2/3}}$	M1 A1	quotient rule or product rule on y – allow one slip correct expression for the derivative	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$=\dfrac{6x-3-2x}{3(2x-1)^{4/3}}=\dfrac{4x-3}{3(2x-1)^{4/3}}$	M1 A1	factorising or multiplying top and bottom by $(2x-1)^{2/3}$	
		ii	$=\dfrac{(4x-3)x^2}{3y^2(2x-1)^{2/3}(2x-1)^{4/3}}=\dfrac{4x^3-3x^2}{3y^2(2x-1)^2}$	A1	establishing equivalence with given answer **NB AG**	
		ii	$y=\left(\dfrac{x^3}{(2x-1)}\right)^{1/3}$ $\Rightarrow \dfrac{dy}{dx}=\dfrac{1}{3}\left(\dfrac{x^3}{(2x-1)}\right)^{-2/3}\dfrac{(2x-1).3x^2-x^3.2}{(2x-1)^2}$	B1 M1A1	$\dfrac{1}{3}\left(\dfrac{x^3}{(2x-1)}\right)^{-2/3}\times\ldots$ $\ldots\times\dfrac{(2x-1).3x^2-x^3.2}{(2x-1)^2}$	
		ii	$=\dfrac{1}{3}\dfrac{4x^3-3x^2}{x^2(2x-1)^{4/3}}=\dfrac{4x-3}{3(2x-1)^{4/3}}$	A1		
		ii	$=\dfrac{(4x-3)x^2}{3y^2(2x-1)^{2/3}(2x-1)^{4/3}}=\dfrac{4x^3-3x^2}{3y^2(2x-1)^2}$	A1	establishing equivalence with given answer **NB AG**	

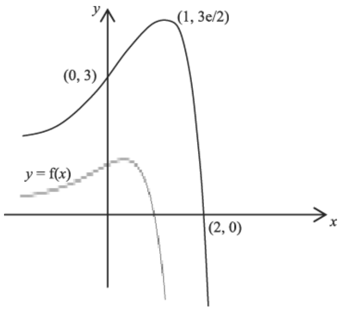
Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$y^3(2x-1)=x^3$ $3y^2\frac{dy}{dx}(2x-1)+y^3.2=3x^2$ $\frac{dy}{dx}=\frac{3x^2-2y^3}{3y^2(2x-1)}$ $=\frac{3x^2-2\frac{x^3}{(2x-1)}}{3y^2(2x-1)}$ $=\frac{3x^2(2x-1)-2x^3}{3y^2(2x-1)^2}=\frac{6x^3-3x^2-2x^3}{3y^2(2x-1)^2}=\frac{4x^3-3x^2}{3y^2(2x-1)^2}$	B1 M1 A1 M1 A1	d/dx(y^3) = $3y^2$(dy/dx) product rule on $y^3(2x-1)$ or $2xy^3$ correct equation subbing for $2y^3$ NB AG ☐<u>Examiner's Comments</u> Candidates tended to score heavily on this part. The implicit differentiation of y^3 was usually correct (albeit introduced into solutions belatedly), and the quotient rule was done well, though occasionally omission of brackets was penalised. Those who cube rooted and differentiated often succeeded in arriving at the given derivative. Another approach was to multiplying across before differentiating implicitly, but with required candidates to substitute for y to deduce the required form for the derivative. Finding x = ¾ for the turning point from the given derivative was straightforward, but some failed to find the correct y-coordinate by omitting the necessary cube root.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$u = 2x - 1 \Rightarrow \mathrm{d}u = 2\mathrm{d}x$ $\int \frac{x}{\sqrt[3]{2x-1}}\mathrm{d}x = \int \frac{\frac{1}{2}(u+1)}{u^{1/3}} \frac{1}{2}\mathrm{d}u$	M1 M1	$\frac{\frac{1}{2}(u+1)}{u^{1/3}}$ if missing brackets, withhold A1 × ½ du condone missing du here, but withhold A1	
		iii	$= \frac{1}{4}\int \frac{u+1}{u^{1/3}}\mathrm{d}u = \frac{1}{4}\int (u^{2/3} + u^{-1/3})\mathrm{d}u$ *	A1	NB AG	
		iii	area $= \int_1^{4.5} \frac{x}{\sqrt[3]{2x-1}}\mathrm{d}x$	M1	correct integral and limits – may be inferred from a change of limits and their attempt to integrate (their) ¼ $(u^{2/3} + u^{-1/3})$	
		iii	when $x = 1$, $u = 1$, when $x = 4.5$, $u = 8$	A1	$u = 1, 8$ (or substituting back to x's and using 1 and 4.5)	
		iii	$= \frac{1}{4}\int_1^8 (u^{2/3} + u^{-1/3})\mathrm{d}u$ $= \frac{1}{4}\left[\frac{3}{5}u^{5/3} + \frac{3}{2}u^{2/3}\right]_1^8$	B1	$\left[\frac{3}{5}u^{5/3} + \frac{3}{2}u^{2/3}\right]$ o.e. e.g. $[u^{5/3}/(5/3) + u^{2/3}/(2/3)]$	
		iii	$= \frac{1}{4}\left[\frac{96}{5} + 6 - \frac{3}{5} - \frac{3}{2}\right]$	A1	o.e. correct expression (may be inferred from a correct final answer)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= 5\dfrac{31}{40} = 5.775$ or $\dfrac{231}{40}$	A1	cao, must be exact; mark final answer **Examiner's Comments** There were plenty of accessible marks here as well. The first three marks, for transforming the integral to the variable *u*, were usually negotiated successfully, although poor notation – omitting d*u*'s or brackets – was sometimes penalised in the A1 mark. The second half involved evaluating the given integral with the correct limits. Some calculated the correct limits, but made errors in the integral (or forgot to integrate altogether). However, a reasonable number of candidates managed to do this work without errors. A rather curious misconception was to cube the correct value of the integral, because the function was presented implicitly in terms of y^3.	
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	(1, 0) and (0, 1)	B1B1	$x = 0$, $y = 1$; $y = 0$, $x = 1$ __Examiner's Comments__ The points of intersection were a write-down for many candidates. Weaker attempts failed to solve $(1 - x)\,e^{2x} = 0$ convincingly.	
		ii	$f'(x) = 2(1 - x)e^{2x} - e^{2x}$	B1	$d/dx(e^{2x}) = 2e^{2x}$	
		ii		M1	product rule consistent with their derivatives	
		ii	$= e^{2x}(1 - 2x)$	A1	correct expression, so $(1 - x)e^{2x} - e^{2x}$ is B0M1A0	
		ii	$f'(x) = 0$ when $x = ½$	M1dep	setting their derivative to 0 dep 1st M1	
		ii		A1cao	$x = ½$	
		ii	$y = ½\,e$	B1	allow $½\,e^1$ isw __Examiner's Comments__ This proved to be an accessible 6 marks for candidates. The derivative of e^{2x} and the product rule were generally correct, and deriving $x = ½$ and $y = e^{1/2}$ was straightforward, though many did not simplify the derivative to $e^{2x} - 2xe^{2x}$ immediately. Some candidates approximated for $e^{1/2}$ and lost a mark.	
		iii	$A = \int_0^1 (1 - x)e^{2x}\,dx$	B1	correct integral and limits; condone no dx (limits may be seen later)	
		iii	$u = (1 - x),\ u' = -1,\ v' = e^{2x},\ v = ½\,e^{2x}$	M1	u, u', v', v, all correct; or if split up $u = x$, $u' = 1$, $v' = e^{2x}$, $v = ½\,e^{2x}$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Rightarrow \quad A=\left[\frac{1}{2}(1-x)e^{2x}\right]_0^1-\int_0^1 \frac{1}{2}e^{2x}.(-1)dx$	A1	condone incorrect limits; or, from above, $\ldots\left[\frac{1}{2}xe^{2x}\right]_0^1-\int_0^1 \frac{1}{2}e^{2x}dx$	
		iii	$=\left[\frac{1}{2}(1-x)e^{2x}+\frac{1}{4}e^{2x}\right]_0^1$	A1	o.e. if integral split up; condone incorrect limits	
		iii	$= ¼\ e^2 - ½ - ¼$ $= ¼\ (e^2 - 3)$ *	A1cao	NB AG __Examiner's Comments__ Most candidates applied integration by parts to either $\int(1-x)\ e^{2x}dx$ or $\int x\ e^{2x}dx$, using appropriate u, v', u' and v. Sign and/or bracket errors sometimes meant they failed to derive the correct result, but many were fully correct.	
		iv	$g(x) = 3f(½\ x) = 3(1 - ½\ x)\ e^x$	B1	o.e; mark final answer	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv		B1 B1dep B1	through (2,0) and (0,3) – condone errors in writing coordinates (e.g. (0,2)). reasonable shape, dep previous B1 TP at (1, 3e/2) or (1, 4.1) (or better). (Must be evidence that x = 1, y = 4.1 is indeed the TP – appearing in a table of values is not enough on its own.) **Examiner's Comments** This part proved to be quite demanding. Deriving the formula for g(x) was rarely correctly done. Common errors were an extra factor of 3 and an incorrect exponent. Most graphs showed the correct points of intersection (0, 3) and (2, 0), but the turning point was quite often incorrect or missing, and the shape failed to convince.	
		v	6 × ¼ (e^2 – 3) [= 3(e^2 – 3)/2]	B1	o.e. mark final answer **Examiner's Comments** Those, of the relatively few candidates, who got this correct just wrote down 2 x 3 x ¼ (e^2 – 3). Some tried to integrate g(x), with little success.	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	18	

Question			Answer/Indicative content	Marks	Part marks and guidance	
25			$\int_0^{\pi/2} \frac{\sin 2x}{3+\cos 2x} dx = \left[-\frac{1}{2}\ln(3+\cos 2x)\right]_0^{\pi/2}$	M1	k ln(3 + cos 2x)	
				A2	½ ln(3 + cos 2x)	
			or u = 3 + cos 2x, du = –2sin2x dx	M1	o.e. e.g. du/dx = –2sin2x or if v = cos2x, dv = –2sin2xdx o.e. condone 2sin2x dx	
			$\int_0^{\pi/2} \frac{\sin 2x}{3+\cos 2x} dx = \int_4^2 -\frac{1}{2u} du$ $= \left[-\frac{1}{2}\ln u\right]_4^2$ $= -$ ½ ln 2 + ½ ln 4 = ½ ln (4/2) = ½ ln 2 *	A1	$\int -\frac{1}{2u} du$, or if $v = \cos 2x$, $\int -\frac{1}{2(3+v)} dv$	
				A1	[– ½ ln$u$] or [– ½ ln(3 + v)] ignore incorrect limits	
				A1	from correct working o.e. e.g. – ½ ln(3 + cos(2.π/2)) + ½ ln(3 + cos(2.0)) o.e. required step for final A1, must have evaluated to 4 and 2 at this stage	
				A1	NB AG ☐<u>Examiner's Comments</u> The error d/dx (cos 2x) = 2sin 2x proved costly here, earning only a consolation M1; many also wrote the limits the wrong way round on the integral, and scored 3 out of 5, unless they 'lost' the negative sign, and scored M1 only. Many candidates seem unaware that swapping limits dealt with the negative sign. We also needed to see some evidence of why ln 4 – ln 2 = 2 to score the final A1.	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	5	

Question	Answer/Indicative content	Marks	Part marks and guidance
26	$y=\ln\left(\sqrt{\frac{2x-1}{2x+1}}\right)=\frac{1}{2}(\ln(2x-1)-\ln(2x+1))$	M1	use of ln(a/b) = ln a – ln b□
		M1	use of ln√c = ½ ln c
	$\Rightarrow \quad \frac{dy}{dx}=\frac{1}{2}\left(\frac{2}{2x-1}-\frac{2}{2x+1}\right)$	A1	o.e.; correct expression (if this line of working is missing, M1M1A0A0)
	$=\frac{1}{2x-1}-\frac{1}{2x+1}$ *	A1	**NB AG** for alternative methods, see additional solutions
	Additional solutions		
	$y=\ln\left(\sqrt{\frac{2x-1}{2x+1}}\right)=\frac{1}{2}\ln\left(\frac{2x-1}{2x+1}\right)$	M1	ln √c = ½ ln c used
	$\Rightarrow \frac{dy}{dx}=\frac{1}{2}\frac{1}{\left(\frac{2x-1}{2x+1}\right)}\frac{(2x+1)2-(2x-1)2}{(2x+1)^2}$ $=\frac{1}{2}\frac{2x+1}{2x-1}\frac{4}{(2x+1)^2}=\frac{2}{(2x-1)(2x+1)}$ $\frac{1}{2x-1}-\frac{1}{2x+1}=\frac{2x+1-(2x-1)}{(2x-1)(2x+1)}$ $=\frac{2}{(2x-1)(2x+1)}$	A2	fully correct expression for the derivative
	$\frac{1}{2x-1}-\frac{1}{2x+1}=\frac{2x+1-(2x-1)}{(2x-1)(2x+1)}=\frac{2}{(2x-1)(2x+1)}$	A1	simplified and shown to be equivalent to $\frac{1}{2x-1}-\frac{1}{2x+1}$
	Additional solutions		
	$y=\ln\left(\sqrt{\frac{2x-1}{2x+1}}\right)=\ln\sqrt{2x-1}-\ln\sqrt{2x+1}$	M1	ln(a/b) = ln a – ln b used
	$\frac{dy}{dx}=\frac{1}{\sqrt{2x-1}}\frac{1}{2}(2x-1)^{-1/2}.2-\frac{1}{\sqrt{2x+1}}\frac{1}{2}(2x+1)^{-1/2}.2$	A2	fully correct expression
	$=\frac{1}{2x-1}-\frac{1}{2x+1}$	A1	simplified and shown to be equivalent to $\frac{1}{2x-1}-\frac{1}{2x+1}$

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Additional solutions			
			$y=\ln\left(\sqrt{\dfrac{2x-1}{2x+1}}\right)$ $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{1}{\sqrt{\dfrac{2x-1}{2x+1}}}\dfrac{1}{2}\left(\dfrac{2x-1}{2x+1}\right)^{-1/2}\dfrac{(2x+1)2-(2x-1)2}{(2x+1)^2}$ or	M1	$\dfrac{1}{u}\times$ their u' where $u=\sqrt{\dfrac{2x-1}{2x+1}}$ or $\dfrac{\sqrt{2x-1}}{\sqrt{2x+1}}$	
			$\dfrac{1}{\dfrac{\sqrt{2x-1}}{\sqrt{2x+1}}}\dfrac{\sqrt{2x+1}.\frac{1}{2}.2(2x-1)^{-1/2}-\sqrt{2x-1}.\frac{1}{2}.2(2x+1)^{-1/2}}{\sqrt{2x+1}^2}$ $=\dfrac{1}{2}\left(\dfrac{2x+1}{2x-1}\right)\dfrac{4}{(2x+1)^2}=\dfrac{2}{(2x-1)(2x+1)}$	A2	(any attempt at u' will do) o.e. any completely correct expression for the derivative or $=\dfrac{\sqrt{2x+1}}{\sqrt{2x-1}}\dfrac{(2x+1)-(2x-1)}{(2x+1)^{3/2}(2x-1)^{1/2}}=\dfrac{2}{(2x+1)(2x-1)}$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\frac{1}{2x-1}-\frac{1}{2x+1}=\frac{(2x+1)-(2x-1)}{(2x-1)(2x+1)}=\frac{2}{(2x-1)(2x+1)}$	A1	simplified and correctly shown to be equivalent to $\frac{1}{2x-1}-\frac{1}{2x+1}$ **Examiner's Comments** Some candidates spotted the trick of simplifying the given function to get $y = ½ \ln(2x – 1) – ½ \ln(2x + 1)$ before differentiating, and thereby made lives considerably easier for themselves! However, writing the answer down from here omitted the vital 2 x ½ working and lost two marks. Those who started differentiating from $y = \ln(\sqrt{(2x – 1)} – \ln(\sqrt{(2x + 1)})$ needed to convince that they were using a chain rule on $\sqrt{u}$, where $u = 2x – 1$. Some tenacious candidates even managed to differentiate the given function correctly without these preliminaries, but made life hard for themselves.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
27			$V = \pi h^2 \Rightarrow \mathrm{d}V/\mathrm{d}h = 2\pi h \Rightarrow$	M1A1	if derivative $2\pi h$ seen without $\mathrm{d}V/\mathrm{d}h = \ldots$ allow M1A0	
			$\mathrm{d}V/\mathrm{d}t = \mathrm{d}V/\mathrm{d}h \times \mathrm{d}h/\mathrm{d}t$	M1	soi; o.e. – any correct statement of the chain rule using V, h and t – condone use of a letter other than t for time here	
			$\mathrm{d}V/\mathrm{d}t = 10$	B1	soi; if a letter other than t used (and not defined) B0	
			$\mathrm{d}h/\mathrm{d}t = 10/(2\pi \times 5) = 1/\pi$	A1	or 0.32 or better, mark final answer <u>Examiner's Comments</u> This proved to be an accessible 5 marks, with many candidates getting the question fully correct. Of those who did not, $\mathrm{d}h/\mathrm{d}t$ = 10 (instead of $\mathrm{d}V/\mathrm{d}t$) was quite a common misconception; some tried to find $\mathrm{d}h/\mathrm{d}V$ but failed to handle the constant of $1/\sqrt{\pi}$ correctly; and a surprising number finished off by saying that $10/10\pi = \pi$ instead of $1/\pi$.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i	Range is $-1 \le y \le 3$	M1	$-1, 3$	
		i		A1	$-1 \le y \le 3$ or $-1 \le f(x) \le 3$ or $[-1, 3]$ (not -1 to 3, $-1 \le x \le 3$, $-1 < y < 3$ etc) <u>**Examiner's Comments**</u> This was generally well done, with one mark awarded for -1 and 3 seen, and one for the correct notation. Some used x instead of y or $f(x)$, and others confused domain and range.	
		ii	$y = 1 - 2\sin x$ $x \leftrightarrow y$		[can interchange x and y at any stage]	
		ii	$x = 1 - 2\sin y \Rightarrow x - 1 = -2\sin y$	M1	attempt to re-arrange	
		ii	$\Rightarrow \sin y = (1 - x)/2$	A1	o.e. e.g. $\sin y = (x - 1)/(-2)$ (or $\sin x = (y - 1)/(-2)$)	
		ii	$\Rightarrow y = \arcsin[(1 - x)/2]$	A1	or $f^{-1}(x) = \arcsin[(1 - x)/2]$, not x or $f^{-1}(y) = \arcsin[1 - y)/2]$ (viz must have swapped x and y for final 'A' mark) $\arcsin[(x - 1)/-2]$ is A0 <u>**Examiner's Comments**</u> Most candidates are well practiced at finding inverses, and were familiar with arcsine, gaining full marks here. Leaving the result as $y = \arcsin((x - 1)/-2)$ lost the final A1. Very occasionally, candidates gave the answer as $1/f(x)$ or $f'(x)$.	
		iii	$f'(x) = -2\cos x$	M1	condone $2\cos x$□	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Rightarrow f'(0) = -2$	A1	cao	
		iii	$\Rightarrow$ gradient of $y = f^{-1}(x)$ at (1, 0) = – ½	A1	not 1/–2 __Examiner's Comments__ Nearly all candidates found f'(x) and f'(0) correctly. The gradient of the inverse function was less successful. Many confused this with the condition for perpendicularity and gave the answer ½ instead of -1/2 . Those who tried to differentiate $f^{-1}(x)$ directly had little success.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i	$xe^{-2x} = mx$	M1	may be implied from 2nd line	
		i	$\Rightarrow e^{-2x} = m$	M1	dividing by x, or subtracting $\ln x$	o.e. e.g. $[\ln x] - 2x = \ln m + [\ln x]$ or factorising: $x(e^{-2x} - m) = 0$
		i	$\Rightarrow -2x = \ln m$			
		i	$\Rightarrow x = -\frac{1}{2}\ln m$*	A1	**NB AG**	
		i	or			
		i	If $x = -\frac{1}{2}\ln m$, $y = -\frac{1}{2}\ln m \times e^{\ln m}$	M1	substituting correctly	
		i	$= -\frac{1}{2}\ln m \times m$	A1		
		i	so P lies on $y = mx$	A1	**Examiner's Comments** This was well answered, even by weaker candidates.	
		ii	let $u = x$, $u' = 1$, $v = e^{-2x}$, $v' = -2e^{-2x}$	M1*	product rule consistent with their derivs	
		ii	$dy/dx = e^{-2x} - 2xe^{-2x}$	A1	o.e. correct expression	
		ii	$= e^{-2.(-\frac{1}{2}\ln m)} - 2.(-\frac{1}{2}\ln m)e^{-2.(-\frac{1}{2}\ln m)}$	M1dep	subst $x = -\frac{1}{2}\ln m$ into their deriv dep M1*	
		ii	$= e^{\ln m} + e^{\ln m}\ln m$ [$= m + m\ln m$]	A1cao	condone e^{1nm} not simplified **Examiner's Comments** The product rule here was generally well done, followed by substituting x = - ½ ln m, where some sign errors occurred. Some left the eln m terms unresolved, which was condoned here. The main error was to get a derivative of -2xe-2x.	but not $-2(-\frac{1}{2}\ln m)$, but mark final ans
		iii	$m + m1n\ m = -m$	M1	their gradient from (ii) $= -m$	
		iii	$\Rightarrow \ln m = -2$			
		iii	$\Rightarrow m = e^{-2}$*	A1	**NB AG**	
		iii	*or*			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$y + ½m\ln m = m(1 + \ln m)(x + ½ \ln m)$ $x = -\ln m$, $y = 0 \Rightarrow ½m\ln m = m(1 + \ln m)(-½ \ln m)$ $\Rightarrow 1 + \ln m = -1$, $\ln m = -2$, $m = e^{-2}$	B2	for fully correct methods finding *x*intercept of equation of tangent and equating to – ln *m*	
		iii	At P, $x = 1$	B1		
		iii	$\Rightarrow y = e^{-2}$	B1	isw approximations **Examiner's Comments** The first two marks here were the least successfully answered, because most candidates were not familiar with the fact that lines equally inclined to the x-axis have gradients m and –m. Only the best candidates found the result successfully. However, many recovered to find the coordinates of P correctly.	not $e^{-2} \times 1$
		iv	Area under curve $= \int_0^1 xe^{-2x}dx$			
		iv	$u = x$, $u' = 1$, $v' = e^{-2x}$, $v = -½ e^{-2x}$	M1	parts, condone $v = k e^{-2x}$, provided it is used consistently in their parts formula	ignore limits until 3rd A1
		iv	$= \left[-\frac{1}{2}xe^{-2x}\right]_0^1 + \int_0^1 \frac{1}{2}e^{-2x}dx$	A1ft	ft their *v*	
		iv	$= \left[-\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}\right]_0^1$	A1	$-\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}$ o.e	
		iv	$= (-½ e^{-2} - ¼ e^{-2}) - (0 - ¼ e^0)$ $[= ¼ - ¾ e^{-2}]$	A1	correct expression	need not be simplified

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	Area of triangle = ½ base × height	M1	ft their 1, e^{-2} or $[e^{-2}x^2/2]$	o.e. using isosceles triangle
		iv	= ½ × 1 × e^{-2}	A1		M1 may be implied from 0.067 …
		iv	So area enclosed = ¼ – $5e^{-2}/4$	A1cao	o.e. must be exact, two terms only **Examiner's Comments** This question tested the more able candidates. The integration by parts required careful control of negative signs and accurate work; the area of the triangle (or integral of the line) were quite often discernable from the working, which was often scrambled and incoherent – perhaps because some candidates were rushing to complete the paper!	isw
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i	$f(-x)=\frac{-x}{\sqrt{2+(-x)^2}}$	M1	substituting $-x$ for x in f (x)	$\frac{-x}{\sqrt{2+-x^2}}, \frac{-x}{\sqrt{2+-(x^2)}}, \frac{-x}{\sqrt{2+(-x^2)}}$ M1A0
		i	$=-\frac{x}{\sqrt{2+x^2}}=-f(x)$	A1	1st line must be shown, must have f ($-x$) = – f (x) oe somewhere	$\frac{-x}{\sqrt{2-x^2}}$ M0A0
		i	Rotational symmetry of order 2 about O	B1	must have 'rotate' and 'O' and 'order 2 or 180 or ½ turn' **Examiner's Comments** Most candidates stated that for an odd function f(-x) = -f(x) or equivalent. It is important when writing f(-x) that brackets are placed round the –x terms: if these were missing, the 'A' mark was lost. The structure of this 'show' was often a bit 'muddy': f(-x) = ̈ = ̈ = -f(x) is clear, but writing f(-x) = -f(x) and then writing expressions for each side of this equation below and showing they are equal is less so, as the direction of the argument, or implications, is not clear. The geometrical description of an odd function required three elements: 'rotational', 'order 2' and 'centre O' or equivalent; reflection in Ox followed by Oy was also allowed.	oe e.g. reflections in both x- and y-axes
		ii	$f'(x)=\frac{\sqrt{2+x^2}.1-x.\frac{1}{2}(2+x^2)^{-1/2}.2x}{(\sqrt{2+x^2})^2}$ $=\frac{2+x^2-x^2}{(2+x^2)^{3/2}}=\frac{2}{(2+x^2)^{3/2}}$ *	M1	quotient or product rule used	QR: condone udv ± vdu, but u, v and denom must be correct
		ii		M1	½ $u^{-1/2}$or – ½ $v^{-3/2}$soi	
		ii		A1	correct expression	$x(-1/2)(2+x^2)^{-3/2}.2x+(2+x^2)^{-1/2}$. $=(2+x^{-3/2})(-x^2+2+x^2)$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	NBAG	
		ii	When $x = 0$, $f'(x) = 2/2^{3/2} = 1/\sqrt{2}$	B1	oe e.g. $\sqrt{2}/2$, $2^{-1/2}$, $1/2^{1/2}$, but not $2/2^{3/2}$ **Examiner's Comments** The difficulty with this sort of product or quotient rule question lies in factorising and hence simplifying the expression, and this was the case here. Many wrote down correct expressions, but then failed to eshowf the printed answer. This difficulty often encouraged multiple attempts, sometimes using a quotient rule, followed by a product rule, etc. A surprising number of candidates muddled up their euf and evf and quotient and product rule, for example using v = (2+x2)– 1/2 in their quotient rule. Often the final answer failed to score because we insisted on this being simplified to 1/√2 or equivalent.	allow isw on these seen
		iii	$A = \int_0^1 \frac{x}{\sqrt{2+x^2}}[\mathrm{d}x]$	B1	correct integral and limits	limits may be inferred from subsequent working, condone no dx
		iii	let $u = 2 + x^2$, $\mathrm{d}u = 2x\,\mathrm{d}x$		or $v = \sqrt{(2 + x^2)}$, $\mathrm{d}v = x(2 + x^2)^{-1/2}\,\mathrm{d}x$	
		iii	$= \int_2^3 \frac{1}{2}\frac{1}{\sqrt{u}}\mathrm{d}u$	M1	$\int \frac{1}{2}\frac{1}{\sqrt{u}}[\mathrm{d}u]$ or $= \int 1[\mathrm{d}v]$ or $k(2+x^2)^{1/2}$	condone no du or dv, but not $\int \frac{1}{2}\frac{1}{\sqrt{u}}\mathrm{d}x$
		iii	$= \left[u^{1/2}\right]_2^3$	A1	$[u^{1/2}]$ o.e. (but not $1/u^{-1/2}$) or $[v]$ or $k = 1$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= \sqrt{3} - \sqrt{2}$	A1cao	must be exact **Examiner's Comments** A substantial minority of candidates thought this integral should be done by parts, and therefore scored nothing after the first B1. Those who tried substituting often got muddled before arriving at $\int 1/2\sqrt{u}\,du$, and some then integrated this incorrectly, e.g as $\ln\sqrt{u}$	isw approximations
		iv	$y^2 = \frac{x^2}{2+x^2}$	M1	squaring (correctly)	must show $[\sqrt{(2 + x^2)}]^2 = 2 + x^2$ (o.e.)
		iv	$\Rightarrow 1/y^2 = (2 + x^2)/x^2 = 2/x^2 + 1$ *	A1	or equivalent algebra **NB AG** **Examiner's Comments** This simple piece of algebra was often over-complicated by round-the-houses methods. An all- too-commonly seen mistake was $x^2/(2+x^2) = x^2/2 + 1$.	If argued backwards from given result without error, SCB1
		iv	$-2y^{-3}dy/dx = -4x^{-3}$	B1B1	LHS, RHS	condone $dy/dx - 2y^{-3}$ unless pursued
		iv	$\Rightarrow dy/dx = -4x^{-3}/-2y^{-3} = 2y^3/x^3$*	B1	**NB AG**	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	Not possible to substitute $x = 0$ and $y = 0$ into this expression	B1	soi (e.g. mention of 0/0) **Examiner's Comments** The implicit differentiation was usually correct, as was the algebra to arrive at the printed result. The exact logic behind why x = 0 and y = 0 could not be substituted into the result expression was often faulty (for example many stated the result would be zero or infinite); we condoned this provided they stated the idea that division by zero is undefined or not possible.	Condone 'can't substitute $x = 0$' o.e. (i.e. need not mention $y = 0$). Condone also 'division by 0 is infinite'
		v	$-2y^{-3}\mathrm{d}y/\mathrm{d}x = -4x^{-3}$ $\Rightarrow \mathrm{d}y/\mathrm{d}x = -4x^{-3}/-2y^{-3} = 2y^3/x^3$* Not possible to substitute $x = 0$ and $y = 0$ into this expression		LHS, RHS **NB AG** soi (e.g. mention of 0/0)	condone $\mathrm{d}y/\mathrm{d}x - 2y^{-3}$ unless pursued Condone 'can't substitute $x = 0$' o.e. (i.e. need not mention $y = 0$). Condone also 'division by 0 is infinite'
			Total	18		

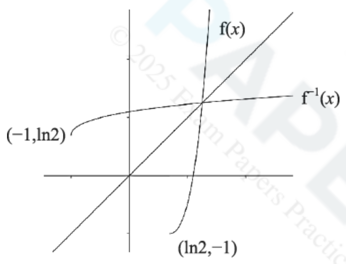
Question			Answer/Indicative content	Marks	Part marks and guidance	
31			$dV/dr = 4\pi r^2$	B1	or $12\pi r^2/3$, condone dr/dV, dV/dR	
			$dV/dt = 10$	B1		Condone use of other letters for t
			$dV/dt = (dV/dr)(dr/dt)$	M1	a correct chain rule soi	o.e. e.g. $dr/dt = (dr/dV)(dV/dt)$
			$\Rightarrow 10 = 4\pi.64.dr/dt$	A1	o.e. (soi) must be correct	
			$\Rightarrow dr/dt = 0.0124 \text{ cm s}^{-1}$	A1	0.012 or better or $10/256\pi$ or $5/128\pi$ **Examiner's Comments** This question proved to be accessible to the overwhelming majority of candidates, and there were many fully correct solutions. Even those who failed to get full marks usually picked up an M1 for a correctly stated chain rule, B1 for $dV/dr = 4\pi r^2$, and a B1 for $dV/dt = 10$. Approximate answers are perhaps preferable in a contextual question, but exact answers were also allowed.	mark final answer
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
32			$y = \ln(1 - \cos 2x)$, let $u = 1 - \cos 2x$			
			$\Rightarrow dy/dx = dy/du.\ du/dx$	M1	$1/(1 - \cos 2x)$soi	
			$= (1/u).\ 2\sin 2x$	M1	$d/dx\,(1 - \cos 2x) = \pm 2\sin 2x$	
			$= \dfrac{2\sin 2x}{1-\cos 2x}$	A1cao		
			When $x = \pi/6$, $\dfrac{dy}{dx} = \dfrac{2\sin(\pi/3)}{1-\cos(\pi/3)}$	M1	substituting π/6 or 30° into their derive	must be in at least two places
			$= 2\sqrt{3}$	A1cao	**Examiner's Comments** Plenty of candidates scored 5 marks here with little difficulty. Some missed out the derivative of 1–cos2x, and some wrote 1/2sin2x instead of 1/(1-cos2x). The substitution of √/6 into the correct derivative was usually done correctly. Some approximation of 2√3 was found, but could usually be condoned by ignoring subsequent working.	isw after correct answer seen
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
33			$\int_0^{\pi/6}(1-\sin 3x)\,\mathrm{d}x = \left[x+\frac{1}{3}\cos 3x\right]_0^{\pi/6}$	M1	$\pm 1/3 \cos 3x$ seen or $\int \frac{1}{3}(1-\sin u)[\mathrm{d}u]$	i.e. condone sign error
				A1	$\left[x+\frac{1}{3}\cos 3x\right]$ or $\left[\frac{1}{3}(u+\cos u)\right]$	condone ' + *c*'
			= π/6 – 1/3	A1cao	o.e., must be exact **Examiner's Comments** This question proved to be a straightforward starter for 3 marks. Sign errors in integrating sin 3x occurred occasionally, as did -3cos3x instead of -1/3 cos 3x.	isw after correct answer seen
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
34		i	At P, $(e^x - 2)^2 - 1 = 0$			
		i	$\Rightarrow e^x - 2 = [\pm]1,$	M1	square rooting – condone no ±	
		i	$e^x = [1 \text{ or}]\ 3$			
		i	*or* $(e^x)^2 - 4e^x + 3 = 0$	M1	expanding to correct quadratic and solve by factorising or using quadratic formula	condone e^*x*^2
		i	$\Rightarrow (e^x - 1)(e^x - 3) = 0$, $e^x = 1$ or 3			
		i	$\Rightarrow x = [0 \text{ or}]\ \ln 3$	A1	*x*-coordinate of P is ln 3; must be exact **Examiner's Comments** Most candidates succeeded in finding x = ln 3, either by square rooting or solving thequadratic in e^x. The second method was somewhat compromised by setting $x = e^x$ (ratherthan a different variable) to get a quadratic in x, though we condoned this for both marks.	condone P = ln 3, but not *y* = ln 3
		ii	$f'(x) = 2(e^x - 2)e^x$	M1	chain rule	e.g. 2 *u* × their deriv of e^x
		ii		A1	correct derivative	$2(e^x - 2)x$ is M0
		ii	$= 0$ when $e^x = 2$, $x = \ln 2$ *	A1	not from wrong working **NB AG**	or verified by substitution
		ii	*or* $f(x) = e^{2x} - 4e^x + 3$	M1	expanding to 3 term quadratic with $(e^x)^2$ or e^{2x}	condone e^*x*^2
		ii	$\Rightarrow f'(x) = 2e^{2x} - 4e^x$	A1	correct derivative, not from wrong working	
		ii	$= 0$ when $2e^{2x} = 4e^x$, $e^x = 2$, $x = \ln 2$ *	A1	or $2e^x(e^x - 2) = 0 \Rightarrow e^x = 2$, $x = \ln 2$	or verified by substitution
		ii			not from wrong working **NB AG**	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	y = f(ln(2)) = – 1	B1	**Examiner's Comments** This provided a simple four marks for most candidates, using a chain rule to find the derivative, setting this to zero and solving to get x = ln 2. A neat alternative method was to recognise that the $(e^x – 2)^2$ term must be non-negative and minimum when $e^x – 2 = 0$, or x = ln 2.	
		iii	$\int_0^{\ln 3}[(e^x-2)^2-1]dx$ $=\int_0^{\ln 3}[(e^x)^2-4e^x+4-1]dx$	M1	expanding brackets must have 3 terms: $(e^x)^2 – 4$ is M0, condone e^x^2	or if $u = e^x$, $\int_1^3[u^2 – 4u + 4 – 1]/u\, du$
		iii	$=\int_0^{\ln 3}[e^{2x}-4e^x+3]dx$	A1	$\int e^{2x} – 4e^x + 3$ [dx] (condone no dx)	= $\int u – 4 + 3/u\, du$
		iii	$=\left[\frac{1}{2}e^{2x}-4e^x+3x\right]_0^{\ln 3}$	B1	$\int e^{2x} = ½ e^{2x}$	= [½ $u^2 – 4u$ + 3ln u
		iii		A1ft	[½ $e^{2x} – 4e^x + 3x$]	
		iii	= (4.5 – 12 + 3ln3) – (0.5 – 4)			
		iii	= 3ln3 – 4 [so area = 4 – 3ln3]	A1	condone 3ln3 – 4 as final ans; mark final ans **Examiner's Comments** This proved to be a rather costly part for candidates unless they recognised the requirement to multiply out $(e^x – 2)^2$ –1 to get $e^{2x} – 4e^x$ + 3 and then integrate term-by-term. Other attempts using substitution or parts usually got nowhere. Although originally we required candidates to give the area as 4 – 3ln3, very few actually did this, so it was decided to condone a (negative) area of 3ln3 – 4.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$y = (e^x - 2)^2 - 1$ $x \leftrightarrow y$			
		iv	$x = (e^y - 2)^2 - 1$			
		iv	$\Rightarrow x + 1 = (e^y - 2)^2$	M1	attempt to solve for *y* (might be indicated by expanding and then taking lns)	or *x* if *x* and *y* not interchanged yet or adding (or subtracting) 1
		iv	$\Rightarrow \pm\sqrt{(x + 1)} = ey - 2$ (+ for $y \geq \ln 2$)	A1	condone no ±	
		iv	$\Rightarrow 2 + \sqrt{(x + 1)} = e^y$			
		iv	$\Rightarrow y = \ln(2 + \sqrt{(x + 1)}) = f^{-1}(x)$	A1	must have interchanged *x* and *y* in final ans	
		iv	Domain is $x \geq -1$	B1	must be ≥ and *x* (not *y*)	if not specified, assume first ans is domain and second range
		iv	Range is $y \geq \ln 2$	B1	or $f^{-1}(x) \geq \ln 2$, must be ≥ (not *x* or f(*x*)) if $x > -1$ and $y > \ln 2$ SCB1	
		iv		M1	recognisable attempt to reflect curve, or any part of curve, in $y = x$	$y = x$ shown indicative but not essential

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv		A1	good shape, cross on $y = x$ (if shown), correct domain and range indicated. [see extra sheet for examples] **Examiner's Comments** Rather more than half of the candidates managed the inverse function well, though a few made errors at the last stage of taking the square root, and concluded with y = ln($\sqrt{(x+1)}$)+ 2, or y = ln($\sqrt{(x+1)}$+ ln 2. Some were perhaps encouraged by the previous part to multiply out $(e^x - 2)^2$ again, though they could still obtain a method mark for a step towards finding y in terms of x. It was not uncommon to see candidates taking logs of individual terms.	e.g. – 1 and ln 2 marked on axes
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
35		i	$f'(x) = \frac{x.2(x-2)-(x-2)^2}{x^2}$	M1	quotient (or product) rule, condone sign errors only	e.g. $\frac{\pm x.2(x-2) \pm (x-2)^2}{x^2}$
		i		A1	correct exp, condone missing brackets here	PR: $(x-2)^2.(-x^{-2}) + (1/x).2(x-2)$
		i	$= \frac{2x^2-4x-x^2+4x-4}{x^2}$			
		i	$= (x^2-4)/x^2 = 1-4/x^2$ *	A1	simplified correctly **NB AG**	with correct use of brackets
		i	*or* $f(x) = (x^2-4x+4)/x$			
		i	$= x-4+4/x$	M1	expanding bracket and dividing each term by x correclty	must be 3 terms: $(x^2-4)/x$ is M0
		i		A1		e.g. $x-4+2/x$ is M1A0
		i	$\Rightarrow f'(x) = 1-4/x2$*	A1	not from wrong working **NB AG**	
		i	$f''(x) 8/x^3$	B1	o.e. e.g. $8x^{-3}$ or $8x/x^4$	
		i	$f'(x) = 0$ when $x^2 = 4$, $x = \pm 2$	M1	$x = \pm 2$ found from $1-4/x^2 = 0$	allow for $x = -2$ unsupported
		i	so at Q, $x = -2$, $y = -8$.	A1	$(-2, -8)$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	f ″ (– 2) [= – 1] < 0 so maximum	B1dep	dep first B1. Can omit – 1, but if shown must be correct. Must state < 0 or negative. **Examiner's Comments** This part was very well-answered, with many getting all 7 marks. The majority of candidates opted to use the quotient rule rather than the slightly easier method of expanding the numerator and dividing through by x. Even so, provided they took care in the use of brackets, they gained the first three marks. The second part was not quite as successful. Some candidates forgot to work out the y-coordinate of Q; others got the second derivative wrong, or failing to state explicitly that for a maximum the second derivative was negative.	must use 2nd derivative test
		ii	f(1) = (– 1)^2/1 = 1			or $(x – 2)^2 = x \Rightarrow x^2 – 5x + 4 = 0$
		ii	f(4) = (2)^2/4 = 1	B1	verifying f(1) = 1 and f(4) = 1	$\Rightarrow (x – 1)(x – 4) = 0$, $x = 1, 4$
		ii	$\int_1^4 \frac{(x-2)^2}{x}\mathrm{d}x = \int_1^4 (x-4+4/x)\mathrm{d}x$	M1	expanding bracket and dividing each term by x 3 terms: $x – 4/x$ is M0	if $u = x – 2$
		ii				$\int \frac{u^2}{u+2}\mathrm{d}u = \int (u-2+\frac{4}{u+2})\mathrm{d}u$
		ii	$=\left[x^2/2-4x+4\ln x\right]_1^4$	A1	$x^2/2 – 4x + 4 \ln x$	$u^2/2 – 2u + 4 \ln (u + 2)$
		ii	= (8 – 16 + 4ln4) – (½ – 4 + 4ln1)			
		ii	= 4ln4 – 4 ½	A1cao		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Area enclosed = rectangle – curve	M1	soi	
		ii	$= 3 \times 1 - (4\ln 4 - 4\tfrac{1}{2}) = 7\tfrac{1}{2} - 4\ln 4$	A1cao	o.e. but must combine numerical terms and evaluate ln 1 – mark final ans	
		ii	*or*			
		ii	$\text{Area} = \int_1^4 [1 - \frac{(x-2)^2}{x}]\,dx$	M1	no need to have limits	
		ii	$= \int_1^4 (5 - x - 4/x)dx$	M1	expanding bracket and dividing each term by x	must be 3 terms in $(x-2)^2$ expansion
		ii	$[5x - x^2/2 - 4\ln x]_1^4$	A1	$5 - x - 4/x$	
		ii	$= 20 - 8 - 4\ln 4 - (5 - \tfrac{1}{2} - 4\ln 1)$	A1	$5x - x^2/2 - 4 \ln x$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	= 7 ½ – 4ln4	A1cao	o.e. but must combine numerical terms and evaluate ln 1 – mark final ans **Examiner's Comments** Most candidates verified that y = 1 when x = 1 or 4, though the majority did this by re-arranging the equation $(x-2)^2 = x$ as a quadratic and solving this. However, for the integration, many candidates failed to spot the need to expand the $(x-2)^2$ term and divide through by x, and most attempts to use substitution (except perhaps for the somewhat fatuous u = x) or parts usually led nowhere. Those who managed this integration successfully often failed to realise they then had to subtract this value from the area of the rectangle, or did this subtraction the wrong way round. Quite a few also calculated the area of the rectangle as 4 × 1 = 4 instead of 3.	
		iii	[g(x) =] f(x + 1) – 1	M1	soi [may not be stated]	
		iii	$= \frac{(x+1-2)^2}{x+1} - 1$	A1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= \frac{x^2-2x+1-x-1}{x+1} = \frac{x^2-3x}{x+1}$ *	A1	correctly simplified – not from wrong working **NB AG** **Examiner's Comments** Fewer than half of the candidates gained full marks for this part of the question. Many did not know how the translation would affect the function algebraically, for example starting with y – 1 = f(x + 1). Of those who derived a correct expression for g(x), many failed to incorporate the '-1' into their fraction.	
		iv	Area is the same as that found in part (ii)	M1	award M1 for ± ans to 8(ii) (unless zero)	
		iv	4ln4 – 7 ½	A1cao	need not justify the change of sign **Examiner's Comments** Very few candidates scored both marks here. A method mark was awarded if their answer indicated recognising the translation of their area from part (ii); but few of those who got part (ii) correct realised that the integral would be the negative of this as the transformed area is below the axis.	
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
36			$y^2 + 2x \ln y = x^2$			
			$1^2 + 2 \times 1 \times \ln 1 = 1^2$ so (1, 1) lies on the curve.	B1	clear evidence of verification needed	at least "1 + 0 = 1"
				M1	$d/dx\,(y^2) = 2y\,dy/dx$	must be correct
				M1	$d/dx\,(2x \ln y) = 2\ln y + 2x/y\ dy/dx$	must be correct
			$2y\frac{dy}{dx} + 2\ln y + 2x.\frac{1}{y}.\frac{dy}{dx} = 2x$	A1cao		condone $dy/dx = \ldots$ unless pursued
			$[\Rightarrow \frac{dy}{dx} = \frac{2x - 2\ln y}{2y + 2x/y}$]			
			when $x = 1, y = 1, \frac{dy}{dx} = \frac{2 - 2\ln 1}{2 + 2}$	M1	substituting both $x = 1$ and $y = 1$ into their dy/dx or their equation in x, y and dy/dx	$2\frac{dy}{dx} + 2\ln 1 + 2\frac{dy}{dx} = 2$
			= ½	A1cao	not from wrong working **Examiner's Comments** Implicit differentiation was well understood, although differentiating the '2xlny' term using the product rule defeated some candidates, and there were some algebraic slips in re-arranging to find dy/dx (which virtually all candidates did before substituting x = 1 and y = 1).	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
37			$h = r$ so $V = \pi h^3/3$	B1	o.e. e.g $\pi h^3 \tan 45°/3$	
			$dV/dt = 5$	B1	soi (can be implied from $V = 5t$)	e.g. from a correct chain rule
			$dV/dh = \pi h^2$	B1ft	must be dV/dh soi, ft their $\pi h^3/3$	but must have substituted for r
			$dV/dt = (dV/dh).dh/dt$	M1	any correct chain rule in V, h and t (soi)	e.g. $dh/dt = dh/dV \times dV/dt$,
			$\Rightarrow 5 = 100\pi\, dh/dt$			
			$\Rightarrow dh/dt = 5/100\pi = 0.016$ cm s^{-1}	A1	0.016 or better; accept $1/(20\pi)$ o.e., but mark final answer	0.01591549 … penalise incorrect rounding penalise incorrect rounding
			or $V = 5t$ so $\pi h^3/3 = 5t$	B1		
			$\Rightarrow \pi h^2 dh/dt = 5$	M1	or $5\, dt/dh = \pi h^2$ o.e.	
			$\Rightarrow dh/dt = 5/\pi h^2 = 5/100\pi = 0.016$ cm s^{-1}	A1	0.016 or better; accept $1/(20\pi)$ o.e., but mark final answer **Examiner's Comments** This question was less well done. Nearly all candidates gained marks for quoting a correct chain rule and using dV/dt = 5. By far the most common error thereafter was to fail to find V as a function of h and instead differentiating $V = \pi r^2 h/3$ to give $dV/dh = \pi r^2/3$. Even when candidates recognised the need to substitute for r, there were a surprising number of trigonometric errors, such as h = r sin 45°. A number of solutions which found dh/dt = 1/20π; then went on to write or evaluate this as π/20.	Penalise incorrect rounding
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
38			let $u = \ln x$, $dv/dx = x^3$, $du/dx = 1/x$, $v = ¼\, x^4$	M1	u, u', v', v all correct	
			$\int_1^2 x^3 \ln x \,dx = \left[\frac{1}{4}x^4 \ln x\right]_1^2 - \int_1^2 \frac{1}{4}x^4 . \frac{1}{x} dx$	A1	$\frac{1}{4}x^4 \ln x - \int \frac{1}{4}x^4 . \frac{1}{x}[dx]$	ignore limits
			$= \left[\frac{1}{4}x^4 \ln x\right]_1^2 - \int_1^2 \frac{1}{4}x^3 dx$	M1dep	simplifying $x^4/x = x^3$ in second term (soi)	dep 1st M1
			$= \left[\frac{1}{4}x^4 \ln x - \frac{1}{16}x^4\right]_1^2$	A1cao	$\frac{1}{4}x^4 \ln x - \frac{1}{16}x^4$ o.e.	
			$= 4\ln 2 – 15/16$	A1cao	o.e. must be exact, but can isw **Examiner's Comments** There was a pleasing response to this question. Integration by parts was well understood by the majority of candidates, many of whom gained full marks. Very occasionally, uand v' were allocated to the wrong parts, and the other most common error was failing to simplify v u' before integrating this.	must evaluate ln 1 = 0 and combine – 1 + 1/16
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
39			let $u = 2x - 1$, $du = 2\,dx$			
			$\int \sqrt[3]{2x-1}\,dx = \int \frac{1}{2}u^{\frac{1}{3}}\,du$	M1	substituting $u = 2x - 1$ in integral	i.e. $u^{1/3}$ or $\sqrt[3]{u}$ seen in integral
				M1	× ½ o.e.	condone no du, or dx instead of du
			$= \frac{3}{8}u^{\frac{4}{3}} + c$	M1	integral of $u^{1/3} = u^{4/3}/(4/3)$ (oe) soi	not $x^{1/3}$
			$= \frac{3}{8}(2x-1)^{\frac{4}{3}} + c$	A1cao	o.e., but must have + c and single fraction mark final answer	so $\frac{3}{4}(2x-1)^{\frac{4}{3}} + c$ is M1M0M1A0
			or			
			$\int \sqrt[3]{2x-1}\,dx = ½ × (2x - 1)^{4/3} ÷ 4/3$	M1	$(2x - 1)^{4/3}$ seen	e.g. correct power of $(2x - 1)$
				M1	÷ 4/3 (oe) soi	e.g. ¾ $(2x - 1)^{4/3}$ seen
				M1	× ½	
			$= \frac{3}{8}(2x-1)^{\frac{4}{3}} + c$	A1cao	o.e., but must have + c and single fraction mark final ans **Examiner's Comments** This question was also answered well, either using substitution or by inspection. However, a surprising number of candidates who substituted left their final answer in terms of u, and a few lost the final mark through omitting the arbitrary constant.	so $\frac{3}{8}(2x-1)^{\frac{4}{3}}$ is M1M1M1A0
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
40			$y = e^{2x}\cos x$	M1	product rule used	consistent with their derivs
			$\Rightarrow dy/dx = 2e^{2x}\cos x - e^{2x}\sin x$	A1	cao – mark final ans	e.g. $2e^{2x} - e^{2x}\tan x$ is A0
			$dy/dx = 0 \Rightarrow e^{2x}(2\cos x - \sin x) = 0$	M1	their derivative = 0	
			$\Rightarrow 2\cos x = \sin x$			
			$\Rightarrow 2 = \sin x/\cos x = \tan x$	M1	$\sin x/\cos x = \tan x$ used	or $\sin^2 x + \cos^2 x = 1$ used
			$\Rightarrow x = 1.11$	A1	1.1 or 0.35 π or better, or arctan 2, not 63.4° but condone ans given in both degrees and radians here	1.1071487 …, 0.352416 … π, penalise incorrect rounding
			$\Rightarrow y = 4.09$	A1cao	art 4.1 **Examiner's Comments** The product rule was done well, and most candidates were successful in arriving at tanx = 2 at the turning point. The most common error was to give x in degrees and then touse this to calculate y, giving a rather alarmingly large result!	no choice
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
41		i	$1 + k$	B1		**Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. This was an easy write-down for virtually all candidates, except those few who did not know that $e^0 = 1$.
		ii	$f'(x) = 2e^{2x} - 2ke^{-2x}$	B1		
		ii	$f'(x) = 0 \Rightarrow 2e^{2x} - 2ke^{-2x} = 0$	M1	their derivative = 0	
		ii	$\Rightarrow e^{2x} = ke^{-2x}$			
		ii	$\Rightarrow e^{4x} = k$, $4x = \ln k$, $x = ¼ \ln k$*	A1	NB AG	
		ii	$y = e^{(½ \ln k)} + ke^{(-½ \ln k)}$	M1	substituting $x = ¼ \ln k$ into $f(x)$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= \sqrt{k} + k/\sqrt{k} = 2\sqrt{k}$	A1cao	or $2k^{1/2}$	**Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The first two marks were pretty universally earned, but deriving $x = ¼ \ln k$, together with the final 'A' mark for getting $2\sqrt{k}$, caused a few problems, with some inaccurate logarithm work. For example, $e^{1/2 \ln k} = 1/2\, k$ was a commonly seen misconception.
		iii	$\text{Area} = \int_0^{\frac{1}{2}\ln k} (e^{2x} + ke^{-2x})[\mathrm{d}x]$	B1	correct integral and limits (soi)	
		iii	$= \int_0^{\frac{1}{2}\ln k} (e^{2x} + ke^{-2x})\,\mathrm{d}x = \left[\frac{1}{2}e^{2x} - \frac{1}{2}ke^{-2x}\right]_0^{\frac{1}{2}\ln k}$	B1	$\left[\frac{1}{2}e^{2x} - \frac{1}{2}ke^{-2x}\right]$	
		iii	$= ½\, k - ½ - ½ + ½\, k$	M1	$e^{\ln k} = k$ or $e^{-\ln k} = 1/k$ (soi)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= k - 1$	A1		**Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The integration was usually correct, but, thereafter, as in part (ii), the simplification to arrive at $k - 1$ proved to be tricky, with similar errors being made.
		iv	(A) $g(x) = e^{2(x + ¼ \ln k)} + ke^{-2(x + ¼ \ln k)}$	M1	Substitute $x + ¼ \ln k$ for x in $f(x)$	condone missing brackets
		iv	$= e^{2x}.e^{½ \ln k} + k\,e^{-2x}.\,e^{-½ \ln k}$	M1	$e^{p+q} = e^p \times e^q$ used	
		iv	$= (e^{\ln k})^{1/2}\,e^{2x} + k.(e^{\ln k})^{-½}\,e^{-2x}$			
		iv	$= k^{1/2}\,e^{2x} + k.k^{-1/2}.\,e^{-2x}$			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$= \sqrt{K}\,(e^{2x} + e^{-2x})$*	A1	**NB AG** – must show enough working	e.g. $k^{1/2}\,e^{2x} + k.k^{-1/2}.\,e^{-2x}$ **Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. Most attempts correctly substituted x + ¼ ln k for x in f(x) to gain the first M mark, but we needed to see clear evidence of how this simplifies to the given result. Often candidates seemed to be working backwards from this without really understanding the process.
		iv	(B) $g(-x) = \sqrt{k}\,(e^{-2x} + e^{2x})$	M1	substituting $-x$ for x	condone ‘f’ used instead of ‘g’ for M1

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	= g(x) so g is even	A1	must include g(–x) = g(x), and either define an even function or conclude that g is even	not f(–x) = f(x) for A1 **Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The definition of an even function was well known, but sometimes the structuring of the proof was indecisively presented. Some used ‘f’ instead of ‘g’ (here, f is indeed not an even function!), and we required to see either a clear statement of the definition of an even function, or a clear conclusion that g is therefore even. The structure ‘g(–x) = … = … = g(…) ⇒ g is even’ is the most transparent formulation to use in such proofs, rather than starting them by stating that g(–x) = g(x), viz the result they are trying to prove!
		iv	(*C*) g(x) is symmetrical about the y-axis, and	B1		
		iv	f(x) is g(x) translated ¼ ln k in x-direction	B1	allow ‘shift’ or ‘move’	or g is f translated – ¼ ln k

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	so f(x) is symmetrical about $x = ¼ \ln k$	B1	allow final B1 even if unsupported	or incorrectly supported **Examiner's Comments** The calculus here was not particularly demanding, requiring only the derivative and integral of e^{kx}; but the simplification of expressions using the laws of logarithms and exponentials proved to be quite testing and found out quite a few candidates. The argument here proved beyond most candidates, with only 20% getting full marks. Many stated that f was an even function, perhaps thinking that any line of symmetry sufficed. Sometimes it was indeed a little difficult to decide whether candidates were referring to f or g in their answers.
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
42		i	$\frac{dy}{dx} = \frac{(x+4)^{1/2}.1 - x.\frac{1}{2}(x+4)^{-1/2}}{[(x+4)^{1/2}]^2}$	M1	quotient rule: v × their u' – u × their v', and correct denominator	or product rule
		i		B1	½ $u^{-1/2}$ soi	or –½ $u^{-3/2}$ (PR)
		i		A1	correct expression	PR: $x(-½)(x + 4)^{-3/2} + (x + 4)^{-1/2}$
		i	$= \frac{x+4-\frac{1}{2}x}{(x+4)^{3/2}} = \frac{\frac{1}{2}x+4}{(x+4)^{3/2}} = \frac{x+8}{2(x+4)^{3/2}}$ *	M1	factoring out $(x + 4)^{-1/2}$ o.e.	$= (x + 4)^{-3/2}(-½ x + x + 4)$
		i		A1	NB AG	**Examiner's Comments** Most candidates scored well on this question, which covered calculus topics such as the product or quotient rule for differentiation and integration by substitution, which are generally well understood by learners. The first three marks here were usually earned, though a minority of weaker candidates mixed up the product and quotient rules, for example using $v = (x + 4)^{-1/2}$ in their quotient rule. The factorisation required to achieve the given result was less successfully done, but just over half the candidates still managed full marks here. There were a lot of repeated attempts at this, for example using the product rule when they got stuck with manipulating their quotient rule expression.
		ii	[asymptote is] $x = -4$	B1	soi	but from correct working
		ii	gradient of tangent at O= $8/(2 \times 4^{3/2})$ = ½	B1	gradient = ½	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	eqn of tangent is $y = ½x$	B1	o.e. e.g. using gradient	
		ii	When $x = –4$, $y = –2$, so $(–4, –2)$	B1		**Examiner's Comments** Most candidates scored well on this question, which covered calculus topics such as the product or quotient rule for differentiation and integration by substitution, which are generally well understood by learners. This proved to be a straightforward 4 marks earned by over 70% of scripts. The asymptote and the gradient and equation of the tangent at the origin were usually correctly found, followed by the coordinates of Q.
		iii	let $u = x + 4$, $du = dx$	B1	or $dx/du = 1$	or $v^2 = x + 4$, $2v dv/dx = 1$ or $2v dv = dx$ oe e.g. $dv/dx = ½(x + 4)^{–1/2}$
		iii	$\int_0^5 \frac{x}{(x+4)^{1/2}} dx = \int_4^9 \frac{u-4}{u^{1/2}} du$	B1	$\int \frac{u-4}{u^{1/2}} [du]$	
		iii	$= \int_4^9 (u^{1/2} - 4u^{-1/2}) du$	B1	$u^{1/2} – 4u^{–1/2}$ or $u^{1/2} – 4/u^{1/2}$, or $\sqrt{u} – 4/\sqrt{u}$	$\int (2v^2 – 8)[dv]$
		iii	$= \left[\frac{2}{3}u^{3/2} - 8u^{1/2}\right]_4^9$	B1	$\left[\frac{2}{3}u^{3/2} - 8u^{1/2}\right]$ o.e.	
		iii	$= (18 – 24) – (16/3 – 16)$	M1	substituting correct limits (upper – lower)	0, 5 for x; 4,9 for u; 2,3 for v
		iii	$= 14/3$	A1cao		
		iii	or (following first 2 marks)			by parts with no substitution:
		iii	let $v = u – 4$, $w' = u^{–1/2}$, $v' = 1$, $w = 2u^{1/2}$	M1		$u = x$, $u' = 1$, $v' = (x + 4)^{–1/2}$, $v = 2(x + 4)^{1/2}$ M1 $= [2x(x + 4)^{1/2}] – \int 2(x + 4)^{1/2}$ A1

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\int_4^9 (u-4)u^{-1/2}\,du = \left[2u^{1/2}(u-4)\right]_4^9 - \int_4^9 2u^{1/2}\,du$	A1		
		iii	$= \left[2u^{1/2}(u-4) - \frac{4}{3}u^{3/2}\right]_4^9$	A1		= 14/3 A1 (so max of 4/6)
		iii	= 14/3	A1cao		
		iii	*y*- coordinate of Q is 2½	B1	(soi)	or $\int_0^5 \frac{1}{2}x\,dx$ M1
		iii	Area of triangle = ½ × 5 × 5/2 = 25/4	B1		$= \left[\frac{1}{4}x^2\right]_0^5 = 25/4$ A1

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Enclosed area = 25/4 – 14/3 = $1\frac{7}{12}$	B1	or 19/12, or $1.58\dot{3}$	isw from correct exact answer **Examiner's Comments** Most candidates scored well on this question, which covered calculus topics such as the product or quotient rule for differentiation and integration by substitution, which are generally well understood by learners. This 9-mark question required careful extended work from candidates, but there was a pleasing response, with just under half the scripts earning full marks. The first six of these were for finding the area under the function using substitution. Here, as usual, notation sometimes left something to be desired, with missing d*u*'s or d*x*'s, integral signs, inconsistent limits, etc. Most of this we condoned, but we did require d*u*/d*x* = 1 or its equivalent to be stated. The final three marks depended upon the correct coordinates for the point Q being found in part (ii). Occasionally the triangle area was found using $\int \frac{1}{2} x \, dx$.
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
43		i	$2\cos 2y\, dy/dx = 1$	M1	$k \cos 2y\, dy/dx = 1$	or $dx/dy = k \cos 2y$, $k \cos 2y dy = dx$
		i	$\Rightarrow dy/dx = 1/(2\cos 2y)$	A1		$dy/dx = k \cos 2y$ is M0
		i	when $x = 1½$, $y = \pi/12$, $dy/dx = 1/(2\cos(\pi/6))$	M1*	substituting $y = \pi/12$ *dep 1st M1	
		i	$= 1/\sqrt{3}$	A1	or $\sqrt{3}/3$	isw from correct exact answer **Examiner's Comments** This question as also very well done, with half the candidates scoring full marks. The implicit differentiation was well understood, though there were the usual blemishes from mixing up the derivative and integral formulae for sin $2y$. A few candidates re-arranged the equation to get x in terms of y, then found dx/dy, and then the reciprocal dy/dx.
		ii	$2y = \arcsin(x - 1)$	M1		
		ii	$\Rightarrow y = ½ \arcsin(x - 1)$	A1	or $½ \sin^{-1}(x - 1)$	
		ii	translation of 1 unit in positive x-direction	B1	or translation $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$	allow 'shift', but not 'move', vector only is B0

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	[one-way] stretch s.f. ½ in *y*-direction	B1	not ‘shrink’, ‘squash’ etc	transformations can be in either order **Examiner's Comments** This question as also very well done, with half the candidates scoring full marks. Re-arranging the given implicit equation to give y = ½ arcsin(x – 1) was well understood, and the transformations were usually accurately described. Note that the preferred terms here are ‘translation’ and ‘one-way stretch’.
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
44		i	$dV/dh = 4.\frac{1}{2}\,(h^3 + 1)^{-1/2}\,.3h^2$	M1	chain rule	their deriv of $4u^{1/2}$ × their deriv of $h^3 + 1$
		i		A1	correct	
		i		M1	substituting $h = 2$ into their derivative	
		i	when $h = 2$, $dV/dh = 8$	A1cao		**Examiner's Comments** This question was extremely well answered, with the majority of candidates scoring full marks. The chain rule on V was successfully negotiated by over half the candidates, and then correctly evaluated at $x = 2$.
		ii	$dV/dt = 0.4$	B1	soi	condone r for t
		ii	$dV/dt = dV/dh \times dh/dt$	M1	o.e.	any correct chain rule in V, h, t (or r)
		ii	$0.4 = 8 \times dh/dt \Rightarrow dh/dt = 0.05$ (m per min)	A1cao	0.05 or 1/20	**Examiner's Comments** This question was extremely well answered, with the majority of candidates scoring full marks. Virtually everyone who scored 4 for part (i) went on to apply the chain rule $dV/dt = dV/dh \times dh/dt$, or some variation of it, to get full marks here. The rest usually earned the first two of the three marks.
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
45			let $u = \ln x$, $u' = 1/x$, $v' = x^{-1/2}$, $v = k\,x^{1/2}$	M1	soi ($k \neq 0$)	
			$\int x^{-1/2} \ln x [\mathrm{d}x] = \left[2x^{1/2} \ln x\right] - \int 2x^{1/2} . \frac{1}{x} [\mathrm{d}x]$	A1		
			$= \left[2x^{1/2} \ln x\right] - \int 2x^{-1/2} [\mathrm{d}x]$	M1	$x^{1/2} / x = x^{-1/2}$ or $1/x^{1/2}$ seen	
			$= \left[2x^{1/2} \ln x - 4x^{1/2}\right]_1^4$	A1	$2x^{1/2} \ln x - 4x^{1/2}$	may be integrated separately
			$= 4 \ln 4 - 8 - (2\ln 1 - 4)$			
			$= 4 \ln 4 - 4$	A1cao	oe (eg $\ln 256 - 4$) but must evaluate $\ln 1 = 0$	mark final answer **Examiner's Comments** Integration by parts was well understood, with just under half candidates scoring full marks for this question. Very occasionally, candidates took $u = x^{-1/2}$ and $v' = \ln x$, and were unable to score any marks. With u and v correct, the next hurdle is to simplify the $2x^{1/2}.1/x$ integrand, and some failed at this stage, and attempted to integrate the product term by term. Having negotiated this successfully, most got full marks, though very occasionally the final answer was spoiled by using $4\ln 4 = \ln 16$.
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
46			$\int_0^{\frac{\pi}{2}}(1+\cos\frac{1}{2}x)\,dx=\left[x+2\sin\frac{1}{2}x\right]_0^{\frac{\pi}{2}}$	B1	$\left[x+2\sin\frac{1}{2}x\right]$	
			$=\frac{\pi}{2}+2\sin\frac{\pi}{4}[-0]$	M1	substituting limits (upper – lower)	allow 1 slip
			$=\frac{\pi}{2}+\sqrt{2}$	A1cao	must be exact, not $2/\sqrt{2}$	isw from correct answer seen **Examiner's Comments** This proved to be a straightforward starter question, with 80% of candidates scoring full marks. Some candidates stopped at π/2 + 2 sinπ/4, presumably because they did not appreciate that 'value of' means numerical. A few weaker candidates confused differentiation and integration, either giving the wrong coefficient or sign for the sin *x*/2 term.
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
47		i	$h = 20$, stops growing	B1	AG need interpretation **Examiner's Comments** Most candidates correctly wrote down the value of h but quite a number failed to give the interpretation that the tree stopped growing when its height was 20m.	
		ii	$h = 20 - 20e^{-t/10}$ $dh/dt = 2e^{-t/10}$	M1A1	differentiation (for M1 need $ke^{-t/10}$, k const)	
		ii	$20e^{-t/10} = 20 - 20(1 - e^{-t/10})$ $= 20 - h$ $= 10dh/dt$	M1		
		ii		A1	oe eg $20 - h = 20 - 20(1 - e^{-t/10}) = 20e^{-t/10}$ $= 10dh/dt$ (showing sides equivalent)	
		ii	when $t = 0$, $h = 20(1 - 1) = 0$	B1	initial conditions	
		ii	---------------------------- OR verifying by integration		----------------------------	
		ii	$\int \frac{dh}{20-h} = \int \frac{dt}{10}$	M1	sep correctly and intending to integrate	
		ii	$\Rightarrow -\ln(20 - h) = 0.1t + c$	A1	correct result (condone omission of c, although no further marks are possible) condone ln (h – 20) as part of the solution at this stage	
		ii	$h = 0, t = 0, \Rightarrow c = -\ln 20$ $\Rightarrow \ln(20 - h) = -0.1t + \ln 20$ $\Rightarrow \ln(\frac{20-h}{20}) = -0.1t$	B1	constant found from expression of correct form (at any stage) but B0 if say $c = \ln(-20)$ (found using ln $(h - 20)$)	
		ii	$\Rightarrow 20 - h = 20e^{-0.1t}$	M1	combining logs and anti-logging (correct rules)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow h = 20(1 - e^{-0.1t})$	A1	correct form (do not award if B0 above) **Examiner's Comments** Those who approached the verification by integration were quite successful. The common errors were:- • omitting the negative sign when integrating 1/(20□*h*) • omitting the constant of integration • giving ln(*h*□20) in their answers (without modulus signs) despite having usually given *h* =20 as a maximum value in (i) • incorrect anti-logging. Those who approached from differentiation usually obtained some marks, particularly the mark for checking the initial conditions but many gave insufficient detail when verifying the given result.	
		iii	$\frac{200}{(20+h)(20-h)} = \frac{A}{20+h} + \frac{B}{20-h}$ $\Rightarrow 200 = A(20 - h) + B(20 + h)$	M1	cover up, substitution or equating coeffs	
		iii	$h = 20 \Rightarrow 200 = 40B$, $B = 5$	A1		
		iii	$h = -20 \Rightarrow 200 = 40A$, $A = 5$ $200\, dh/dt = 400 - h^2$	A1		
		iii	$\Rightarrow \int \frac{200}{400-h^2} dh = \int dt$	M1	separating variables and intending to integrate (condone sign error)	
		iii	$\Rightarrow \int (\frac{5}{20+h} + \frac{5}{20-h}) dh = \int dt$		substituting partial fractions	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Rightarrow 5\ln(20+h) - 5\ln(20-h) = t + c$	A1	ft their *A*, *B*, condone absence of *c*, Do not allow ln (h-20) for A1.	
		iii	When $t = 0$, $h = 0 \Rightarrow 0 = 0 + c \Rightarrow c = 0$	B1	cao need to **show** this. *c* can be found at any stage. **NB** $c = \ln(-1)$ (from ln $(h - 20)$) or similar **scores B0.**	
		iii	$\Rightarrow \quad 5\ln\frac{20+h}{20-h} = t$			
		iii	$\Rightarrow \quad \frac{20+h}{20-h} = e^{t/5}$	M1	anti-logging an equation of the correct form . Allow if *c* = 0 clearly stated (provided that *c* = 0) even if B mark is not awarded, but do not allow if *c* omitted. Can ft their *c*.	
		iii	$\Rightarrow 20 + h = (20-h)e^{t/5} = 20e^{t/5} - he^{t/5}$ $\Rightarrow h + he^{t/5} = 20e^{t/5} - 20$ $\Rightarrow h(e^{t/5} + 1) = 20(e^{t/5} - 1)$	DM1	making *h* the subject, dependent on previous mark	
		iii	$\Rightarrow \quad h = \frac{20(e^{t/5}-1)}{e^{t/5}+1}$		**NB** method marks can be in either order, in which case the dependence is the other way around.(In which case, 20 + *h* is divided by 20 – *h* first to isolate *h*).	
		iii	$\Rightarrow \quad h = \frac{20(1-e^{-t/5})}{1+e^{-t/5}}$ *	A1	**AG must have obtained B1** (for *c*) in order to obtain final A1. **Examiner's Comments** There were a few completely correct solutions to this part. However, many different errors were seen from the majority of candidates. There was also a lot of confused work. Those who started with the correct partial fractions, from 200/(20☐h)(20+*h*) or 1/(20☐*h*)(20+*h*), usually	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					obtained the first three marks and then integrated having scored M1A1A1M1 thus far. Common errors then included omitting the negative sign when integrating 5/(20☐*h*) (ie giving 5ln(20☐*h*) and hence A0) or failing to state and then evaluate a constant. Those who had no constant were unable to score further marks. Those who did score the first 5 or 6 marks (dependent upon when the constant was evaluated) often used the laws of logarithms correctly and anti-logged although some fiddled the signs when subsequently making h the subject. Some candidates thought that 1/(400☐h^2)=1/(h☐20)(*h*+20). Marks were scored for using partial fractions on 1/(*h*☐20)(*h*+20) but logarithms such as ln(h☐20) for h<20 and constants such as ln(☐1) could not obtain accuracy marks although the marks for anti-logging and making h the subject were still available. There were also a number who felt that 1/(400☐h^2)=1/(200☐*h*)(200+*h*). The use of modulus signs was rarely seen.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	As $t \to \infty$, $h \to 20$. So long-term height is 20m.	B1	www **Examiner's Comments** Usually correct.	
		v	1st model $h = 20(1 - e^{-0.1}) = 1.90..$	B1	Or 1st model $h = 2$ gives $t = 1.05..$	
		v	2nd model $h = 20(e^{1/5} - 1)/(e^{1/5} + 1) = 1.99..$	B1	2nd model $h = 2$ gives $t = 1.003..$	
		v	so 2nd model fits data better	B1 dep	dep previous B1s correct **Examiner's Comments** Most candidates scored all three marks.	
			Total	19		

Question			Answer/Indicative content	Marks	Part marks and guidance	
48		i	$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{2\cos 2\theta}{\cos\theta}$			
		i		M1 A1	their dy/dθ / their dx/dθ www correct (can isw)	
		i	When $\theta = \pi/6$ $= \frac{dy}{dx} = \frac{2\cos(\pi/3)}{\cos(\pi/6)}$	DM1	subst θ = π/6 in theirs	
		i	$= \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$	A1	oe exact only, www (but not 1/√3/2)	
		i	---------------------------- OR		----------------------------	
		i	$y = 2x\sqrt{(1-x^2)}$	M1	full method for differentiation including product rule and function of a function oe	
		i	$\frac{dy}{dx} = -2x^2(1-x^2)^{-1/2} + 2(1-x^2)^{1/2}$	A1	oe cao (condone lack of consideration of sign)	
		i	*at*θ = π/6, sin π/6 = 1/2	DM1	subst sin π/6 = ½ in theirs	
		i	$\frac{dy}{dx} = \frac{-2}{4}(1-\frac{1}{4})^{-1/2} + 2(\frac{3}{4})^{1/2} = \frac{2}{\sqrt{3}}$	A1	oe, exact only, www (but not 1/√3/2) **Examiner's Comments** This question was successfully answered by most candidates. Some failed to give their answers in exact form.	
		ii	y = sin 2θ = 2 sinθ cosθ	M1	using sin 2θ = 2 sinθcosθ	
		ii	⇒ y^2 = 4 $\sin^2$θ $\cos^2$θ = 4x^2(1 – x^2)	M1	using $\cos^2$θ = 1 –$\sin^2$θ to eliminate cosθ	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= 4x^2 - 4x^4$*	A1	**AG** need to see sufficient working or A0. **Examiner's Comments** Most candidates used y = sin 2θ = 2sinθ cosθ and many squared this but not all candidates subsequently used $\cos^2\theta = 1 - \sin^2\theta$ to continue to the required result.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
49		i	$v\mathrm{d}v/\mathrm{d}x + 4x = 0$ $\int v\mathrm{d}v = -\int 4x\,\mathrm{d}x$	M1	separating variables and intending to integrate	
		i	$\frac{1}{2}v^2 = -2x^2 + c$	A1	oe condone absence of *c*. [Not immediate $v^2 = -4x^2$ (+c)]	
		i	When $x = 1$, $v = 4$, so $c = 10$	B1	finding *c*, must be convinced as AG, need to see at least the statement given here oe (condone change of c)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	so $v^2 = 20 - 4x^2$ *	A1	**AG** following finding c convincingly Alternatively, **SC** $v^2 = 20 - 4x^2$, by differentiation, $2v\,\mathrm{d}v/\mathrm{d}x = -8x$ $v\mathrm{d}v/\mathrm{d}x + 4x = 0$ scores B2 if, in addition, they check the initial conditions a further B1 is scored (ie 16 = 20 – 4). Total possible 3/4. <u>Examiner's Comments</u> Most candidates scored all four marks when solving the differential equation. It was pleasing to see so few candidates failing to include the constant of integration. Some candidates, however, tried to work backwards from the answer, or wrote $v^2 = -4x^2 + c$ without showing from where it came. The answer was given in this case so stages of working were needed. Whilst, on this occasion, examiners condoned the change of constant candidates should be encouraged to change their constant when appropriate in the future and not use c twice to mean different things within the same question.	
		ii	$x = \cos 2t + 2\sin 2t$ when $t = 0$, $x = \cos 0 + 2\sin 0 = 1$*	B1	**AG** need some justification	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$v = dx/dt = -2\sin 2t + 4\cos 2t$	M1	differentiating, accept ±2, ±4 as coefficients but not ±1, ±2 and not ±1/2, ±1 from integrating	
		ii		A1	cao	
		ii	$v = 4\cos 0 - 2\sin 0 = 4$*	A1	www AG **Examiner's Comments** Most candidates obtained the mark for verifying that $x = 1$. Many others also scored the following three marks but some had the incorrect coefficients when differentiating and only had the correct coefficient in the second term when working backwards from the answer, 4.	
		iii	$\cos 2t + 2\sin 2t = R\cos(2t - \alpha) = R(\cos 2t\cos\alpha + \sin 2t\sin\alpha)$ $R = \sqrt{5}$	B1	or 2.24 or better (not ± unless negative rejected)	
		iii	$R\cos\alpha = 1$, $R\sin\alpha = 2$	M1	correct pairs soi	
		iii	$\tan\alpha = 2$,	M1	correct method	
		iii	$\alpha = 1.107$	A1	cao radians only, 1.11 or better (or multiples of π that round to 1.11)	
		iii	$x = \sqrt{5}\cos(2t - 1.107)$ $v = -2\sqrt{5}\sin(2t - 1.107)$	A1	differentiating or otherwise, ft their numerical R, α (not degrees) required form **SC** B1 for $v = \sqrt{20}\cos(2t + 0.464)$ oe	
		iii	EITHER $v^2 = 20\sin^2(2t - \alpha)$ $20 - 4x^2 = 20 - 20\cos^2(2t - \alpha)$	M1	squaring their v (if of required form with same α as x), and x, and attempting to show $v^2 = 20 - 4x^2$ ft their R, α (incl. degrees) [α may not be specified].	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= 20(1 - \cos^2(2t - \alpha)) = 20\sin^2(2t - \alpha)$ so $v^2 = 20 - 4x^2$	A1	cao www (condone the use of over-rounded α (radians) or degrees)	
		iii	OR multiplying out $v^2 = (-2\sin 2t + 4\cos 2t)^2$ $= 4\sin^2 2t - 16\sin 2t\cos 2t + 16\cos^2 2t$ and $4x^2 = 4(\cos^2 2t + 4\sin 2t\cos 2t + 4\sin^2 2t)$ $= 4\cos^2 2t + 16\sin 2t\cos 2t + 16\sin^2 2t$ (need middle term) and attempting to show that $v^2 + 4x^2 = 4(\sin^2 2t + \cos^2 2t) + 16(\cos^2 2t + \sin^2 2t)$ $= 4 + 16 = 20$ (or $20 - 4x^2 = v^2$) oe	M1	differentiating to find v (condone coefficient errors), squaring v and x and multiplying out (need attempt at middle terms) and attempting to show $v^2 = 20 - 4x^2$	
		iii	so $v^2 = 20 - 4x^2$	A1	cao www	
		iii			$R\cos\alpha = 1$, $R\sin\alpha = 2$, $R = \sqrt{5}$, $\alpha = 1.107$ 1) Missing R Ie $\cos\alpha = 1$, $\sin\alpha = 2$. We reluctantly condone this provided that it is followed by working that suggests R was implied such as $\tan\alpha = 2$, $\alpha = 1.107$ M1M1A1. Other methods are possible. We do not award M1 for $\cos\alpha = 1$, $\sin\alpha = 2$ if it is followed by $\alpha = \text{inv} \cos 1$ as R is not implied. B1 is still available. 2) Incorrect pairs eg $R\sin\alpha = 1$, $R\cos\alpha = 2$ scores M0 but would obtain the second M1ft if it was followed by $\tan\alpha = 1/2$. M0M1A0. B1 is possible. 3) Incorrect method $R\sin\alpha = 2$, $R\cos\alpha = 1$ followed by $\tan\alpha = 1/2$ scores M1M0A0. B1 is possible.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					4) Incorrect pairs and incorrect method Rsinα = 1, Rcosα = 2, tanα = 2 is M0M0A0. B1 is possible. This is easily over-looked and is a double error leading to an apparently correct answer. 5) Incorrect signs (all could score B1) (a) Rcosα = 1, Rsinα = –2, tanα = –2, M1, M1ft, A0 (b) Rcosα = 1, Rsinα = –2, tanα = 2, M1 M0ft, A0 sign error (c) Rcosα = 1, Rsinα = –2, R = √5, sinα = –2/√5, M1 M1 A0 sign error (d) Rcosα = 1, Rsinα = –2, sinα = 2/√5, M1M0 sign error A0 (e) Rcosα = 1, Rsinα = –2, cosα = 1/√5, α = 1.107 M1, M1, A0 sign error (even though not used) 6) Incorrect R a) Rcosα = 1, Rsinα = 2, R = 5 (say), cosα = 1/5, scores B0 M1M1ftA0 b) Rcosα = 1, Rsinα = 2, R = 5 (say), tanα = 2, α = 1.107 scores M1M1B0A1 (allow) 7) Missing Working a) tanα = 2, α = 1.107, R = √5, scores M1M1A1B1 soi b) tanα = 1/2, R = √5 scores M1M0B1A0 (either correct pairs or correct method but not both) c) Rcosα = 1, R = √5, α = 1.107 M1M1B1A1 soi Other options are possible. Examiners should consult their Team Leaders if in doubt.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					__Examiner's Comments__ This part was rarely answered completely successfully. Most candidates understood that the '*R*' method was needed and scored the first three marks. This was the modal mark. Calculus was needed in this question. The candidates were asked to find the constant *α*, and *t* is a time to be combined in (2*t*-*α*) so answers given in radians were required. The use of degrees here was a very common mistake. Many candidates then differentiated their angles in degrees and obtained no marks for *v*. The last two marks were obtained only rarely but they were a good differentiator for the able candidates. Use of unspecified α, or in degrees or radians was allowed in this last part. Some candidates had difficulty as they had found both *x* and *v* separately using the '*R*' method and so had different values for their angles. Others realised the problem and were able to use trigonometric identities to change their *v* (or *x*) to have the same angle.	
		iv	$x = \sqrt{5}\cos(2t - \alpha)$ or otherwise x max = $\sqrt{5}$	B1	ft their *R*	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	when $\cos(2t - \alpha) = 1$, $2t - 1.107 = 0$,	M1	oe (say by differentiation) ft their α in radians or degrees for method only	
		iv	$2t = 1.107$ $t = 0.55$	A1	cao (or answers that round to 0.554) <u>Examiner's Comments</u> The majority of candidates scored the first two marks.. It was disappointing that candidates did not realise their mistake in part (ii) when they obtained an answer of time, t= 31.7 degrees.	
			Total	18		

50		i	$\frac{x}{(1+x)(1-2x)} = \frac{A}{1+x} + \frac{B}{1-2x}$			
		i	$\Rightarrow x = A(1 - 2x) + B(1 + x)$	M1	expressing in partial fractions of correct form (at any stage) and attempting to use cover up, substitution or equating coefficients Condone a single sign error for M1 only.	
		i	$x = ½ \Rightarrow ½ = B(1 + ½\ ) \Rightarrow B = 1/3$	A1	www cao	
		i	$x = -1 \Rightarrow -1 = 3A \Rightarrow A = -1/3$	A1	www cao (accept A/(1 + x) + B/(1 – 2x), A = –1/3, B = 1/3 as sufficient for full marks without needing to reassemble fractions with numerical numerators)	
		i			**Examiner's Comments** Whilst almost all candidates knew the general method for expressing the given fraction in partial fractions, there were a surprising number of numerical errors.	
		ii	$\frac{x}{(1+x)(1-2x)} = \frac{-1/3}{1+x} + \frac{1/3}{1-2x}$ $= \frac{1}{3}\left[(1-2x)^{-1} - (1+x)^{-1}\right]$ $= \frac{1}{3}[1 + (-1)(-2x) + \frac{(-1)(-2)}{2}(-2x)^2 +$			

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$= \frac{1}{3}[1 + 2x + 4x^2 + \ldots - (1 - x + x^2 + \ldots)]$	A1 A1	$1 + 2x + 4x2$ $1 - x + x^2$ (or 1/3/ –1/3 of each expression, ft their *A*/ *B*) If $k(1 - x + x^2)$ (A1) not clearly stated separately, condone absence of inner brackets (i.e. $1 + 2x + 4x^2 - 1 - x + x^2$) **only if** subsequently it is clear that brackets were assumed, otherwise A1A0. [i.e. $-1 - x + x^2$ is A0 unless it is followed by the correct answer] Ignore any subsequent incorrect terms	
		ii	$= \frac{1}{3}(3x + 3x^2 + \ldots) = x + x^2 + \ldots$ so $a = 1$ and $b = 1$	A1	or from expansion of $x(1 - 2x)^{-1}(1 + x)^{-1}$ www cao	
		ii	OR $x(1 - x - 2x^2) = x(1 - (x + 2x^2))$ $= x(1 + x + 2x^2 + (-1)(-2)(x + 2x^2)^2/2 + \ldots\ldots\ldots)$	M1	correct binomial coefficients throughout for $(1 - (x + 2x^2))$ oe (i.e. 1, –1), at least as far as necessary terms $(1 + x)$ (NB third term of expansion unnecessary and can be ignored)	
		ii	$= x(1 + x + 2x^2 + x^2 \ldots\ldots\ldots)$	A2	$x(1 + x)$ www	
		ii	$= x + x^2 \ldots$ so $a = 1$ and $b = 1$	A1	www cao	
		ii	Valid for $-½ < x < ½$ or $\|x\| < ½$	B1	independent of expansion. Must combine as one overall range. condone $\leq$ s (although incorrect) or a combination. Condone also, say $-½ < \|x\| < ½$ but not $x < ½$ or $-1 < 2x < 1$ or $-½ > x > ½$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii			<u>Examiner's Comments</u> Most candidates were able to use the binomial expansion correctly although there were sign errors - often from using (-2x) as (2x). The most common error- which was very common- was using $\frac{1}{3(1+x)} = 3(1+x)^{-1}{}_{=3(1-x+x^2...)=3-3x+3x^2}$ and similarly for $\frac{1}{3(1-2x)}$. The other frequent error was in the validity. Some candidates omitted this completely but many others failed to combine the validities from the two expansions, or failed to choose the more restrictive option.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
51		i	Either $h = (1 - \frac{1}{2}At)^2 \Rightarrow$ $dh/dt = -A(1 - \frac{1}{2}At)$	M1	Including function of a function, need to see middle step	
		i	$= -A\sqrt{h}$	A1	AG	
		i	when $t = 0$, $h = (1 - 0)^2 = 1$ as required	B1		
		i	Or $\int \frac{dh}{\sqrt{h}} = \int -A\,dt$	M1	Separating variables correctly and integrating	
		i	$2h^{1/2} = -At + c$	A1	Including c. [Condone change of c.]	
		i	$h = \left(\frac{-At + c}{2}\right)^2$ at $t = 0$, $h = 1$, $1 = (c/2)^2 \Rightarrow c$ $= 2$, $h = (1 - At/2)^2$			
		i		B1	Using initial conditions AG **Examiner's Comments** The method of separating the variables and integrating was more popular than verification, and was more successful. Those that verified usually forgot to use the initial conditions. When integrating there was sometimes confused work when the arbitrary constant was changed but continued to be used as c.	
		ii	When $t = 20$, $h = 0$	M1	Subst and solve for A	
		ii	$\Rightarrow 1 - 10A = 0$, $A = 0.1$	A1	cao	
		ii	When the depth is 0.5 m, $0.5 = (1 - 0.05t)^2$	M1	substitute $h = 0.5$ and their A and solve for t	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow 1 - 0.05t = \sqrt{0.5}$, $t = (1 - \sqrt{0.5})/0.05 = 5.86$s	A1	www cao accept 5.9 **Examiner's Comments** Good marks were scored in this part by all candidates. Some made the question more difficult when finding A by using a quadratic equation. The most common error was in using $\sqrt{0.5}$ as 0.25 when finding *t*.	
		iii	$\frac{dh}{dt} = -B\frac{\sqrt{h}}{(1+h)^2}$ $\Rightarrow \int \frac{(1+h)^2}{\sqrt{h}} dh = -\int B \ dt$	M1	separating variables correctly and intend to integrate both sides (may appear later) [NB reading $(1 + h)^2$ as $1 + h^2$ eases the question. Do not mark as a MR] In cases where $(1 + h)^2$ is MR as $1 + h^2$ or incorrectly expanded, as say $1 + h + h^2$ or $1 + h^2$, allow first M1 for correct separation and attempt to integrate and can then score a max of M1M0A0A0A1 (for $-Bt + c$) A0A0, max 2/7.	
		iii	Either, LHS			
		iii	$\int \frac{1 + 2h + h^2}{\sqrt{h}} dh$	M1	expanding $(1 + h)^2$ and dividing by $\sqrt{h}$ to form a one line function of *h* (indep of first M1) with each term expressed as a single power of *h* eg must simplify say $1/\sqrt{h} + 2h/\sqrt{h} + h^2\sqrt{h}$,condone a single error for M1 (do not need to see integral signs)	
		iii	$= \int (h^{-1/2} + 2h^{1/2} + h^{3/2})dh$	A1	$h^{-1/2} + 2h^{1/2} + h^{3/2}$ cao dep on second M only -do not need integral signs	
		iii	Or, LHS either,	M1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$(1 + 2h + h)^2h^{1/2} - \int 2h^{1/2}\,(2 + 2h)dh$		using $\int udv + uv + \int vdu$ correct formula used correctly, indep of first M1 condone a single error for M1 if intention clear	
		iii	**or,** $h^{1/2} + h^{3/2} + \frac{h^{5/2}}{3} + \int \frac{1}{2} h^{-3/2} (h + h^2 + \frac{h^3}{3}) dh$	A1	cao oe	
		iii	$2h^{1/2} + \frac{4h^{3/2}}{3} + \frac{2h^{5/2}}{5}$	A1	cao **oe, both sides dependent on first M1 mark**	
		iii	$= -Bt + c$	A1	cao need $-Bt$ and c for second A1 but the constant may be on either side	
		iii	$\Rightarrow 2h^{1/2} + 4h^{3/2}/3 + 2h^{5/2}/5 = -Bt + c$			
		iii	When $t = 0$, $h = 1 \Rightarrow c = 56/15$	A1	from correct work only (accept 3.73 or rounded answers here but not for final A1) or c = -56/15 if constant on opposite side.	
		iii	$\Rightarrow h^{1/2}(30 + 20h + 6h^2) = 56 - 15Bt$*	A1	**NB AG must be from all correct exact work including exact *c*.** **Examiner's Comments** A pleasing number of candidates scored full marks here. Most separated the variables correctly and successfully integrated the RHS, including the inclusion of $+c$. Those candidates who realised to expand the bracket and divide often were able to score all the remaining marks. A few used the approach from integration by parts but usually did not reach the end.	
		iv	$h = 0$ when $t = 20$	M1	Substituting $h = 0, t = 20$	
		iv	$\Rightarrow$ B = 56/300 = 0.187	A1	Accept 0.187	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	When $h = 0.5$ $56 - 2.8t = 29.3449\ldots$	M1	Subst their $h = 0.5$, ft their B and attempt to solve	
		iv	$\Rightarrow t = 9.52$s	A1	Accept answers that round to 9.5s www. **Examiner's Comments** Many good scores were achieved here when substituting to find B and t. There were a lot of numerical errors from others.	
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
52		i	EITHER	B1	soi	
		i	$x = e^{3t}$, $y = te^{2t}$	M1	Their $dy / dt \div dx / d$ in terms of t	
		i	$dy / dt = 2te^{2t} + e^{2t}$ $\Rightarrow dy/dx = (2te^{2t} + e^{2t})/3e^{3t}$	A1	oe cao allow for unsimplified form even if subsequently cancelled incorrectly iecan isw	
		i	when $t = 1$, $dy/dx = 3e^2/3e^3 = 1/e$	A1	cao www must be simplified to 1/e oe	
		i	OR			
		i	$3t = \ln x$, $y = \frac{\ln x}{3} e^{2/3 \ln x} = \frac{x^{2/3} \ln x}{3}$	B1	Any equivalent form of y in terms of x only	
		i	$dy / dx = \frac{1}{3} x^{2/3} \frac{1}{x} + \ln x \frac{2}{9} x^{-1/3}$	M1	Differentiating their y provided not eased ie need a product including	
		i	$= \frac{1}{3e^t} + \frac{2t}{3e^t}$	A1	ln kx and x^p and subst $x = e^{3t}$ to obtain dy/dx in terms of t oe cao	
		i	$dy/dx = 1/3 + 2/3e = 1/e$	A1	www cao **exact only** must be simplified to 1/e or 1/e or e^{-1} **Examiner's Comments** Scores here tended to be 4 or 1. Much depended upon whether the candidates used the product rule when finding dy/dt. The very common error was to state that $dy/dt=2te^{2t}$. The majority of candidates did understand the general method of using $dy/dt \div dx/dt$. A number also failed to cancel their final answer, often leaving it as $\frac{e^2}{e^3}$.	
		ii	$3t = \ln x \Rightarrow t = (\ln x)/3$	B1	Finding t correctly in terms of x	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	ii. $y = (\ln x) / 3e^{(2\ln x)/3}$	M1	Subst in y using their t	
		ii	iii. $= \frac{1}{3}x^{\frac{2}{3}}\ln x$	A1	Required form $a\,x^b \ln x$ only NB If this work was already done in 5(i), marks can only be scored in 5(ii) if candidate specifically refers in this part to their part (i). Examiner's Comments Most candidates found t correctly in terms of x and then substituted it into y. The simplification was not always then complete.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
53			$\frac{3x}{(2-x)(4+x^2)} = \frac{A}{2-x} + \frac{Bx+C}{4+x^2}$	M1	correct form of partial fractions (condone additional coeffs eg $\frac{(Ax+B)}{2-x} + \frac{(Cx+D)}{4+x^2}$* for M1 **BUT** $\frac{A}{2-x} + \frac{B}{4+x^2}$ ** is M0)	
			$\Rightarrow 3x = A(4+x^2) + (Bx+C)(2-x)$	M1	Multiplying through oe and substituting values or equating coeffs at **LEAST AS FAR AS FINDING A VALUE** for one of their unknowns (even if incorrect) Can award in cases * and ** above Condone a sign error or single computational error for M1 but not a conceptual error Eg $3x = A(2-x) + (Bx+C)(4+x^2)$ is M0 $3x(2-x)(4+x^2) = A(4+x^2) + (Bx+C)(2-x)$ is M0 Do not condone missing brackets unless it is clear from subsequent work that they were implied. Eg $3x = A(4+x^2) + Bx + C(2-x) = 4A + Ax^2 + Bx + 2C - Cx$ is M0 $= 4A + Ax^2 + 2Bx - Bx^2 + 2C - Cx$ is M1	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$x = 2 \Rightarrow 6 = 8A$, $A = ¾$	A1	oe www [SC B1 A = 3/4 from cover up rule can be applied, then the M1 applies to the other coefficients] NB $\frac{A}{2-x} + \frac{B}{4+x^2} \Rightarrow$ **A=3/4** is A0 ww (wrong working)	
			x^2 coeffs: $0 = A - B \Rightarrow B = ¾$	A1	oe www	
			constants: $0 = 4A + 2C \Rightarrow C = -1½$	A1	oe www [In the case of * above, all 4 constants are needed for the final A1] Ignore subsequent errors when recompiling the final solution provided that the coeffs were all correct. Examiner's Comments Most candidates understood the method of expressing the fraction in partial fractions. Many were completely successful and most errors were arithmetic. $\frac{A}{(2-x)} + \frac{B}{(4+x^2)}$ A few incorrectly used	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
54		i	$\frac{1}{(1+2x)(1-x)}=\frac{A}{1+2x}+\frac{B}{1-x} \Rightarrow 1=A(1-x)+B(1+2x)$	Enter text here.	Enter text here.	
		i	Enter text here.	M1	Cover up, substitution or equating coefficients	
		i	$x = 1 \Rightarrow 3B = 1$, $B = 1/3$	A1	Enter text here.	
		i	$x = -½ \Rightarrow 1 = 3A/2$, $A = 2/3$	A1	isw after correct A and B stated **Examiner's Comments** Was answered extremely well with nearly all candidates correctly expressing $\frac{1}{(1+2x)(1-x)}$ in partial fractions.	
		ii	$1 + x - 2x^2 = (1 + 2x)(1 - x)$	B1	May be seen in separation of variables (may be implied by later working) – implied by the use of factors $(1 + 2x)$ and $(1 - x)$	
		ii	$\Rightarrow \quad \frac{1}{3}\int[\frac{2}{(1+2x)}+\frac{1}{1-x}]dx=\int k dt$	M1	Separating variables and substituting partial fractions. If no subsequent work integral signs needed, but allow omission of dx or dt, but must be correctly placed if present	
		ii	$\lambda \ln(1 + 2x) + \mu \ln(1 - x) = kt$ $(+ c)$	A1	Any non-zero constant λ, μ	
		ii	$\Rightarrow \ln(1 + 2x) - \ln(1 - x) = 3kt$ $(+ c)$	A1	www oe (condone absence of c)	
		ii	When $t = 0$, $x = 0 \Rightarrow c = 0$	B1	cao (**must** follow previous A1) need to **show** (at some stage) that $c = 0$. As a minimum $t = 0$, $x = 0$, $c = 0$. Note that $c = \ln(-1)$ (usually from incorrect integration of $(1 - x)$) or similar scores B0	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\Rightarrow \quad \ln\left(\frac{1+2x}{1-x}\right)=3kt$	M1	Combining both their log terms correctly. Follow through their c. Allow if c = 0 clearly stated (provided that c = 0) even if B mark is not awarded, but do not allow if c omitted	
		ii	$\Rightarrow \frac{1+2x}{1-x}=e^{3kt}$ *	A1	**AG** www must have obtained all previous marks in this part **Examiner's Comments** In part (ii) the majority of candidates were able to separate the variables and substitute their partial fractions correctly. There were, however, frequent errors in the integration usually when candidates forgot to divide by 2 when integrating $\frac{1}{1+2x}$ and/or when they forgot to do the same process with the –1 when integrating $\frac{1}{1-x}$. Many candidates did not include a constant of integration or, if included, it was either subsequently ignored or set to zero without any mathematical justification. Most candidates were able to combine theirlogarithmic terms correctly, though examiners noted the high volume of cases in which the ‘correct’ printed answer was seen following earlier incorrect working.	
		iii	(1 + 2(0.75)) / (1 – 0.75) = e^{3k}	M1	substituting t = 1, x = 0.75 at any stage	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$k = (1/3)\ln 10$ (= 0.768 (3 s.f.))	A1	3sf or better	
		iii	$t = \ln(2.8/0.1)/3k = 1.45$ hours	A1	1.45 (or better) or 1 hr 27 mins **Examiner's Comments** Most candidates achieved the first two marks in part (iii) for finding the valu of *k* , but many made errors in handling the logarithms to find the time taken for the drug concentration to reach 90% of its maximum value. Examiners noted the large variation in the accuracy of candidates'final answers in this part.	
		iv	$1 + 2x = e^{3kt} - xe^{3kt}$	Enter text here.	Enter text here.	
		iv	$\Rightarrow 2x + xe^{3kt} = e^{3kt} - 1$	M1*	Multiplying out and collecting *x* terms (condone one error)	
		iv	$\Rightarrow x(2 + e^{3kt}) = e^{3kt} - 1$	M1dep*	Factorising their *x* terms correctly	
		iv	$\Rightarrow x = (e^{3kt} - 1) / (2 + e^{3kt})$	A1	Enter text here.	
		iv	$= (1 - e^{-3kt})/(1 + 2e^{-3kt})$*	A1	www (**AG**) – as AG must be an indication of how previous line leads to the required result (eg stating or showing multiplying by e^{-3kt})	
		iv	when $t \to \infty$ $e^{-3kt} \to 0$ $x = (1 - e^{-3kt})/(1 + 2e^{-3kt}) \to 1/1 = 1$	B1	clear indication that $e^{-3kt} \to 0$ so, for example, accept as a minimum $(x \to)\frac{1-0}{1+0} = 1$ **or** $e^{-3kt} \to 0 \Rightarrow (x \to)\ 1$ (NB substitution of large values of *t* with no further explanation is B0)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	OR $\dfrac{1-x}{1+2x}=e^{-3kt}$	B1	Enter text here.	
		iv	$1 - x = e^{-3kt} + 2xe^{-3kt}$	M1*	Multiplying up and expanding (condone one error)	
		iv	$x(1 + e^{-3kt}) = 1 - e^{-3kt}$	M1dep*	Factorising their x terms correctly	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$x = (1 - e^{-3kt})/(1 + 2e^{-3kt})$*	A1	www (AG) – final B mark as in scheme above **Examiner's Comments** In part (iv) most candidates multiplied up by $1–x$, collected and factorised the x terms correctly. The main problem seemed to be how to get the negative exponent. These often appeared when candidates divided $e^{3kt} – 1$ by $2 + e^{3kt}$, losing the final accuracy mark in the process. It was also common for candidates to simply not show how $\frac{e^{3kt}-1}{2+e^{3kt}}$ was equal to the given answer. Those candidates who started by taking the reciprocal of the answer given in part(ii) part (ii) were usually far more successful in deriving the required result in this part. The majority of candidates who attempted to verify that the drug concentration approached its maximum value in the long term recognised that as $t \to \infty, e^{-3kt} \to 0$although some candidates simply substituted a large value of t to show that x was close to 1, this approach was not sufficient to earn the final mark in this part.	
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
55			$\Rightarrow 5x(x + 1) - 3(2x + 1) = (2x + 1)(x + 1)$	M1*	Multiplying throughout by $(2x + 1)(x + 1)$ or combining fractions and multiplying up oe (eg can retain denominator throughout) Condone a single numerical error, sign error or slip provided that there is no conceptual error in the process involved Do not condone omission of brackets unless it is clear from subsequent work thatthey were assumed eg $5x(x + 1) - 3(2x + 1) = (2x + 1)(x - 1)$ gets M1 $5x(x + 1) - 3(2x + 1) = 1$ gets M0 $5x(x + 1)(2x + 1) - 3(2x + 1)(x + 1) = (x + 1)(2x + 1)$ gets M0 $5x(x + 1) - 3(2x + 1) = (2x + 1)$ gets M1, just, for slip in omission of $(x + 1)$	
				M1dep*	Multiplying out, collecting like terms and forming quadratic (= 0). Follow through from their equation provided the algebra is not significantly eased and it isa quadratic. Condone a further sign or numerical error or a minor slip when rearranging	
			$\Rightarrow 3x^2 - 4x - 4 = 0$	A1	oe www (not fortuitously obtained – check for double errors)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\Rightarrow (3x + 2)(x - 2) = 0$	M1	Solving their three term quadratic (= 0) provided $b^2 - 4ac \geq$ Use of correct quadratic equation formula (if formula isquoted correctly then only one sign slip is permitted, if the formula is quoted incorrectly M0, if not quoted at all substitution must be completely correct to earn the M1) or factorising (giving their x^2 term and one other term when factors multiplied out) or comp. the square (must get to the square root stage involving ± and arithmetical errors may be condoned provided their $3(x - 2/3)^2$ seen or implied)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			⇒ $x = -2/3$ or 2	A1	cao for both obtained www (condone – 0.667 or better) (If no factorisation (oe) seen B1 for each answer stated following correct quadratic) **Examiner's Comments** Common errors included: • $5(x + 1) - 3(2x + 1) = (2x + 1)(x + 1)$ (and not the correct $5x(x + 1) -\ldots$) • Expanding $-3(2x+1)$ as either $-6x+3$ or $-\{6x-1$ or $-\{6x+1$ • There were some candidates who did not multiply up on the right-hand side and so obtained $5x(x+1)$ $-3(2x1) = 1$ • Some lost the final two marks for not applying the quadratic formula correctly However, this question was generally done well with most candidates scoring ful marks and demonstrating sound basic algebraic manipulation skills. It was commonto see the use of the quadratic formula as much as factorising to solve the final quadratic equation. Very few completed the square but those that did were mainly successful.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
56		i		B1	**NB AG** - must be at least one intermediate step before given answer - correct application of partial fractions is fine **Examiner's Comments** Nearly all candidates correctly showed the required result in part (i) although a few attempted to use partial fractions with varying degrees of success. In part (ii) many candidates incorrectly verified that when $x = 0$, $t = 0$ rather than the required result of showing that when $t = 0$, $x = 0$.Those that did begin by setting $t = 0$ usually went on to score both marks in this part. Part (iii) proved to be quite discriminating with many candidates unable to show that the rate of change of x was proportional to the given product. The most common method seen was to write t as ln(2 + x) – ln(2 – x) and then to differentiate this expression with respect to x and obtain $\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{2+x} + \frac{1}{2-x}$ and then $\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{4}{(2+x)(2-x)} \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{(2+x)(2-x)}{4}$ used part (i) to show that and hence the constant of proportionality is clearly $\frac{1}{4}$. The most common error was a failure to differentiate t	

Question				Answer/Indicative content	Marks	Part marks and guidance	
						correctly with many retaining the negative between the two terms. A number of candidates attempted instead to derive the given result by starting with the differential equation $\frac{dx}{dt}=k(2+x)(2-x)$ and attempting to solve this using the method of separation of variables. While a number were successful in obtaining the required constant many failed to deal with $\int \frac{dx}{(2+x)(2-x)}$ correctly or forgot to include the required constant of integration. Part (iv) was answered well with many correctly starting by either writing $e^t=\frac{2+x}{2-x}$ or $e^{-t}=\frac{2-x}{2+x}$, , the latter of these two usually lead to the correct given answer while the former lead to $x=\frac{2(e^t-1)}{1+e^t}$ with the vast majority of candidates being unable to explain clearly why this would lead to the given result. As this was a show that question there needed to be a clear indication of how this result would leads	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					to $x = \dfrac{2(1-e^{-t})}{1+e^{-t}}$. Finally in this part many candidates correctly stated that the long-term mass of the substance was 2 mg. Part (v) was answered with varying degrees of success with the vast majority correctly separating the variables to obtain $\int \dfrac{dx}{(2-x)(2+x)} = k\int e^{-t}dt$ - however, from this point it was all too clear that a number of candidates did not, as requested, show by integration the given result, but simply wrote down the given answer (or an answer only a single step away from the given answer) without clearly showing how either side of the given equation was obtained. In many cases candidates failed to include a constant of integration that needed to be found using the given initial conditions. Part (vi) was answered extremely well with many candidates obtaining the correct answer of 0.811 which was achieved by setting e^{-t} equal to zero and substituting 1.85 for *x*. The most common error seen by examiners was to set	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					$\ln\left(\frac{2+x}{2-x}\right)$ equal to 1.85 and solve for *k* with e^{-t} equal to zero.	
		ii		B1	or = e^0 or $\ln(2+x) = \ln(2-x)$	
		ii	$2 + x = 2 - x \Rightarrow x = 0$	B1	If only this line seen then award B0B1 **SC: Allow B1 only for verifying that when $x = 0$, $t = 0$**	
		iii		B1		
		iii		M1	**Correct** differentiation of their *t* **OR for first two marks** - If no subtraction law of logs seen e.g. $\frac{dt}{dx} = \left(\frac{1}{\left(\frac{2+x}{2-x}\right)}\right)\left(\frac{(2-x)(1)-(2+x)(-1)}{(2-x)^2}\right)$ award B1 for correct first bracket (reciprocal expression) and B1 for second correct bracket (quotient/chain rule)(oe) – if additional constant(s) added (e.g. $t = k\ln(\ldots)$) then award B1 only for a constant times a fully correct derivative	
		iii		A1	$\frac{dt}{dx}$ and $\frac{dx}{dt}$ must be correctly attributed to the correct expression for this mark	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$k = \frac{1}{4}$ See next page for an alternative solution	A1	Explicitly stating (that the constant of proportionality is) $\frac{1}{4}$ - therefore it is possible to score A0A1in this part	
		iii	OR $\frac{dx}{dt} = k(2+x)(2-x) \Rightarrow \int \frac{dx}{(2+x)(2-x)} = k\int dt$	B1	Allow omission of dx and/or dt	
		iii	$\lambda \ln(2 + x) + \mu \ln(2 - x) = kt(+c)$	M1	Any non-zero constant λ, μ - further guidance in (v) for this and the next mark	
		iii	$\frac{1}{4}[\ln(2+x) - \ln(2-x)] = kt(+c)$ $x = 0,\ t = 0 \Rightarrow c = 0$	A1	www oe (condone absence of c)	
		iii		A1	www - must include c and show that $c = 0$. Must explicitly state that $k = \frac{1}{4}$	
		iv	$e^t = \frac{2+x}{2-x}$ $(2 - x)e^t = 2 + x$	B1		
		iv	$\Rightarrow 2e^t - xe^t = 2 + x$ $\Rightarrow x(1 + e^t) = 2e^t - 2$	M1	Multiplying out, collecting x terms (condone sign slips and numerical errors (eg loss of a 2) only but M0 if e^t incorrectly replaced with e^{-t}) **and** factorising their x terms correctly	
		iv		A1	www **NB AG –** as AG must be an indication of how previous line leads to the required result (eg stating or showing multiplying by e^{-t})	
		iv		B1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	OR (for first three marks) $e^{-t} = \dfrac{2-x}{2+x}$ $(2 + x)e^{-t} = 2 - x$	B1		
		iv	$\Rightarrow 2e^{-t} + xe^{-t} = 2 - x$ $\Rightarrow x(1 + e^{-t}) = 2 - 2e^{-t}$	M1	Multiplying out, collecting x terms (condone sign slips as above) and factorising their x terms correctly	
		iv		A1	www **NB AG**	
		v		M1*	Separating variables - condone sign slips and issues with placement of k but M0 for $\int(2 - x)(2 + x)dx$ =… or equivalent algebraic error in separating variables unless recovered. If no subsequent work integral signs needed, but allow omission of dx and/or dt but must be correctly placed if present	
		v	$\alpha \ln(2 + x) + \beta \ln(2 - x) = \gamma e^{-t}(+c)$	A1	Any non-zero constants α, β, γ - **this line must be seen and cannot be implied by later working (as this is an AG)** – condone absence of c or if a constant present condone the use of k for their constant. Do not condone invisible brackets e.g. ln 2 + x unless recovered before subtraction law of logs applied – all of these points apply to the next A mark too	
		v	$\ln(2 + x) - \ln(2 - x) = -4ke^{-t}(+c)$	A1	www oe	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	When $t = 0$, $x = 0 \Rightarrow c = 4k$	M1dep*	Substituting $x = 0$, $t = 0$ into each term in an attempt to find their c (must get $c = \ldots$) - if they integrate and use k as their constant they must use $x = 0$, $t = 0$ to find this single k term only	
		v		A1	www **NB AG** must have obtained all previous marks in this part	
		v	**OR (for first 3 marks) – final M1A1 as above** $\int \frac{1}{(2-x)(2+x)}\mathrm{d}x = k\int \mathrm{e}^{-t}\mathrm{d}t$	M1*	Separating variables. If no subsequent work integral signs needed, but allow omission of dx or dt, but must be correctly placed if present	
		v		A2	**Must see** $1/(4 - x^2)$ on lhs – please note that one A mark cannot be awarded	
		vi	as $t \to \infty$, $x \to 1.85 \Rightarrow \ln 3.85/0.15 = 4k$	M1	Sets e^{-t} to 0 and substitutes $x = 1.85$ – condone substitution of a 'large' value of t only if it leads to the correct value of k	
		vi	$\Rightarrow k = 0.811$	A1	$k = 0.25 \ln (77 / 3)$ or 0.81 or better	
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
57			$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-2/t^2}{2}\left(= -\frac{1}{t^2}\right)$	M1*	M1 for their (dy / dt) / their (dx / dt) in terms of t with at least one term correct	
				A1	A1 cao (oe) – allow unsimplified even if subsequently cancelled incorrectly i.e. can isw	
				M1dep*	$y - \frac{2}{t} = f(t)(x - 2t)$ with any non-zero gradient expressed as a function of t - or any equivalent form (e.g. $y = mx + c$) but must have used the correct point $\left(2t, \frac{2}{t}\right)$ - if using $y = mx + c$ must explicitly have $c = \ldots$ before M1 can be awarded	
			$y - \frac{2}{t} = -\frac{1}{t^2}(x - 2t)$	A1	oe - need not be simplified	
			When $x = 0,\ y = \frac{4}{t}$	A1ft	Must be a function of t	
			When $y = 0$, $x = 4t$	A1ft	Must be a function of t	
			So area of $\text{triangle} = \frac{1}{2} \times \frac{4}{t} \times 4t = 8$ (which is independent of t)	A1	No ft on this mark – an answer of 8 (www) with no additional comment is sufficient to award this mark	
			OR (for the first two marks) $y = \frac{2}{\left(\frac{x}{2}\right)} = \frac{4}{x} \Rightarrow \frac{dy}{dx} = -\frac{4}{x^2}$	M1*	Attempt to eliminate t and correctly differentiates their Cartesian equation	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\Rightarrow\left(\frac{dy}{dx}\right)=-\frac{4}{(2t)^2}$	A1	**Examiner's Comments** Candidates found this unstructured question on parametric equations quite demanding with the modal mark scored being 2 out of a possible 7. Most candidates scored these two marks by correctly obtaining $\frac{dy}{dx}=-\frac{1}{t^2}$ although the vast majority failed to make any further progress worthy of merit with many trying to argue that the area of triangle OQR is independent of t without any further working of a mathematical nature. Of those that failed to obtain the correct derivative a number incorrectly stated that $\frac{dy}{dt}=2\ln t$ or implied that $\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dx}{dt}$. All that was required from those candidates who had obtained the correct gradient function in terms of t was to write down the equation of the tangent $\left(y-\frac{2}{t}=-\frac{1}{t^2}(x-2t)\right)$, substitute x = 0 and y = 0 to obtain $R\left(0,\frac{4}{t}\right)$ and Q(4t,0) respectively and hence calculate the area of the triangle as	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					$8\left(=\frac{1}{2}\times 4t\times\frac{4}{t}\right)$ which is clearly independent of t. A number of candidates began by finding the Cartesian equation of the curve and correctly obtaining $\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{4}{x^2}$ but they then incorrectly used this gradient function in their equation of the tangent.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
58		i	$kx^{\frac{1}{3}-1}$ oe $4x^{\frac{-2}{3}}$ isw cao	M1	*k* is any non-zero constant ignore + *c*	allow any equivalent exact simplified form **Examiner's Comments** This was done well. A small minority of candidates failed to score: most problems were caused by a failure to put the original function into index form correctly.
		ii	kx^{-3+1} oe $-3x^{-2}$ isw + c	A1	*k* is any non-zero constant	allow any equivalent exact simplified form **Examiner's Comments** A few candidates differentiated or tried to integrate both the numerator and the denominator independently, but most knew what to do here and went on to score 2 or 3 marks. A significant minority of candidates neglected to add '+ c', thereby losing an easy mark.
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
59		a	Correct shape for $y=\frac{1}{x}$, translated vertically upward Crosses x-axis at $x=-\frac{1}{a}$ Asymptotes x = 0 and y = a	B1(AO1.1) B1(AO2.2a) B2(AO1.1) (AO2.2a) [4]	B1 for one asymptote correct	
		b	$\frac{dy}{dx}=-\frac{1}{x^2}$ When x = 2 , gradient = $-\frac{1}{4}$ Gradient of normal = 4 FT their gradient $y-\frac{5}{2}=4(x-2)$FT their gradient 2y = 8x –11 oe	M1(AO1.1a) M1(AO1.1) M1(AO2.1) M1(AO1.1) A1(AO2.1) [5]	Or gradient of given line is 4 Or $y=\frac{5}{2}$ at the point on the curve where x = 2 At least one correct interim step or clear check that $\left(2, \frac{5}{2}\right)$ is on given line Any simplified form	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$2\left(\frac{1}{x}+2\right)=8x-11$ $8x^2-15x-2=0$ Other point is $x=-\frac{1}{8}$, $y=-6$	M1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]	Substitution Forming quadratic, condone one error BC $x = 2$ not needed in this case		
			Total	12			
60			$\int_0^{\frac{\pi}{12}}\cos 3x\,dx=\left[\frac{\sin 3x}{3}\right]_0^{\frac{\pi}{12}}$ $=\frac{1}{3}\left(\sin\frac{\pi}{4}-0\right)$ $=\frac{\sqrt{2}}{6}$ o.e.	B1(AO1.1) M1(AO1.1) A1(AO1.1) [3]	$\frac{\sin 3x}{3}$ Must be in exact form		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance		
61		a	$6x^2 + 3y^2\frac{dy}{dx} = 5\frac{dy}{dx}\left[\Rightarrow \frac{dy}{dx} = \frac{6x^2}{5-3y^2}\right]$ when $x = 1$, $y = 2$, $6 + 12\frac{dy}{dx} = 5\frac{dy}{dx}$ $\Rightarrow \frac{dy}{dx} = -\frac{6}{7}$	M1(AO1.1a) A1(AO1.1) M1(AO1.1) A1cao(AO2.1) [4]	implicit differentiation correct substituting $x = 1$, $y = 2$		
		b	$\frac{dy}{dx} = 0$ so $6x^2 = 0$ $x = 0$ so all stationary points lie on y-axis	B1(AO1.2) E1(AO2.1) [2]	Substitute $\frac{dy}{dx} = 0$ into their differentiated expression Completion of argument		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance		
62			let $u=1+\sqrt{x} \quad du=\frac{1}{2}x^{-\frac{1}{2}}\,dx$	M1(AO3.1a)	substituting $u=1+\sqrt{x}$ or $w=\sqrt{x}$	Evidence of method must be seen	
			$\Rightarrow dx = 2(u-1)du$	A1(AO1.1)	$dx = 2(u-1)\,du$ or $dx = 2w\,dw$		
			$\Rightarrow \int_0^1 \frac{1}{1+\sqrt{x}}dx=\int_1^2 \frac{2(u-1)}{u}du$	A1(AO1.1)	$\frac{2(u-1)}{u}(du)$ or $\frac{2w}{(w+1)}(dw)$		
			$=\int_1^2\left(2-\frac{2}{u}\right)du$	M1(AO3.1a)	splitting fraction or dividing to get $2-\frac{2}{(w+1)}$ (or substituting $u = w + 1 \Rightarrow \left(\frac{2(u-1)}{u}\right)$ and then splitting fraction)	Evidence of method must be seen	
			$=\left[2u-2\ln u\right]_1^2$	A1(AO1.1)	or $\left[2w-2\ln(w+1)\right]_0^1$ if still in terms of w		
			$= 4 - 2\ln 2 - 2 = 2 - 2\ln 2$ or $2 - \ln 4$	A1cao(AO1.1)			
				[6]			
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance	
63		a	$\int \frac{\mathrm{d}m}{m} = \int \frac{\mathrm{d}t}{t(1+2t)}$ $\frac{1}{t(1+2t)} \equiv \frac{A}{t} + \frac{B}{1+2t}$ $\Rightarrow 1 \equiv A(1 + 2t) + Bt$ $t = 0 \Rightarrow A = 1$ $t = -½ \Rightarrow 1 = -½ B \Rightarrow B = -2$ $\Rightarrow \int \frac{\mathrm{d}m}{m} = \int \left(\frac{1}{t} - \frac{2}{1+2t}\right)\mathrm{d}t$ $\Rightarrow \ln m = \ln t - \ln(1 + 2t) + c$ $t = 1,\ m = 1 \Rightarrow c = \ln 3$ $\Rightarrow \ln m = \ln\left(\frac{3t}{1+2t}\right)$ $\Rightarrow m = \frac{3t}{(1 + 2t)}$	M1(AO1.1a) M1(AO3.1b) M1(AO1.1) A1A1(AO1.1 1.1) B1FT(AO2.1) M1(AO1.1) E1(AO2.1) [8]	separating variables using partial fractions substituting values, equating coeffs or cover up $A = 1$, $B = -2$ FT their i, ii, condone no c evaluating constant of integration AG	
		b	i $1.25 = \frac{3t}{(1+2t)}$ $\Rightarrow 1.25 + 2.5t = 3t$ $\Rightarrow t = 1.25 \div 0.5 = 2.5$ minutes	M1(AO1.1a) A1(AO1.1) [2]	Enter text here.	
		b	ii $m = \frac{3}{\left(\frac{1}{t} + 2\right)}$ $\rightarrow 1.5$ [grams]	M1(AO3.1b) A1(AO2.2a) [2]	Enter text here.	
			Total	12		

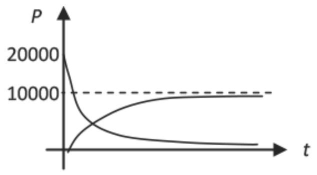
Question			Answer/Indicative content	Marks	Part marks and guidance	
64			$\frac{1}{3}x^3$ $-\frac{1}{x}$ oe $+ c$	B1(AO1.1) B1(AO1.1) B1(AO1.1) [3]		
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance		
65			$\int_1^3 3x^{-\frac{3}{2}}\,dx$ $\left[-6x^{-\frac{1}{2}}\right]_1^3$ $\frac{-6}{\sqrt{3}}-\frac{-6}{\sqrt{1}}$ $\frac{-6}{\sqrt{3}}+6$ $6-2\sqrt{3}$ AG	M1(AO1.1a) A1(AO1.1) A1(AO1.1) M1(AO1.1) E1(AO2.1) [5]	Attempt to integrate (ignore missing limits) Correct integration Correct limits seen at some point Substitution of limits (condone one error) Correct intermediate step using surds which follows from the substitution of limits and is not identical to given answer and completion	Do not award any A- marks if M0 is given Given answer must be seen to score E1	
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
66		a	$-0.03e^{-0.03t}$	B1(AO1.2) [1]		
		b	Decreasing function because $e^{-0.03t}$ is positive [for all values of t] so the gradient is negative.	E1(AO2.2a) [1]	Explanation may include a sketch graph of the function $70e^{-0.03t}$ but it must be clear that the graph is of the function and the answer must clearly refer to the gradient of the function and not the trend in the data	
		c	A 70	B1(AO1.1) [1]		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	B 38.[4168…]	B1(AO1.1) [1]		
		d	Data values decreasing so decreasing function is suitable At t = 0 , calculated D = 70 and this matches the data At t = 20, data value is 40 which is not exact but close	E1(AO3.5a) B1(AO3.5a) B1(AO3.5b) [3]		
			Total	7		
67		a	$\frac{dy}{dx} = \frac{1}{2}\left(1-3x^2\right)^{-½}.(-6x)$ $= \frac{-3x}{\sqrt{(1-3x^2)}}$	B1(AO1.1) M1(AO1.1) A1(AO1.1) [3]	$\frac{1}{2}u^{-\frac{1}{2}}$ soi Chain rule oe, but must simplify $\frac{1}{2} \times 6$	
		b	$\frac{dy}{dx} = \frac{(3x+2).2x - x^2.3}{(3x+2)^2}$ $= \frac{3x^2+4x}{(3x+2)^2}$	M1(AO1.1) A1(AO1.1) A1(AO1.1) [3]	Quotient rule or product rule oe, but must simplify numerator	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
68			let $u = x^2$, $u' = 2x$, $v' = e^{2x}$, $v = \frac{1}{2}e^{2x}$ $\int x^2e^{2x}dx = \frac{1}{2}x^2e^{2x} - \int 2x.\frac{1}{2}e^{2x}dx = \frac{1}{2}x^2e^{2x} - \int xe^{2x}dx$ let $u = x$, $u' = 1$, $v' = e^{2x}$, $v = \frac{1}{2}e^{2x}$ $\int xe^{2x}dx = \frac{1}{2}xe^{2x} - \int \frac{1}{2}e^{2x}dx$ $= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x}(+c)$ so $\int x^2e^{2x}dx = \frac{1}{2}x^2e^{2x} - \frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + c$	M1A1(AO1.1a 1.2) A1(AO1.1) M1(AO1.1a) A1(AO1.1) A1(AO1.1) A1(AO2.5) [7]	Do not award if no '+ c'	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance		
69		a	*A* 	M1(AO1.1) M1(AO1.1)	P_G shape through O P_R shape through (0, 20000), [condone graphs for –ve *t*]		
		a	*B* asymptote for P_G = 10 000 Asymptote for P_R = 0	A1(AO2.2a) A1(AO2.2a) [4]	Or p = 10 000 Or p = 0		
		b	Red squirrels zero Grey 10 000	B1(AO3.4) B1(AO3.4) [2]			
		c	One relevant comment evaluating the validity of the model	B1(AO3.5a)	E.g. One of • Grey population increases as would be expected [since grey squirrels are larger and more successful] • Red population decreases as would be expected [since red		

Question			Answer/Indicative content	Marks	Part marks and guidance		
				[1]	squirrels have to compete with the larger grey squirrels for food] • Number of squirrels tends to a limit as would be expected [since there is limited food and space] • Would expect grey population to grow slower at first • Would expect red population to fall slower at first		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d		M1(AO3.1b)	Attempts to differentiate either or both	
			$\frac{dP_G}{dt} = 10\ 000ke^{-kt}$	A1(AO1.1)		
			$\frac{dP_R}{dt} = -20\ 000ke^{-kt}$	A1(AO1.1)		
			so $\frac{dP_R}{dt} = -2\frac{dP_G}{dt}$	E1(AO2.1)	Or in words	
				[4]		
		e	$10\ 000\ (1 - e^{-3k}) = 20\ 000\ e^{-3k}$	M1(AO1.1a)		
			$\Rightarrow 1 - e^{-3k} = 2\ e^{-3k}$			
			$\Rightarrow e^{-3k} = \frac{1}{3}$	A1(AO1.1)		
			$\Rightarrow -3k = \ln\left(\frac{1}{3}\right)$	M1(AO1.1)	Taking natural logs of both sides	
			$\Rightarrow k = -\frac{1}{3}\ln\left(\frac{1}{3}\right) = 0.366$ or $\frac{1}{3}\ln 3$	A1cao(AO2.1)		
				[4]		
			Total	15		

Question			Answer/Indicative content	Marks	Part marks and guidance	
70		a	P is $(\sqrt{2}, \frac{\sqrt{2}}{2})$ $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$ $= \frac{\cos\theta}{-2\sin\theta}$ When $\theta = \frac{\pi}{4}$, $\frac{dy}{dx} = -\frac{1}{2}$ Equation of tangent is $(y - \frac{\sqrt{2}}{2}) = -\frac{1}{2}(x - \sqrt{2})$ $\Rightarrow \quad y = -\frac{1}{2}x + \frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}$ $\Rightarrow \quad x + 2y = 2\sqrt{2}$	B1(AO1.1) M1(AO3.1a) A1(AO1.1) A1(AO1.1) B1(AO2.1) E1(AO1.1) **[6]**	oe AG	
		b	When x = 0, y = $\sqrt{2}$ so A is (0, $\sqrt{2}$) When y = 0, x = $2\sqrt{2}$ so B is ($2\sqrt{2}$, 0) Area of triangle $= \frac{1}{2}\sqrt{2} \times 2\sqrt{2} = 2$units^2	B1(AO1.1) B1(AO1.1) B1(AO1.1) **[3]**		
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance	
71			$\frac{dy}{dx} = 4x^3 - 12x + 4$ $\frac{d^2y}{dx^2} = 12x^2 - 12 = 0$ $x = \pm 1$ (–1, –4) and (1, 4)	M1(AO1.1) A1(AO1.1) M1(AO1.2) A1(AO1.1) A1(AO2.1) [5]	Differentiating once First derivative Differentiating a second time and equating to zero	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
72		a	$f(-1) = (-1)^4 + (-1)^3 - 2(-1)^2 - 4(-1) - 2$ $= 1 - 1 - 2 + 4 - 2 = 0$	E1(AO1.1) [1]		
		b	$f(1) = 1 + 1 - 2 - 4 - 2 = -6$ or 'negative' $f(2) = 16 + 8 - 8 - 8 - 2 = 6$ or 'positive' change of sign $\Rightarrow$ root between 1 and 2	B1(AO1.1) E1(AO2.4) [2]	both correct allow no mention of continuity of f AG	
		c	long division or equating coeffts $\Rightarrow g(x) = x^3 - 2x - 2$ so $a = -2$, $b = -2$	M1(AO1.1) A1A1(AO2.2a 1.1) [3]		
		d	Clear explanation E.g. $f(x) = (x + 1)g(x)$ For the root of $f(x) = 0$ between 1 and 2, RHS is also zero hence $g(x) = 0$	E1(AO2.4) [1]		
		e	$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)}$ $= x_n - \frac{x_n^3 - 2x_n - 2}{3x_n^2 - 2}$ $= \frac{3x_n^3 - 2x_n - x_n^3 + 2x_n + 2}{3x_n^2 - 2}$ $= \frac{2x_n^3 + 2}{3x_n^2 - 2}$ Root 1.769 (4sf)	M1(AO1.1) E1(AO2.4) A1(AO2.2a) [3]	AG BC	
			Total	10		

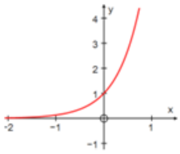
Question			Answer/Indicative content	Marks	Part marks and guidance	
73		a	(i) $f'(x) = e^{-x}\cos x - e^{-x}\sin x$ $f'(x) = 0$ and $e^{-x} \neq 0 \Rightarrow$ $\cos x = \sin x$ $\Rightarrow \tan x = 1$ $\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$	M1(AO3.1a) A1(AO1.1) E1(AO2.2a) M1(AO1.1) A1(AO1.1)	product rule correct Use of $\frac{\sin}{\cos} = \tan$ $x = \frac{\pi}{4}$ (condone 45°) $\frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4},$ …	
		a	So an AP with $d = \pi$ (ii) $y = \frac{\sqrt{2}}{2}e^{-\frac{\pi}{4}}, -\frac{\sqrt{2}}{2}e^{-\frac{5\pi}{4}}, \frac{\sqrt{2}}{2}e^{-\frac{9\pi}{4}}, -\frac{\sqrt{2}}{2}e^{-\frac{13\pi}{4}}$ This is a GP with $r = -e^{-\pi}$	E1FT(AO2.1) M1(AO3.10a) A1(AO1.1) E1FT(AO2.1) [9]	must state the common difference substituting one value of x into $f(x)$ must state common ratio, www	FT their values of x FT their values of y
		b	Yes with explanation that values of x would continue to be separated by pi and so values of y would continue to have same common ratio.	E1(AO2.2a) [1]		
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance		
74			$-2 \times 5x^{-2-1}$	M1(AO1.1)	soi		
			6	B1(AO1.1)	or correct differentiation of $6x + 3$		
			$\left[\frac{dy}{dx} = \right] 6 + 10x^{-3}$	A1(AO1.1)		Allow equivalent form, but constant must be simplified to 10.	
				[3]			
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance		
75			$3x^2 - 6x + 6$ $6x - 6 = 0$ $x = 1$ gradient has a minimum value of 3 at x =1. This is a minimum value because the gradient function is parabolic and the coefficient of x^2 is positive. As the gradient is always positive the function is always increasing oe	M1(AO2.1) A1(AO1.1) M1(AO2.1) A1(AO1.1) E1(AO2.2a) [5]	Differentiation All correct Differentiates again or completes the square or uses the discriminant *alternatively* gradient is never zero since discriminant is negative / can't solve from completing square, and must therefore be always positiyve since term in x^2 is positive oe	NB $3(x - 1)^2 + 3$ NB – 36	
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
76			$[y=]x-\frac{4x^{\frac{1}{2}}}{\frac{1}{2}}+\frac{6x^3}{3}$	M1(AO2.1)	Must be three terms	at least two terms correct	
			$[y=]x-8\sqrt{x}+2x^3+c$	A1(AO1.1)	$8\sqrt{x}$ or $8x^{½}$		
			Substitution of y = 122 and x = 4 in their	A1(AO1.1)	All correct including + c		
			$y=x-8\sqrt{x}+2x^3+c$	M1(AO1.1)			
			$y=x-8\sqrt{x}+2x^3+6$	A1(AO1.1) [5]			
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance
77			DR Consider $\int_{-1}^{0} x^3(x-3)\,dx$ and $\int_{0}^{3} x^3(x-3)\,dx$ $\int x^3(x-3)\,dx = \int (x^4 - 3x^3)\,dx$ $\frac{x^5}{5} - \frac{3x^4}{4} (+c)$ $\left[\frac{x^5}{5} - \frac{3x^4}{4}\right]_{-1}^{0} = 0 - \left(\frac{(-1)^5}{5} - \frac{3(-1)^4}{4}\right) = \frac{19}{20}$ $\left[\frac{x^5}{5} - \frac{3x^4}{4}\right]_{0}^{3} = \left(\frac{3^5}{5} - \frac{3\times3^4}{4}\right) - 0 = -\frac{243}{20}$ $\frac{19}{20} + \frac{243}{20} = \frac{131}{10}$	M1(AO 3.1a) M1(AO 1.1a) A1(AO 1.1a) A1(AO 1.1b) A1(AO 1.1b) B1(AO 1.1b) [6]	Splitting the integral into positive and negative regions with correct limits Attempting to integrate expanded form Correct indefinite integral seen Must be seen to use limits Must be seen to use limits www; not dependent on previous marks
			Total	6	

Question			Answer/Indicative content	Marks	Part marks and guidance	
78		a		G1(AO 1.2) G1(AO 1.2) [2]	Correct shape with x -axis as asymptote (0, 1) clearly shown	
		b	Stretch in the x-direction Stretch scale factor $\frac{1}{2}$	B1(AO 1.1a) B1(AO 1.1a) [2]	'Stretch' must be seen at least once for any marks to be awarded, but the word needn't be repeated	
		c	$\frac{dy}{dx} = 2e^{2x}$ When $x = 3$, $\frac{dy}{dx} = 2e^6$ and $y = e^6$ Tangent is $y - e^6 = 2e^6 (x - 3)$ $y = 2e^6x - 5e^6$ $y = e^6 (2x - 5)$	M1(AO 1.1a) A1(AO 1.1b) A1(AO 1.1a) B1(AO 1.1a) M1(AO 1.1a) A1(AO 2.1) [6]	Allow for any ke^{2x} Correct $2e^{2x}$ Allow method if 403 or better used for e^6 Must be in this form	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance		
79			$\frac{dy}{dx} = 4(3x^2+5)^3 \times 6x$ $\frac{dy}{dx} = 24x(3x^2+5)^3$	M1(AO1.1a) A1(AO 1.1b) A1(AO 1.1b) [3]	Use of chain rule attempted 6x soi		
			Total	3			
80			DR $\int_0^{\frac{1}{3}\pi}(3\sin 3x - \sin 3x)\,dx$ $\left[-\frac{2}{3}\cos 3x\right]_0^{\frac{1}{3}\pi}$ $= -\frac{2}{3}(\cos\pi - \cos 0)$ $= -\frac{2}{3}(-1-1) = \frac{4}{3}$ **AG**	M1(AO 1.1a) B1(AO 1.1b) B1(AO 3.1a) A1(AO 2.1) [4]	Integrating the difference of the functions $\int \sin 3x\,dx = -\frac{1}{3}\cos 3x$ oe soi Limits used or indicated on a diagram Must be clearly shown		
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance	
81		a	$\int \frac{1}{y(1+y)}\,dy = \int (1-x)\,dx$ $\frac{1}{y(1+y)} = \frac{1}{y} - \frac{1}{1+y}$ $\ln y - \ln(1+y) = x - \frac{1}{2}x^2 + c$ $\ln 1 - \ln(1+1) = 1 - \frac{1}{2} + c$ $c = -\frac{1}{2} - \ln 2$ $\ln y - \ln(1+y)$ $= x - \frac{1}{2}x^2 - \frac{1}{2} - \ln 2$ $\ln\left(\frac{2y}{1+y}\right) = x - \frac{1}{2}x^2 - \frac{1}{2}$ $\frac{2y}{1+y} = e^{x-\frac{1}{2}x^2-\frac{1}{2}}$ $2y = (1+y)e^{x-\frac{1}{2}x-\frac{1}{2}x^2}$ $y = \frac{e^{x-\frac{1}{2}x^2-\frac{1}{2}}}{2-e^{x-\frac{1}{2}x^2-\frac{1}{2}}}$	M1(AO 1.1a) M1(AO 3.1a) A1(AO 1.1b) A1(AO 1.1b) M1(AO 1.1a) A1(AO 1.1b) M1(AO 1.1a) M1(AO 1.1a) A1(AO 1.1b) [9]	Separating the variables Correct form $\frac{A}{y} + \frac{B}{1+y}$ used Correct partial fractions oe, e.g. RHS $-\frac{1}{2}(1-x)^2 + c$ Use of (1, 1) to find c Writing in non-logarithmic form Making y the subject Correct $y = f(x)$	

Question			Answer/Indicative content	Marks	Part marks and guidance
		b	At, (1, 1),	B1(AO 3.1a)	
			$\frac{dy}{dx} = 1 \times 2(1-1) = 0$	M1(AO 1.1b)	Determining sign of gradient on each side
			Near (1, 1), $y(1 + y)$ is positive and $(1 - x)$ is positive for $x < 1$ but negative for $x > 1$	E1(AO 2.2a)	Clear deduction seen
			$\frac{dy}{dx} > 0$ for $x < 1$ and $\frac{dy}{dx} < 0$ for $x > 1$ so		
			maximum point	B1	
			Alternative method	M1	
			At, (1, 1),		Differentiation using product rule and evaluation at (1, 1)
			$\frac{dy}{dx} = 1 \times 2(1-1) = 0$	E1	Condone –2 not seen if clearly negative
			$\frac{d^2y}{dx^2} = (1+2y)\frac{dy}{dx}(1-x) + (y+y^2)(-1)$	[3]	
			$\frac{d^2y}{dx^2} = 3 \times 0 + 2(-1) = -2 < 0$		Must follow both 1st and 2nd derivatives
			(1, 1) is a maximum point		
			Total	12	

Question			Answer/Indicative content	Marks	Part marks and guidance		
82			Gradient of chord $= \frac{(x+h)^3 - x^3}{h}$ oe $= \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$ Derivative is limit of gradient of chord as $h \to 0$ $3x^2 + 3xh + h^2 \to 3x^2$ AG	M1(AO 2.1) M1(AO 1.1a) A1(AO 1.1) E1(AO 2.4) B1(AO 2.2a) [5]	Expansion of cubic attempted, with at least some terms correct Completely correct expansion	Not just $x^3 + h^3$	
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance
83		a	DR $\frac{dy}{dx} = 4x^3 - 6x$ When x = 1, $\frac{dy}{dx} = -2$ Gradient of normal is $\frac{1}{2}$ Equation is $y = \frac{1}{2}(x-1)$	M1(AO 1.1) A1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [4]	Attempt to differentiate with at least one term correct Use of $m_1 m_2 = -1$ oe
		b	DR Curve is part **(a)** translated up by $\frac{1}{2}$ c = 2.5 **Alternative method** Equation of normal is $y = \frac{1}{2}x$ Point on curve is $\left(1, \frac{1}{2}\right)$ c = 2.5	M1(AO 3.1a) M1(AO 1.1) A1(AO 2.2a) M1 M1 A1 [3]	
			Total	7	

Question			Answer/Indicative content	Marks	Part marks and guidance		
84			$\frac{e^{3x}}{3}$ $4e^{3x} + c$	M1(AO 1.1) A1(AO 1.1) [2]	soi		
			Total	2			
85			$\frac{du}{dx} = 3$ substitution of $3x = u - 1$ $\int 2(u-1)u^7 \, du$ $\frac{2u^9}{9} - \frac{2u^8}{8} (+c)$ $\frac{2}{9}(3x+1)^9 - \frac{1}{4}(3x+1)^8 + c$	B1(AO 1.1) M1(AO 1.1) A1(AO 1.1) A1(AO 1.1) A1(AO 1.1) A1(AO 1.1) [6]	for either term correctly integrated both correct	Other correct methods eg integration by parts, are acceptable	
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance		
86			DR $\frac{dx}{dt} = -\sin t - 3$ $\frac{dy}{dt} = 3 + 4\sin t - 2\cos 2t$ $\frac{dy}{dx} = \frac{3 + 4\sin t - 2\cos 2t}{-\sin t - 3}$ denominator is always negative since $-1 \le \sin t \le 1$ Substitution of $\cos 2t = 1 - 2\sin^2 t$ in numerator $4\sin^2 t + 4\sin t + 1$ seen in numerator Numerator is $(2\sin t + 1)^2$ so numerator is always positive and denominator is always negative, so $\frac{dy}{dx}$ is always negative	B1(AO 3.1a) B1(AO 1.1) M1(AO 2.1) E1(AO 2.4) M1(AO 1.1) A1(AO 1.1) E1(AO 2.4) [7]			
			Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance	
87		a	$k = 5$	B1(AO 3.3) [1]	$36 = 9k - 9$	
		b	$x = 7.5$, model gives 140.25 ≈ 140 so model works well	B1(AO 3.4) B1(AO 3.5a) [2]		
		c	*A* $V = 88.5$ substitution of $x = 5.20$ and 5.21 to obtain 88.33 and 88.58 respectively	B1(AO 3.1a) B1(AO 1.1) [2]	NB 88.496 878236	
		c	*B* $\frac{dV}{dt} = 28e^{-0.2t}$ = 10.3 when $t = 5$ $\frac{dV}{dt} = (10x - x^2) \times \frac{dx}{dt}$ $\frac{dx}{dt} = \frac{10.3}{10 \times 5.21 - 5.21^2} = 0.413$ [cms^{-1}]	M1(AO 1.1) A1(AO 2.1) M1(AO 3.4) A1(AO 1.1) [4]	NB 10.300 6243528 Accept awrt 0.412 or 0.413	
		c	*C* Forever for a full glass or more than 70 cm^3 full after 5 seconds from (*A*) so better to fill the two half glasses	B1(AO 3.5b) B1(AO 3.4) [2]		
			Total	11		

Question			Answer/Indicative content	Marks	Part marks and guidance
88		a	f(0.1) = 3.897…and f(0.9) = –1.194… sign change so 0.1 < α < 0.9	B1(AO 1.1) E1(AO 1.2) [2]	
		b	$\left[\frac{dy}{dx}=\right]2-\frac{1}{x^2}+\frac{1}{x}$ $x_{r+1}=x_r-\frac{2x_r+\frac{1}{x_r}+\ln x_r-4}{2-\frac{1}{x_r^2}+\frac{1}{x_r}}$ $x_{r+1}=x_r-\frac{2x_r^3+x_r+x_r^2(\ln x_r-4)}{2x_r^2-1+x_r}$	B1(AO 1.1) M1(AO 1.1) A1(AO 2.2a) [3]	AG
		c	(A) x_1 = – 0.52359… which is negative so the iteration stops because f(x) is undefined for x < 0	B1(AO 1.1) E1(AO 2.4) [2]	
		c	(B) gradient is zero at 0.5 so tangent never touches x-axis	B1(AO 1.1) E1(AO 2.4) [2]	
		c	(C) 2.4497, 1.4667, 1.4675, 1.4675 *or* 0.6 is to the right of the turning point so converges to larger root	B1(AO 1.1) E1(AO 2.4) [2]	
			Total	11	

Question			Answer/Indicative content	Marks	Part marks and guidance		
89		i	$2x^2$ oe F[5] – F[1] 48 cao	B1 M1 A1 [3]	where F[x] = kx^2	ignore + c for the first two marks no marks for 48 unsupported **A0** for 48 + c	
					Examiner's Comments Most candidates successfully integrated and went on to obtain the correct answer. A few spoiled this by leaving "+ c" in the final answer, and a small number either differentiated or simply evaluated the integrand.		
		ii	$kx^{\frac{1}{2}+1}$ seen $4x^{\frac{3}{2}}+c$ or $4\sqrt{x^3}+c$ or $4\left(\sqrt{x}\right)^3+c$ isw	M1 A1 [2]			
					Examiner's Comments Nearly all candidates achieved the method mark by integrating, but a surprising number omitted the constant of integration thereby losing an easy mark.		
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
90		i	$\frac{\log_{10} 0.2 - \log_{10} 0.1}{0.2 - 0.1}$ or eg $\frac{-0.7 - -1}{0.2 - 0.1}$ seen 3.01 to 3.0103 isw or 10 $\log_{10} 2$ isw oe	M1 A1 [2]	**NB** $\frac{\log_{10} 2}{0.1}$ or $\frac{0.3}{0.1}$ allow –0.69 to –0.7 for $\log_{10} 2$ in gradient formula for **M1** **Examiner's Comments** Most knew what to do, but many slipped up by making a sign error in the numerator or by working with a rounded or truncated value of $\log_{10}$ 0.2, thus losing the accuracy mark.	condone omission of base 10; **B2** for 3.01 …unsupported	
		ii	one point C marked on curve between A and B or before A	B1 [1]	 **Examiner's Comments** Nearly all candidates correctly identified a suitable point on the curve. A few guessed wrongly and placed C to the right of B, and a very small number placed C off the curve altogether.	condone omission of label of C	
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance		
91			$\left[\frac{dy}{dx}=\right] kx^2$ soi	M1	$k > 0$	NB $6x^2$	
			When $x = 2$, $\left[\frac{dy}{dx}=\right]$ 24	A1			
			$-\frac{1}{their\ 24}$	M1	their 24 must come from evaluating their derivative	M0 if their 24 from elsewhere eg integration	
			$x = 2$, $y = 16$	B1			
			$x + 24y = 386$ oe	A1	NB $y-16=-\frac{1}{24}(x-2)$ coefficients in any exact form eg $\frac{1}{24}x+y=\frac{193}{12}$ but not rounded or truncated decimals		
				[5]	**Examiner's Comments** Most candidates were familiar with this sort of question and obtained the first four marks without difficulty. A few slipped up with the arithmetic, and a similar number found the equation of the tangent. A very small number of candidates integrated or went straight to working with $y = mx + c$		

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	5		
92			kx^4	M1	$k > 0$	must **not** follow from use of $y = mx + c$
			$3x^4$	A1	may be seen later	
			$-7x + c$	B1	must follow from integration	
			$10 = (\text{their } 3) \times 2^4 - 7 \times 2 + c$ oe	M1	must be 3 terms on RHS including term in x^4, term in x and "c";	must **not** follow from use of $y = mx + c$
			$y = 3x^4 - 7x - 24$	A1 [5]	or $y = 3x^4 - 7x + c$ and $c = -24$ stated isw	must see "y =" or "f(x) = " at some point for A1
					Examiner's Comments The vast majority of candidates tackled this question successfully. A few slipped up with the arithmetic in finding c, and a small minority worked with $y = mx + c$ with $m = 12 x 3 - 7$ and failed to score.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance		
93		i	correct rearrangement of 400 = $\pi r^2 h$ seen, where h is not in the denominator	B1	eg $h=\frac{400}{\pi r^2}, rh=\frac{400}{\pi r}, \pi rh=\frac{400}{r}$ or $2\pi rh=\frac{2\times 400}{r}$ if **B0B0 allow SC2** for eg 400 $\pi r^2 h$ used	allow embedded versions of these must see	
			substitution seen to obtain given answer $A=2\pi r^2+\frac{800}{r}$ not from wrong working	B1 [2]	$\frac{800}{r}=\frac{2\times 400}{r}$ $\left(\text{or } \frac{2V}{r}\right)=\frac{2\times \pi r^2 h}{r}$ used to obtain A $2\pi r^2$ + $2\pi rh$ **Examiner's Comments** Most candidates scored full marks here, but poor algebra let some candidates down. A wide variety of solutions were seen, some of which very elaborate.	all the steps if starting from $A=2\pi r^2+\frac{800}{r}$	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	$\left(\frac{dA}{dr}=\right)4\pi r-\frac{800}{r^2}$ oe $\left(\frac{d^2A}{dr^2}=\right)4\pi+\frac{1600}{r^3}$ oe	B1 B1 B1 B1 [4]	for first term for second term **FT** to give non-zero first term **FT** negative power of r to give non-zero second term Examiner's Comments In spite of the correct expression being given in part (i), some candidates worked with an expression involving h, which inhibited much further progress. Some candidates worked with 800^{-r} and some disregarded π or treated it as a variable. The majority, however, differentiated successfully to obtain full marks.	A maximum of **B1B0B1B0** is available if 2nd term left in terms of h	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	their $\frac{dA}{dr} = 0$ seen	M1		
			$r = \sqrt[3]{\frac{200}{\pi}}$ or 3.99… isw	A1	A0 for two or more values eg r = 0, 3.99 or ± 3.99	NB 3.9929 4542466
			$\frac{d^2A}{dr^2} > 0$ justified so minimum oe or check gradient either side of *their* positive r	B1	eg $4\pi > 0$ and $\frac{1600}{r^3} > 0$ NB 12π or 37.699… to 38	simply stating that second derivative is positive is insufficient
			A = 300 to 301	A1 [4]	NB 300.53 0027931	ignore units
					Examiner's Comments A sizeable minority of candidates failed to score any marks in this part, beginning with an inequality in the second derivative. A good number of candidates started on the right track by setting the first derivative to zero, but then failed to make progress. Only rarely did candidates successfully find r and A and then use the second derivative correctly to establish that they had indeed found the minimum surface area.	
			Total	10		

Question			Answer/Indicative content	Marks	Part marks and guidance	
94			$\frac{1}{(5-2x^3)^2}=(5-2x^3)^{-2}$ $\frac{\mathrm{d}}{\mathrm{d}x}(5-2x^3)^{-2}=(-6x^2)(-2)(5-2x^3)^{-3}$ $= 12x^2(5-2x^3)^{-3}$ isw	M1 A1 A1cao [3]	Chain rule on $(5-2x^3)^{-2}$ correct expression, allow $(-6x^2)(-2)u^{-3}$ o.e. or $\frac{12x^2}{(5-2x^3)^3}$ isw	or quotient (or product) e.g. $\frac{(5-2x^3)^2.0-1.2(5-2x^3)(-6x^2)}{(5-2x^3)^4}$ M1A1 [must have correct denom for M1] $u\,v' - v\,u'$ in QR is M0
					Examiner's Comments This was a straightforward test of the chain rule, in which over three quarters of the candidates scored full marks. Occasionally we saw a quotient rule used, which required to be simplified to gain full marks. Another occasional error was to get the wrong sign, e.g. $-12x^2(5-2x^3)^{-3}$.	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
95		i	$2x^{-\frac{1}{3}}+\frac{2}{3}y^{-\frac{2}{3}}\frac{\mathrm{d}y}{\mathrm{d}x}=0$ $\Rightarrow\frac{\mathrm{d}y}{\mathrm{d}x}=-3x^{-\frac{1}{3}}y^{\frac{2}{3}}$ o.e.	M1 A1 A1 [3]	$\frac{\mathrm{d}}{\mathrm{d}x}(y^{\frac{1}{3}})=\frac{1}{3}y^{-\frac{2}{3}}\frac{\mathrm{d}y}{\mathrm{d}x}$ seen correct equation must simplify 2 / (2/3) = 3 **Examiner's Comments** The implicit derivative here was a straightforward example, and virtually all the candidates got the derivative equation correct. However, simplifying the fractional expression to get the final mark was often missing or incorrect: in particular, many learners made mistakes when dividing 2 by 2/3.	mark final answer
		ii	when $x = 1$, $y = 8$, $\Rightarrow\frac{\mathrm{d}y}{\mathrm{d}x}=-3\times4=-12$	M1 A1cao [2]	substituting both $x = 1$ and $y = 8$ into their dy/dx NB check power of x is correct in part (i) **Examiner's Comments** There were two easy marks here, and virtually all candidates achieved the 'M' mark for substituting for x and y in their derivative.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
96		i	$\frac{1}{2}x = \sin y$ $\Rightarrow x = 2\sin y$ $\frac{dx}{dy} = 2\cos y$	B1cao B1cao [2]	Examiner's Comments The majority of candidates scored both these marks. Occasionally they found dy/dx instead of dx/dy and lost a mark.	
		ii	$\frac{dx}{dt} = \frac{dx}{dy} \times \frac{dy}{dt}$ When $x = 1$, $y = \arcsin\frac{1}{2} = \frac{\pi}{6}$ so $\frac{dx}{dy} = 2\cos\frac{\pi}{6} = \sqrt{3}$ or $\sin y = ½$ $\Rightarrow \cos y = \sqrt{1-\sin^2 y}$ $= \sqrt{(1 - ¼)} = \frac{\sqrt{3}}{2}$ $\Rightarrow \frac{dx}{dy} = \sqrt{3}$ or $\frac{dy}{dx} = \frac{1}{\sqrt{1-\frac{1}{4}x^2}}.\frac{1}{2}$	M1 M1 A1 A1 M1 A1 A1 M1A1 A1 A1	o.e. substituting $x = 1$ into $y = \arcsin ½ x$ π/6 or. $\frac{dy}{dx} = \frac{1}{\sqrt{3}}$ $\sqrt{(1 - ¼)}$ soi, e.g. $\frac{dy}{dt} = \frac{2}{\sqrt{3}}$	condone 30° soi e.g. by $\frac{dy}{dt} = \frac{2}{\sqrt{3}}$ must be

Question			Answer/Indicative content	Marks	Part marks and guidance		
			when $x = 1$, $\frac{dy}{dx} = \frac{1}{\sqrt{3}}$ $\Rightarrow 2 = \sqrt{3}.\frac{dy}{dt}$, $\frac{dy}{dt} = \frac{2}{\sqrt{3}}$	[5]	or $\frac{2\sqrt{3}}{3}$ **Examiner's Comments** Virtually all candidates wrote down a chain rule and scored 1 mark. Thereafter, many scored all the remaining marks. Errors were caused by muddling derivatives like dy/dx and dx/dy. Occasionally candidates attempted to use the derivative of arcsin x, though this was often incorrect through missing out the '½' factor.	exact, but isw if appro ximated	
			Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance	
97		i	$(\frac{\pi}{2}, 0)$, $(0, \frac{1}{2})$	B1B1 [2]	or $y = 0 \Rightarrow x = \pi/2$; $x = 0 \Rightarrow y = ½$ (isw) | Ignore incorrect labelling **Examiner's Comments** Some candidates lost marks here from working in degrees rather than radians. It is important that candidates state both the coordinates the right way round, so 'A = π/2 ' and 'B = ½ ' scored zero.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{dy}{dx} = \frac{(2-\sin x)(-\sin x) - \cos x(-\cos x)}{(2-\sin x)^2}$ when $\frac{dy}{dx}=0$ $(2-\sin x)(-\sin x)-\cos x(-\cos x)=0$ $\Rightarrow 1 - 2\sin x = 0$ $\Rightarrow x = \frac{\pi}{6},\ y = \frac{\sqrt{3}}{3}$ o.e.	M1 A1 M1 M1 A1 A1 A1 [7]	correct quotient or product rule correct expression (isw) setting (only) their numerator to zero use of $\sin^2 x + \cos^2 x = 1$ must be exact, isw	denom must be correct at some stage missing brackets may be inferred from subsequent work not denominator withhold if denom is set to zero

Examiner's Comments

Over half got full marks here. The quotient rule was well answered, and the subsequent simplification using $\sin^2 x + \cos^2 x = 1$ was good. The most common error was in the sign of the derivatives of sin x and cos x, which could fortuitously lead to the correct turning point – the final 'A' marks here being withheld in this case.

Question			Answer/Indicative content	Marks	Part marks and guidance		
		iii	(A) $\int_0^{\frac{\pi}{2}} \frac{\cos x}{2-\sin x}[\mathrm{d}x] = [-\ln(2-\sin x)]_0^{\frac{\pi}{2}}$ or let $u = 2 - \sin x$, $\mathrm{d}u/\mathrm{d}x = -\cos x$ $= \int_2^1 -\frac{1}{u}\mathrm{d}u$ $= [-\ln u]_2^1$ = ln 2	B1ft M1 A1 M1 A1 A1 [4]	correct integral and limits $c \ln(2 - \sin x)$ $c = -1$ $\int -\frac{1}{u}[\mathrm{d}u]$ (ignore limits) or $\int_1^2 \frac{1}{u}[\mathrm{d}u]$ [–ln *u*] (ignore limits) or $[\ln u]_1^2$ –ln (½) is A0, isw after ln 2	ft their π/2, not 90°, limits may be implied from subsequent work or $u = \sin x$, $\mathrm{d}u/\mathrm{d}x = \cos x$ $\int \frac{1}{2-u}[\mathrm{d}u]$ [–ln(2 – *u*)] not ln 2 – ln 1	**Examiner's Comments** Most candidates used a substitution u = 2 – sin *x*. Errors thereafter were du / dx = cos *x*, or getting the limits the wrong way round, perhaps under the misconception that the larger number must be the upper limit of the integral).
		iii	(B) $\int_0^k \frac{\cos x}{2-\sin x}[\mathrm{d}x] = \frac{1}{2}\ln 2$	M1	equating integral from 0 to *k* or from *k* to π/2 to ½	or equating integral from 0 to *k* to integral from *k* to	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\Rightarrow \ln 2 - \ln(2 - \sin k) = \frac{1}{2} \ln 2$	A1	their area	$\pi/2$
				M1	o.e. e.g. $\ln(2 - \sin k) = \frac{1}{2} \ln 2$	$\ln 2 - \ln(2 - \sin k) = \ln(2 - \sin k)$ dep first M1
			$\ln (2 - \sin k) = \frac{1}{2} \ln 2 = \ln \sqrt{2}$	A1	eliminating logarithms correctly	
			$\Rightarrow 2 - \sin k = \sqrt{2}$		o.e. e.g $(2 - \sin k)^2 = 2$	
			$\Rightarrow \sin k = 2 - \sqrt{2}$	A1cao		
			$\Rightarrow k = \arcsin(2 - \sqrt{2})$*	M1	NB AG	
			or $\int_0^{\arcsin(2-\sqrt{2})} \frac{\cos x}{2-\sin x}[\mathrm{d}x] = [-\ln(2-\sin x)]_0^{\arcsin(2-\sqrt{2})}$		SC: verifying **(max 3 marks out of 5)**: attempt to find integral from 0 to $\arcsin(2-\sqrt{2})$	or from $\arcsin(2-\sqrt{2})$ to $\pi/2$
			$= \ln 2 - \ln(2 - 2 + \sqrt{2}) = \ln 2 - \ln\sqrt{2}$	A1	correct expression	
			$= \ln 2 - \frac{1}{2} \ln 2 = \frac{1}{2} \ln 2$*	A1	N.B AG	
				[5]		
					Examiner's Comments This question was the most demanding in the paper, with nearly half the candidates scoring zero marks. Often the problem seemed to lie with getting expressions consistent with the limits of the integral, either in terms of *x* or *u*.	
			Total	18		